Instituto Superior Técnico – DEEC
Control of Cyber-Physical Systems
Mock Assessment 2

Modules 3.1–3.6

Duration:	30 minutes
Number of problems:	4
Total:	20 points
Recommended time:	approximately 7 minutes per problem
Aiding material:	simple calculator

Instructions

- Answer all four problems.
- Justify your answers. Unrelated statements or unexplained formulas may receive no credit.
- Show the main algebraic steps used to obtain numerical results.
- Unless otherwise stated, all quantities are real-valued.
- Use the notation and conventions of the course.

Mock MAP2 – Modules 3.1–3.6

Duration: 30 minutes **Total:** 20 points

Instructions. Answer all four problems. Justify your answers and show the main algebraic steps. A simple calculator is allowed.

Problem 1 [5 points]

Consider the scalar model

$$y(k) = a\,x(k) + e(k),$$

where a is an unknown constant parameter.

An experiment provides the following measurements:

k	1	2	3	4
$x(k)$	1	2	3	4
$y(k)$	1.1	1.9	3.2	3.8

(a) Write the least-squares criterion used to estimate a. **[1.0 points]**

(b) Derive the normal equation satisfied by the least-squares estimate $\hat{a}$. **[1.5 points]**

(c) Compute the numerical value of $\hat{a}$. **[1.5 points]**

(d) State the condition on the data $x(k)$ that guarantees uniqueness of the estimate. **[1.0 points]**

Problem 2 [5 points]

Consider the dynamic model

$$y(k) = a\,y(k-1) + b\,u(k-1) + e(k),$$

where a and b are unknown parameters.

For three observations, the measured data are

k	1	2	3
$y(k-1)$	1	2	1
$u(k-1)$	0	1	2
$y(k)$	0.8	2.1	1.9

(a) Write the model in regression form

$$y(k) = \boldsymbol{\phi}^\top(k-1)\boldsymbol{\theta} + e(k),$$

explicitly identifying $\boldsymbol{\phi}(k-1)$ and $\boldsymbol{\theta}$. **[1.0 points]**

(b) Construct the regression matrix Φ and observation vector $\mathbf{y}$. **[1.0 points]**

(c) Compute $\Phi^\top\Phi$ and determine whether the two parameters are uniquely identifiable from these data. **[1.5 points]**

(d) Compute the least-squares estimate

$$\hat{\boldsymbol{\theta}} = (\Phi^\top\Phi)^{-1}\Phi^\top\mathbf{y}.$$

[1.5 points]

Problem 3 [5 points]

A recursive least-squares estimator is used for the scalar model

$$y(k) = \phi(k-1)\theta + e(k).$$

At time $k-1$,

$$\hat{\theta}(k-1) = 1, \qquad P(k-1) = 4.$$

The new measurement is

$$\phi(k-1) = 2, \qquad y(k) = 3.$$

Assume no forgetting, $\lambda = 1$.

Use

$$K(k) = \frac{P(k-1)\phi(k-1)}{1 + \phi^2(k-1)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \phi(k-1)\hat{\theta}(k-1)\right],$$

and

$$P(k) = [1 - K(k)\phi(k-1)]\,P(k-1).$$

(a) Compute the RLS gain $K(k)$. **[1.5 points]**

(b) Compute the prediction error

$$\varepsilon(k) = y(k) - \phi(k-1)\hat{\theta}(k-1).$$

[1.0 points]

(c) Compute the updated parameter estimate $\hat{\theta}(k)$. **[1.0 points]**

(d) Compute $P(k)$ and state whether the estimation uncertainty increased or decreased after incorporating the new measurement. **[1.5 points]**

Problem 4 [5 points]

Consider the identification model

$$y(k) = a\,u(k-1) + v(k),$$

where

$$v(k) = e(k) + c\,e(k-1),$$

and $e(k)$ is a zero-mean white-noise sequence with variance σ_e^2.

(a) For $c \neq 0$, determine whether $v(k)$ is white noise. Compute $\mathrm{Cov}(v(k), v(k-1))$ and justify your answer. **[1.5 points]**

(b) Suppose that $u(k-1)$ is uncorrelated with $v(k)$. Explain why the least-squares estimate of a can be unbiased under this assumption. **[1.5 points]**

(c) Suppose instead that the experiment is performed in closed loop and $u(k-1)$ becomes correlated with $e(k-1)$. Explain why this can lead to a biased least-squares estimate. **[1.0 points]**

(d) Give one practical way of improving parameter identifiability when the input does not provide sufficient excitation. **[1.0 points]**

Solutions

Solution 1

(a) Least-squares criterion

The residual for a given parameter a is

$$\varepsilon(k) = y(k) - a\,x(k).$$

Therefore, the least-squares criterion is

$$J(a) = \sum_{k=1}^{4} \left(y(k) - a\,x(k)\right)^2 .$$

This is the standard scalar least-squares criterion. *(Module 3.1, slides 4–22.)*

(b) Normal equation

Differentiate $J(a)$ with respect to a:

$$\frac{dJ}{da} = -2\sum_{k=1}^{4} x(k)\left[y(k) - a\,x(k)\right].$$

At the minimum,

$$\frac{dJ}{da} = 0.$$

Hence

$$\sum_{k=1}^{4} x(k)y(k) - a\sum_{k=1}^{4} x^2(k) = 0.$$

Therefore,

$$\boxed{\left(\sum_{k=1}^{4} x^2(k)\right)\hat{a} = \sum_{k=1}^{4} x(k)y(k)}$$

and consequently

$$\boxed{\hat{a} = \frac{\sum_{k=1}^{4} x(k)y(k)}{\sum_{k=1}^{4} x^2(k)}}.$$

This is the scalar form of the least-squares normal equation. *(Module 3.1, slides 4–22.)*

(c) Numerical estimate

First compute

$$\sum_{k=1}^{4} x(k)y(k) = 1(1.1) + 2(1.9) + 3(3.2) + 4(3.8).$$

Thus

$$\sum_{k=1}^{4} x(k)y(k) = 1.1 + 3.8 + 9.6 + 15.2 = 29.7.$$

Also,

$$\sum_{k=1}^{4} x^2(k) = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30.$$

Hence

$$\boxed{\hat{a} = \frac{29.7}{30} = 0.99}.$$

(d) Uniqueness

The scalar information matrix is

$$\Lambda = \sum_{k=1}^{4} x^2(k).$$

The estimate is unique provided

$$\Lambda > 0.$$

Therefore,

$$\boxed{\sum_{k=1}^{4} x^2(k) > 0}$$

is required, meaning that the input data cannot be identically zero.

This is the scalar version of the requirement that the information matrix be nonsingular. *(Module 3.2, slides 3–20.)*

Solution 2

(a) Regression form

The model is

$$y(k) = a\, y(k-1) + b\, u(k-1) + e(k).$$

Define

$$\boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad \boldsymbol{\phi}(k-1) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}.$$

Then

$$\boxed{y(k) = \boldsymbol{\phi}^{\top}(k-1)\boldsymbol{\theta} + e(k)}.$$

This is the standard regression representation used for dynamic-model parameter estimation. *(Module 3.1, slides 4–22.)*

(b) Regression matrix

Using the three observations,

$$\Phi = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 0.8 \\ 2.1 \\ 1.9 \end{bmatrix}.$$

Therefore,

$$\boxed{\Phi = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 0.8 \\ 2.1 \\ 1.9 \end{bmatrix}}.$$

(c) Information matrix and identifiability

Compute

$$\Phi^{\top}\Phi = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore,

$$\Phi^\top\Phi = \begin{bmatrix} 1+4+1 & 0+2+2 \\ 0+2+2 & 0+1+4 \end{bmatrix} = \boxed{\begin{bmatrix} 6 & 4 \\ 4 & 5 \end{bmatrix}}.$$

Its determinant is

$$\det(\Phi^\top\Phi) = 6(5) - 4(4) = 30 - 16 = 14.$$

Since

$$14 \neq 0,$$

the matrix is invertible.

Hence the two parameters are uniquely identifiable from these data:

$$\boxed{\text{The parameters } a \text{ and } b \text{ are uniquely identifiable.}}$$

This follows from the requirement that the information matrix be nonsingular. *(Module 3.2, slides 3–20.)*

(d) Least-squares estimate

First compute

$$\Phi^\top\mathbf{y} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0.8 \\ 2.1 \\ 1.9 \end{bmatrix}.$$

Thus

$$\Phi^\top\mathbf{y} = \begin{bmatrix} 0.8+4.2+1.9 \\ 2.1+3.8 \end{bmatrix} = \begin{bmatrix} 6.9 \\ 5.9 \end{bmatrix}.$$

The inverse of the information matrix is

$$(\Phi^\top\Phi)^{-1} = \frac{1}{14}\begin{bmatrix} 5 & -4 \\ -4 & 6 \end{bmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\theta}} = \frac{1}{14}\begin{bmatrix} 5 & -4 \\ -4 & 6 \end{bmatrix}\begin{bmatrix} 6.9 \\ 5.9 \end{bmatrix}.$$

For a,

$$\hat{a} = \frac{1}{14}\left(5(6.9) - 4(5.9)\right)$$
$$= \frac{34.5 - 23.6}{14} = \frac{10.9}{14} \approx 0.7786.$$

For b,

$$\hat{b} = \frac{1}{14}\left(-4(6.9) + 6(5.9)\right)$$
$$= \frac{-27.6 + 35.4}{14} = \frac{7.8}{14} \approx 0.5571.$$

Hence

$$\boxed{\hat{\boldsymbol{\theta}} = \begin{bmatrix} 0.7786 \\ 0.5571 \end{bmatrix}}.$$

Solution 3

(a) RLS gain

The scalar gain is

$$K(k) = \frac{P(k-1)\phi(k-1)}{1+\phi^2(k-1)P(k-1)}.$$

Substitute

$$P(k-1) = 4, \qquad \phi(k-1) = 2.$$

Therefore,

$$K(k) = \frac{4(2)}{1+2^2(4)} = \frac{8}{1+16} = \boxed{\frac{8}{17}}.$$

Numerically,

$$K(k) \approx 0.4706.$$

(Module 3.3, slides 3–15.)

(b) Prediction error

The prediction error is

$$\varepsilon(k) = y(k) - \phi(k-1)\hat{\theta}(k-1).$$

Thus

$$\varepsilon(k) = 3 - 2(1) = \boxed{1}.$$

(c) Parameter update

The RLS update is

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k).$$

Hence

$$\hat{\theta}(k) = 1 + \frac{8}{17}(1) = \frac{17}{17} + \frac{8}{17} = \boxed{\frac{25}{17}}.$$

Therefore,

$$\boxed{\hat{\theta}(k) \approx 1.4706}.$$

(d) Covariance update

Using

$$P(k) = [1 - K(k)\phi(k-1)]\, P(k-1),$$

we obtain

$$P(k) = \left[1 - \frac{8}{17}(2)\right] 4.$$

Thus

$$P(k) = \left(1 - \frac{16}{17}\right) 4 = \frac{1}{17}4.$$

Therefore,

$$\boxed{P(k) = \frac{4}{17} \approx 0.2353}.$$

Since

$$P(k) = 0.2353 < 4 = P(k-1),$$

the estimation uncertainty decreases after incorporating the new measurement.

This is consistent with the interpretation of $P(k)$ as the estimation error covariance. *(Module 3.3, slides 3–15.)*

Solution 4

(a) Is $v(k)$ white?

We have

$$v(k) = e(k) + c\,e(k-1).$$

Consider the covariance between consecutive samples:

$$\mathrm{Cov}(v(k), v(k-1)) = E[v(k)v(k-1)]$$

because the process has zero mean.

Since

$$v(k-1) = e(k-1) + c\,e(k-2),$$

we obtain

$$E[v(k)v(k-1)] = E[(e(k) + ce(k-1))(e(k-1) + ce(k-2))].$$

Expanding,

$$E[e(k)e(k-1)] + cE[e(k)e(k-2)] + cE[e^2(k-1)] + c^2E[e(k-1)e(k-2)].$$

Because $e(k)$ is white noise, different-time samples are uncorrelated. Therefore,

$$E[e(k)e(k-1)] = 0,$$

$$E[e(k)e(k-2)] = 0,$$

and

$$E[e(k-1)e(k-2)] = 0.$$

Only

$$E[e^2(k-1)] = \sigma_e^2$$

remains. Hence

$$\boxed{\mathrm{Cov}(v(k), v(k-1)) = c\sigma_e^2}.$$

For $c \neq 0$,

$$\mathrm{Cov}(v(k), v(k-1)) \neq 0.$$

Therefore,

$$\boxed{v(k) \text{ is colored noise, not white noise, when } c \neq 0.}$$

This is the type of colored-noise situation discussed in Modules 3.5 and 3.6. *(Module 3.5, slides 3–10; Module 3.6, slides 3–22.)*

(b) Uncorrelated input and noise

For the scalar model

$$y(k) = a\,u(k-1) + v(k),$$

the least-squares estimate can be written as

$$\hat{a} = a + \frac{\sum_k u(k-1)v(k)}{\sum_k u^2(k-1)}.$$

Taking expectations,

$$E[\hat{a}] = a + \frac{E\left[\sum_k u(k-1)v(k)\right]}{\sum_k u^2(k-1)}.$$

If

$$E[u(k-1)v(k)] = 0,$$

then the expected cross-term is zero. Consequently,

$$\boxed{E[\hat{a}] = a}.$$

Thus the estimator is unbiased under the stated uncorrelatedness condition.

(Module 3.6, slides 3–22.)

(c) Closed-loop correlation and bias

The disturbance contains

$$v(k) = e(k) + c\,e(k-1).$$

If the closed-loop experiment makes $u(k-1)$ correlated with $e(k-1)$, then generally

$$E[u(k-1)v(k)] = E[u(k-1)e(k)] + cE[u(k-1)e(k-1)].$$

Even if

$$E[u(k-1)e(k)] = 0,$$

the second term may be nonzero:

$$cE[u(k-1)e(k-1)] \neq 0.$$

Therefore,

$$E[u(k-1)v(k)] \neq 0,$$

and the noise term does not disappear in the expected least-squares estimate.

Hence,

$$\boxed{E[\hat{a}] \neq a}$$

in general, meaning that the LS estimate can be biased.

This is precisely the closed-loop/colored-noise identifiability issue emphasized in Module 3.6. *(Module 3.6, slides 3–22.)*

(d) Improving excitation

A practical solution is to introduce an additional sufficiently rich external excitation signal into the experiment, while maintaining the system within safe operating limits.

For example, an external signal can be injected into the input so that the regressor contains sufficient independent information for the parameters to be distinguished.

Thus one valid answer is

inject an external excitation signal into the closed-loop experiment.

The key requirement is that the resulting data provide sufficient excitation/information for the parameters to be identifiable. *(Module 3.2, slides 3–20.)*

Where to Study Next

- **Module 3.1, slides 4–22:** least-squares criterion, normal equations, regressors, and dynamic-model parameter estimation.
- **Module 3.2, slides 3–20:** information matrix, excitation, identifiability, and closed-loop experiment design.

- **Module 3.3, slides 3–15:** RLS, gain, covariance, and recursive parameter updates.
- **Module 3.4, slides 3–18:** forgetting factors, adaptation, and colored-noise extensions.
- **Module 3.5, slides 3–10:** stochastic processes, stationarity, white noise, and covariance functions.
- **Module 3.6, slides 3–22:** LS bias, white versus colored noise, consistency, and maximum likelihood.

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Mock MAP2 – Modules 3.1–3.6

Duration: 30 minutes **Total:** 20 points

Instructions. Answer all four problems. Justify your answers and show the main algebraic steps. A simple calculator is allowed.

Problem 1 [5 points]

Consider the scalar identification model

$$y(k) = a\,x(k) + e(k),$$

where a is an unknown parameter.

The following measurements are available:

k	1	2	3	4
$x(k)$	1	2	3	4
$y(k)$	0.9	2.2	2.8	4.1

(a) Write the least-squares cost function $J(a)$ and derive the normal equation satisfied by $\hat{a}$. **[1.5 points]**

(b) Compute $\hat{a}$ numerically. **[1.5 points]**

(c) Compute the residuals

$$\varepsilon(k) = y(k) - \hat{a}x(k)$$

and the residual sum of squares

$$J(\hat{a}) = \sum_{k=1}^{4} \varepsilon^2(k).$$

[1.0 points]

(d) Assuming the noise variance is estimated using $N-1$ degrees of freedom, compute

$$\hat{\sigma}^2 = \frac{J(\hat{a})}{N-1}$$

and give the corresponding estimate of the variance of $\hat{a}$. **[1.0 points]**

Problem 2 [5 points]

Consider the dynamic model

$$y(k) = a\,y(k-1) + b\,u(k-1) + e(k).$$

For an experiment, the following sums have been computed:

$$\sum_{i=1}^{N} y^2(i-1) = 30,$$

$$\sum_{i=1}^{N} u^2(i-1) = 50,$$

$$\sum_{i=1}^{N} y(i-1)u(i-1) = 10,$$

$$\sum_{i=1}^{N} y(i)y(i-1) = 15,$$

$$\sum_{i=1}^{N} y(i)u(i-1) = 30.$$

(a) Write the regression vector $\boldsymbol{\phi}(i-1)$ and parameter vector $\boldsymbol{\theta}$. **[1.0 points]**

(b) Construct the information matrix

$$\Phi^\top \Phi$$

and the vector

$$\Phi^\top \mathbf{y}$$

directly from the given sums. **[1.5 points]**

(c) Compute the batch least-squares estimates of a and b. **[1.5 points]**

(d) State the condition that the information matrix must satisfy for the estimate to be unique. Verify this condition for the given experiment. **[1.0 points]**

Problem 3 [5 points]

A recursive least-squares estimator with exponential forgetting is applied to

$$y(k) = \boldsymbol{\phi}^\top(k-1)\boldsymbol{\theta} + e(k).$$

At time $k-1$,

$$\hat{\boldsymbol{\theta}}(k-1) = \begin{bmatrix} 0.4 \\ -0.1 \end{bmatrix}, \qquad P(k-1) = 10 I_2,$$

and the new data are

$$\boldsymbol{\phi}(k-1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y(k) = 1.0.$$

Use the forgetting factor

$$\lambda = 0.98.$$

Use the RLS equations

$$K(k) = \frac{P(k-1)\boldsymbol{\phi}(k-1)}{\lambda + \boldsymbol{\phi}^\top(k-1)P(k-1)\boldsymbol{\phi}(k-1)},$$

$$\hat{\boldsymbol{\theta}}(k) = \hat{\boldsymbol{\theta}}(k-1) + K(k)\left[y(k) - \boldsymbol{\phi}^\top(k-1)\hat{\boldsymbol{\theta}}(k-1)\right],$$

and

$$P(k) = \frac{1}{\lambda}\left[I - K(k)\boldsymbol{\phi}^\top(k-1)\right]P(k-1).$$

(a) Compute the gain vector $K(k)$. **[1.5 points]**

(b) Compute the prediction error and the updated parameter vector $\hat{\boldsymbol{\theta}}(k)$. **[1.5 points]**

(c) Compute the updated covariance matrix $P(k)$. **[1.0 points]**

(d) Explain qualitatively the role of $\lambda < 1$. What happens to the effective memory of the estimator when λ decreases? **[1.0 points]**

Problem 4 [5 points]

Let $\{e(k)\}$ be a zero-mean white-noise sequence with

$$E[e(k)^2] = \sigma_e^2.$$

Define the process

$$v(k) = e(k) + c\,e(k-1).$$

(a) Compute the mean $E[v(k)]$. **[0.75 points]**

(b) Compute the variance

$$\mathrm{Var}(v(k)).$$

[1.0 points]

(c) Compute the covariance

$$\mathrm{Cov}(v(k), v(k-1)).$$

[1.25 points]

(d) For $c \neq 0$, is $v(k)$ white noise? Justify using your result from part (c). **[1.0 points]**

(e) Explain briefly why the distinction between white and colored noise is relevant when using least squares for parameter estimation. **[1.0 points]**

Solutions

Solution 1

(a) Cost function and normal equation

The residual is

$$\varepsilon(k) = y(k) - a\,x(k).$$

Therefore,

$$J(a) = \sum_{k=1}^{4} \left[y(k) - a\,x(k)\right]^2.$$

Differentiating,

$$\frac{dJ}{da} = -2\sum_{k=1}^{4} x(k)\left[y(k) - a\,x(k)\right].$$

At the minimum,

$$\frac{dJ}{da} = 0.$$

Hence

$$\sum_{k=1}^{4} x(k)y(k) = \hat{a}\sum_{k=1}^{4} x^2(k),$$

so

$$\boxed{\hat{a} = \frac{\sum_{k=1}^{4} x(k)y(k)}{\sum_{k=1}^{4} x^2(k)}}.$$

(Module 3.1, slides 4–22.)

(b) Numerical estimate

First,

$$\sum_{k=1}^{4} x(k)y(k) = 1(0.9) + 2(2.2) + 3(2.8) + 4(4.1).$$

Thus

$$\sum x(k)y(k) = 0.9 + 4.4 + 8.4 + 16.4 = 30.1.$$

Also,

$$\sum_{k=1}^{4} x^2(k) = 1 + 4 + 9 + 16 = 30.$$

Therefore,

$$\boxed{\hat{a} = \frac{30.1}{30} = 1.00333}.$$

(c) Residuals and residual sum of squares

Using

$$\hat{a} = 1.00333,$$

the residuals are

$$\varepsilon(1) = 0.9 - 1.00333(1) = -0.10333,$$

$$\varepsilon(2) = 2.2 - 1.00333(2) = 0.19333,$$

$$\varepsilon(3) = 2.8 - 1.00333(3) = -0.21000,$$
$$\varepsilon(4) = 4.1 - 1.00333(4) = 0.08667.$$

Therefore,

$$J(\hat{a}) = (-0.10333)^2 + (0.19333)^2 + (-0.21000)^2 + (0.08667)^2.$$

Hence

$$\boxed{J(\hat{a}) \approx 0.10297}.$$

(d) Noise variance and parameter variance

There are $N = 4$ measurements. Using $N - 1 = 3$,

$$\hat{\sigma}^2 = \frac{0.10297}{3} \approx \boxed{0.03432}.$$

For the scalar parameter,

$$\widehat{\mathrm{Var}}(\hat{a}) = \hat{\sigma}^2 \left(\sum_{k=1}^{4} x^2(k) \right)^{-1}.$$

Therefore,

$$\widehat{\mathrm{Var}}(\hat{a}) = \frac{0.03432}{30} \approx \boxed{1.144 \times 10^{-3}}.$$

(Module 3.1, slides 4–22; Module 3.2, slides 3–20.)

Solution 2

(a) Regression representation

The model

$$y(k) = a\, y(k-1) + b\, u(k-1) + e(k)$$

can be written as

$$y(k) = \boldsymbol{\phi}^\top(k-1)\boldsymbol{\theta} + e(k),$$

with

$$\boxed{\boldsymbol{\phi}(k-1) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}.}$$

(Module 3.1, slides 4–22.)

(b) Information matrix and data vector

The information matrix is

$$\Phi^\top \Phi = \sum_{i=1}^{N} \boldsymbol{\phi}(i-1)\boldsymbol{\phi}^\top(i-1).$$

Therefore,

$$\Phi^\top \Phi = \begin{bmatrix} \sum y^2(i-1) & \sum y(i-1)u(i-1) \\ \sum y(i-1)u(i-1) & \sum u^2(i-1) \end{bmatrix}.$$

Using the supplied values,

$$\boxed{\Phi^\top \Phi = \begin{bmatrix} 30 & 10 \\ 10 & 50 \end{bmatrix}.}$$

Similarly,

$$\Phi^\top \mathbf{y} = \begin{bmatrix} \sum y(i)y(i-1) \\ \sum y(i)u(i-1) \end{bmatrix} = \boxed{\begin{bmatrix} 15 \\ 30 \end{bmatrix}}.$$

(c) Least-squares estimate

The normal equations are

$$\begin{bmatrix} 30 & 10 \\ 10 & 50 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 15 \\ 30 \end{bmatrix}.$$

The determinant is

$$30(50) - 10(10) = 1500 - 100 = 1400.$$

Thus

$$(\Phi^\top \Phi)^{-1} = \frac{1}{1400} \begin{bmatrix} 50 & -10 \\ -10 & 30 \end{bmatrix}.$$

Hence

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \frac{1}{1400} \begin{bmatrix} 50 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 15 \\ 30 \end{bmatrix}.$$

For $\hat{a}$,

$$\hat{a} = \frac{750 - 300}{1400} = \frac{450}{1400} = \boxed{0.3214}.$$

For $\hat{b}$,

$$\hat{b} = \frac{-150 + 900}{1400} = \frac{750}{1400} = \boxed{0.5357}.$$

Therefore,

$$\boxed{\hat{\boldsymbol{\theta}} = \begin{bmatrix} 0.3214 \\ 0.5357 \end{bmatrix}}.$$

(d) Uniqueness

The least-squares estimate is unique if

$$\boxed{\Phi^\top \Phi \text{ is nonsingular.}}$$

Here,

$$\det(\Phi^\top \Phi) = 1400 \neq 0.$$

Therefore,

$$\boxed{\text{the estimate is unique.}}$$

The experiment contains sufficient independent information in the regressors to distinguish the two parameters. *(Module 3.2, slides 3–20.)*

Solution 3

(a) Gain vector

First compute

$$P(k-1)\boldsymbol{\phi}(k-1) = 10 I_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \end{bmatrix}.$$

The denominator is

$$\lambda + \boldsymbol{\phi}^\top P \boldsymbol{\phi}.$$

Since

$$P\boldsymbol{\phi} = \begin{bmatrix} 10 \\ 20 \end{bmatrix},$$

we have

$$\boldsymbol{\phi}^\top P\boldsymbol{\phi} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \end{bmatrix} = 50.$$

Therefore,

$$K(k) = \frac{1}{0.98 + 50} \begin{bmatrix} 10 \\ 20 \end{bmatrix}.$$

Hence

$$\boxed{K(k) = \begin{bmatrix} 0.19608 \\ 0.39216 \end{bmatrix}.}$$

(Module 3.3, slides 3–15; Module 3.4, slides 3–18.)

(b) Prediction error and parameter update

The predicted output is

$$\hat{y}(k) = \boldsymbol{\phi}^\top(k-1)\hat{\boldsymbol{\theta}}(k-1).$$

Thus

$$\hat{y}(k) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 0.4 \\ -0.1 \end{bmatrix} = 0.4 - 0.2 = 0.2.$$

Therefore,

$$\varepsilon(k) = y(k) - \hat{y}(k) = 1.0 - 0.2 = \boxed{0.8}.$$

The parameter update is

$$\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 0.4 \\ -0.1 \end{bmatrix} + \begin{bmatrix} 0.19608 \\ 0.39216 \end{bmatrix} (0.8).$$

Therefore,

$$\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 0.4 + 0.15686 \\ -0.1 + 0.31373 \end{bmatrix}.$$

Hence

$$\boxed{\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 0.55686 \\ 0.21373 \end{bmatrix}.}$$

(c) Covariance update

We have

$$P(k) = \frac{1}{0.98} \left[I - K\boldsymbol{\phi}^\top \right] 10 I_2.$$

First,

$$K\boldsymbol{\phi}^\top = \begin{bmatrix} 0.19608 \\ 0.39216 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.19608 & 0.39216 \\ 0.39216 & 0.78431 \end{bmatrix}.$$

Therefore,

$$I - K\boldsymbol{\phi}^\top = \begin{bmatrix} 0.80392 & -0.39216 \\ -0.39216 & 0.21569 \end{bmatrix}.$$

Multiplying by 10/0.98 gives

$$\boxed{P(k) \approx \begin{bmatrix} 8.203 & -4.002 \\ -4.002 & 2.201 \end{bmatrix}.}$$

(d) Role of the forgetting factor

With exponential forgetting, older measurements receive progressively smaller weight.

The effective memory is approximately

$$N_{\text{eff}} \approx \frac{1}{1-\lambda}.$$

For $\lambda = 0.98$,

$$N_{\text{eff}} \approx \frac{1}{0.02} = \boxed{50}.$$

Therefore, decreasing λ shortens the effective memory. This makes the estimator more responsive to parameter changes, but generally increases the estimation variance.

(Module 3.4, slides 3–18.)

Solution 4

(a) Mean

Since $e(k)$ is zero mean,

$$E[e(k)] = 0, \qquad E[e(k-1)] = 0.$$

Therefore,

$$\begin{aligned} E[v(k)] &= E[e(k) + ce(k-1)] \\ &= E[e(k)] + cE[e(k-1)] = 0. \end{aligned}$$

Hence

$$\boxed{E[v(k)] = 0}.$$

(b) Variance

Because the white-noise samples at different times are uncorrelated,

$$E[e(k)e(k-1)] = 0.$$

Therefore,

$$\text{Var}(v(k)) = E[v^2(k)].$$

Expanding,

$$E[v^2(k)] = E[(e(k) + ce(k-1))^2].$$

Thus

$$E[v^2(k)] = E[e^2(k)] + 2cE[e(k)e(k-1)] + c^2E[e^2(k-1)].$$

Hence

$$\text{Var}(v(k)) = \sigma_e^2 + c^2\sigma_e^2.$$

Therefore,

$$\boxed{\text{Var}(v(k)) = (1 + c^2)\sigma_e^2.}$$

(Module 3.5, slides 3–10.)

(c) Covariance between consecutive samples

We have

$$v(k) = e(k) + ce(k-1)$$

and

$$v(k-1) = e(k-1) + ce(k-2).$$

Since the mean is zero,

$$\mathrm{Cov}(v(k), v(k-1)) = E[v(k)v(k-1)].$$

Therefore,

$$E[(e(k) + ce(k-1))(e(k-1) + ce(k-2))].$$

Expanding,

$$E[e(k)e(k-1)] + cE[e(k)e(k-2)] + cE[e^2(k-1)] + c^2E[e(k-1)e(k-2)].$$

All cross-time terms vanish for white noise, leaving

$$\boxed{\mathrm{Cov}(v(k), v(k-1)) = c\sigma_e^2.}$$

(Module 3.5, slides 3–10.)

(d) Whiteness

A white-noise sequence must have zero covariance between distinct samples.

For $c \neq 0$,

$$\mathrm{Cov}(v(k), v(k-1)) = c\sigma_e^2 \neq 0.$$

Therefore,

$$\boxed{v(k) \text{ is not white noise for } c \neq 0.}$$

It is a colored-noise process.

(Module 3.5, slides 3–10.)

(e) Relevance to least squares

For least squares, the relationship between the regressor and the disturbance is important.

If the disturbance is correlated with the regressors, the cross-term does not vanish in the least-squares estimate. In particular, for

$$y(k) = \boldsymbol{\phi}^\top(k)\boldsymbol{\theta}_0 + v(k),$$

the estimate is affected by

$$E[\boldsymbol{\phi}(k)v(k)].$$

If

$$\boxed{E[\boldsymbol{\phi}(k)v(k)] \neq 0,}$$

the LS estimate can be biased.

Thus colored noise is not automatically sufficient to imply LS bias; what matters is its relationship with the regressors. This distinction is central to the course treatment of LS bias and identification. *(Module 3.6, slides 3–22.)*

Where to Study Next

- **Module 3.1, slides 4–22:** least-squares criteria, normal equations, regressors, uniqueness, and dynamic-model parameter estimation.
- **Module 3.2, slides 3–20:** information matrices, excitation, identifiability, and experiment design.
- **Module 3.3, slides 3–15:** recursive least squares, gain, covariance, and recursive updates.
- **Module 3.4, slides 3–18:** forgetting factors, effective memory, tracking, and covariance effects.
- **Module 3.5, slides 3–10:** means, variances, stationarity, white noise, and covariance functions.
- **Module 3.6, slides 3–22:** LS bias, white versus colored noise, consistency, likelihood, and maximum likelihood.

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Mock MAP2 – Modules 3.1–3.8

Duration: 30 minutes **Total:** 20 points

Instructions. Answer all four problems. Justify your answers and show the main algebraic steps. A simple calculator is allowed.

Problem 1 [5 points]

Consider the scalar model

$$y(k) = a\,x(k) + e(k),$$

where a is an unknown parameter.

The measured data are

k	1	2	3	4
$x(k)$	1	2	3	4
$y(k)$	1.0	2.1	2.9	4.2

(a) Write the least-squares functional $J(a)$ and derive the normal equation satisfied by $\hat{a}$. **[1.5 points]**

(b) Compute the least-squares estimate $\hat{a}$. **[1.0 points]**

(c) Compute the residual sum of squares

$$J(\hat{a}) = \sum_{k=1}^{4} \left[y(k) - \hat{a}x(k)\right]^2 .$$

[1.0 points]

(d) Assuming zero-mean white noise with unknown variance and using $N - 1$ degrees of freedom, compute $\hat{\sigma}^2$ and $\widehat{\mathrm{Var}}(\hat{a})$. **[1.5 points]**

Problem 2 [5 points]

Consider the ARX-type model

$$y(k) = a\,y(k-1) + b\,u(k-1) + e(k).$$

For the experiment, the following quantities have been calculated:

$$\sum_{k=1}^{N} y^2(k-1) = 40, \qquad \sum_{k=1}^{N} u^2(k-1) = 25,$$

$$\sum_{k=1}^{N} y(k-1)u(k-1) = 10,$$

$$\sum_{k=1}^{N} y(k)y(k-1) = 20, \qquad \sum_{k=1}^{N} y(k)u(k-1) = 15.$$

(a) Define the regression vector $\boldsymbol{\phi}(k-1)$ and parameter vector $\boldsymbol{\theta}$. **[1.0 points]**

(b) Construct $\Phi^\top\Phi$ and $\Phi^\top\mathbf{y}$ from the given sums. **[1.0 points]**

(c) Compute the least-squares estimate

$$\hat{\boldsymbol{\theta}} = (\Phi^\top\Phi)^{-1}\Phi^\top\mathbf{y}.$$

[2.0 points]

(d) State the condition for unique parameter identification and verify it for this experiment. **[1.0 points]**

Problem 3 [5 points]

A recursive least-squares estimator with exponential forgetting is used for

$$y(k) = \boldsymbol{\phi}^\top(k-1)\boldsymbol{\theta} + e(k).$$

At time $k-1$,

$$\hat{\boldsymbol{\theta}}(k-1) = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix}, \qquad P(k-1) = 10I_2,$$

and the new observation is

$$\boldsymbol{\phi}(k-1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad y(k) = 1.1.$$

The forgetting factor is

$$\lambda = 0.98.$$

Use

$$K(k) = \frac{P(k-1)\boldsymbol{\phi}(k-1)}{\lambda + \boldsymbol{\phi}^\top(k-1)P(k-1)\boldsymbol{\phi}(k-1)},$$

$$\hat{\boldsymbol{\theta}}(k) = \hat{\boldsymbol{\theta}}(k-1) + K(k)\left[y(k) - \boldsymbol{\phi}^\top(k-1)\hat{\boldsymbol{\theta}}(k-1)\right],$$

and

$$P(k) = \frac{1}{\lambda}\left[I - K(k)\boldsymbol{\phi}^\top(k-1)\right]P(k-1).$$

(a) Compute the gain vector $K(k)$. **[1.5 points]**

(b) Compute the prediction error and the updated parameter estimate $\hat{\boldsymbol{\theta}}(k)$. **[1.5 points]**

(c) Compute the updated covariance matrix $P(k)$. **[1.0 points]**

(d) The desired effective memory is approximately $N = 50$ samples. Using

$$N \simeq \frac{1}{1-\lambda},$$

determine an appropriate value of λ. **[1.0 points]**

Problem 4 [5 points]

An engineer is identifying a discrete-time plant from input-output data. Three candidate models are considered:

$$\begin{aligned}
M_1: &\quad y(k) = a_1 y(k-1) + b_1 u(k-1) + e(k),\\
M_2: &\quad y(k) = a_1 y(k-1) + a_2 y(k-2) + b_1 u(k-1) + b_2 u(k-2) + e(k),\\
M_3: &\quad y(k) = a_1 y(k-1) + a_2 y(k-2) + b_1 u(k-1) + b_2 u(k-2) + c_1 e(k-1) + e(k).
\end{aligned}$$

The engineer obtains the following validation results:

Model	M_1	M_2	M_3
AIC	120	108	110
BIC	125	118	123

(a) Identify the model structures represented by M_1, M_2 and M_3 (ARX/ARMAX terminology). **[1.0 points]**

(b) According to AIC and BIC separately, which model would each criterion select? **[1.0 points]**

(c) Explain why the model with the lowest training error is not necessarily the best model for prediction or validation. **[1.0 points]**

(d) Give two appropriate validation checks that could be performed after estimating a model. **[1.0 points]**

(e) Suppose the input signal used during the experiment contains almost no variation over the frequency band of interest. Explain why this can make reliable parameter estimation difficult. **[1.0 points]**

Solutions

Solution 1

(a) Least-squares functional and normal equation

The residual is

$$\varepsilon(k) = y(k) - a\,x(k).$$

Therefore,

$$J(a) = \sum_{k=1}^{4} \left[y(k) - a\,x(k)\right]^2 .$$

Differentiating with respect to a,

$$\frac{dJ}{da} = -2\sum_{k=1}^{4} x(k)\left[y(k) - a\,x(k)\right].$$

At the minimum,

$$\frac{dJ}{da} = 0.$$

Hence

$$\sum_{k=1}^{4} x(k)y(k) = \hat{a}\sum_{k=1}^{4} x^2(k),$$

and therefore

$$\boxed{\hat{a} = \frac{\sum_{k=1}^{4} x(k)y(k)}{\sum_{k=1}^{4} x^2(k)}}.$$

(Module 3.1, slides 4–22; Problem Series S3P2–S3P4.)

(b) Numerical estimate

Compute

$$\sum x(k)y(k) = 1(1.0) + 2(2.1) + 3(2.9) + 4(4.2).$$

Thus

$$\sum x(k)y(k) = 1 + 4.2 + 8.7 + 16.8 = 30.7.$$

Also,

$$\sum x^2(k) = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

Therefore,

$$\boxed{\hat{a} = \frac{30.7}{30} = 1.02333.}$$

(c) Residual sum of squares

The residuals are

$$\varepsilon(1) = 1.0 - 1.02333(1) = -0.02333,$$

$$\varepsilon(2) = 2.1 - 1.02333(2) = 0.05333,$$

$$\varepsilon(3) = 2.9 - 1.02333(3) = -0.17000,$$

$$\varepsilon(4) = 4.2 - 1.02333(4) = 0.10667.$$

Hence

$$J(\hat{a}) = (-0.02333)^2 + (0.05333)^2 + (-0.17)^2 + (0.10667)^2.$$

Therefore,

$$\boxed{J(\hat{a}) \approx 0.04187.}$$

(d) Noise variance and parameter variance

Using $N = 4$ and $N - 1 = 3$,

$$\hat{\sigma}^2 = \frac{J(\hat{a})}{N-1} = \frac{0.04187}{3}.$$

Thus

$$\boxed{\hat{\sigma}^2 \approx 0.01396.}$$

For the scalar parameter,

$$\widehat{\mathrm{Var}}(\hat{a}) = \hat{\sigma}^2 \left(\sum_{k=1}^{4} x^2(k) \right)^{-1}.$$

Therefore,

$$\widehat{\mathrm{Var}}(\hat{a}) = \frac{0.01396}{30} \approx \boxed{4.65 \times 10^{-4}}.$$

(Module 3.1–3.2, slides 4–22 and 3–20.)

Solution 2

(a) Regression representation

The model is

$$y(k) = a\, y(k-1) + b\, u(k-1) + e(k).$$

Define

$$\boxed{\boldsymbol{\phi}(k-1) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}.}$$

Then

$$y(k) = \boldsymbol{\phi}^\top(k-1)\boldsymbol{\theta} + e(k).$$

(Module 3.1, slides 4–22.)

(b) Information matrix and data vector

The information matrix is

$$\Phi^\top \Phi = \begin{bmatrix} \sum y^2(k-1) & \sum y(k-1)u(k-1) \\ \sum y(k-1)u(k-1) & \sum u^2(k-1) \end{bmatrix}.$$

Hence

$$\boxed{\Phi^\top \Phi = \begin{bmatrix} 40 & 10 \\ 10 & 25 \end{bmatrix}.}$$

Furthermore,

$$\Phi^\top \mathbf{y} = \begin{bmatrix} \sum y(k)y(k-1) \\ \sum y(k)u(k-1) \end{bmatrix},$$

so

$$\boxed{\Phi^\top \mathbf{y} = \begin{bmatrix} 20 \\ 15 \end{bmatrix}.}$$

(c) Least-squares estimate

The inverse is

$$(\Phi^\top \Phi)^{-1} = \frac{1}{40(25) - 10^2} \begin{bmatrix} 25 & -10 \\ -10 & 40 \end{bmatrix}.$$

Since

$$40(25) - 10^2 = 1000 - 100 = 900,$$

we have

$$(\Phi^\top \Phi)^{-1} = \frac{1}{900} \begin{bmatrix} 25 & -10 \\ -10 & 40 \end{bmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\theta}} = \frac{1}{900} \begin{bmatrix} 25 & -10 \\ -10 & 40 \end{bmatrix} \begin{bmatrix} 20 \\ 15 \end{bmatrix}.$$

For a,

$$\hat{a} = \frac{25(20) - 10(15)}{900} = \frac{500 - 150}{900} = \boxed{0.3889}.$$

For b,

$$\hat{b} = \frac{-10(20) + 40(15)}{900} = \frac{-200 + 600}{900} = \boxed{0.4444}.$$

Thus

$$\boxed{\hat{\boldsymbol{\theta}} = \begin{bmatrix} 0.3889 \\ 0.4444 \end{bmatrix}.}$$

(d) Identifiability

For a unique least-squares estimate,

$$\boxed{\Phi^\top \Phi \text{ must be nonsingular.}}$$

Here,

$$\det(\Phi^\top \Phi) = 900 \neq 0.$$

Therefore,

$$\boxed{\text{the parameters are uniquely identifiable from these data.}}$$

(Module 3.2, slides 3–20.)

Solution 3

(a) Gain vector

First,

$$P(k-1)\boldsymbol{\phi}(k-1) = 10 I_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix}.$$

Next,

$$\boldsymbol{\phi}^\top P \boldsymbol{\phi} = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \end{bmatrix} = 20.$$

Therefore,

$$K(k) = \frac{1}{0.98 + 20} \begin{bmatrix} 10 \\ 10 \end{bmatrix}.$$

Hence

$$\boxed{K(k) = \begin{bmatrix} 0.47687 \\ 0.47687 \end{bmatrix}.}$$

(Module 3.3, slides 3–15; Module 3.4, slides 3–18.)

(b) Prediction error and parameter update

The predicted output is

$$\hat{y}(k) = \boldsymbol{\phi}^\top(k-1)\hat{\boldsymbol{\theta}}(k-1).$$

Therefore,

$$\hat{y}(k) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} = 0.3.$$

Thus

$$\varepsilon(k) = 1.1 - 0.3 = \boxed{0.8}.$$

The parameter update is

$$\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} + 0.8 \begin{bmatrix} 0.47687 \\ 0.47687 \end{bmatrix}.$$

Hence

$$\boxed{\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 0.88150 \\ 0.18150 \end{bmatrix}.}$$

(c) Covariance update

First,

$$K\boldsymbol{\phi}^\top = \begin{bmatrix} 0.47687 \\ 0.47687 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.47687 & 0.47687 \\ 0.47687 & 0.47687 \end{bmatrix}.$$

Thus

$$I - K\boldsymbol{\phi}^\top = \begin{bmatrix} 0.52313 & -0.47687 \\ -0.47687 & 0.52313 \end{bmatrix}.$$

Using

$$P(k) = \frac{1}{0.98}\left(I - K\boldsymbol{\phi}^\top\right) 10 I_2,$$

we obtain

$$\boxed{P(k) \approx \begin{bmatrix} 5.338 & -4.866 \\ -4.866 & 5.338 \end{bmatrix}.}$$

The matrix remains symmetric, as required for a covariance matrix.

(d) Memory design

The effective-memory approximation is

$$N \simeq \frac{1}{1-\lambda}.$$

For

$$N = 50,$$

we obtain

$$50 \simeq \frac{1}{1-\lambda}.$$

Therefore,

$$1 - \lambda = \frac{1}{50} = 0.02,$$

so

$$\boxed{\lambda \simeq 0.98}.$$

This agrees with the value used in the update.

A smaller λ gives shorter memory and faster tracking, but increases the variance/tracking trade-off. *(Module 3.4, slides 3–18; previous MAP2, Problem 4.)*

Solution 4

(a) Model structures

The three models are:

$$M_1: \qquad y(k) = a_1 y(k-1) + b_1 u(k-1) + e(k).$$

This is an

$$\boxed{\text{ARX model}}$$

with one output lag and one input lag.

For M_2,

$$y(k) = a_1 y(k-1) + a_2 y(k-2) + b_1 u(k-1) + b_2 u(k-2) + e(k),$$

which is also

$$\boxed{\text{ARX}},$$

but of higher order.

Finally,

$$M_3$$

contains the additional noise model

$$c_1 e(k-1) + e(k),$$

so it is an

$$\boxed{\text{ARMAX model}}.$$

(Module 3.7, slides 3–22; Problem Series S3P9.)

(b) AIC and BIC selection

Both criteria are minimised.

For AIC,

$$120, \qquad 108, \qquad 110$$

are obtained for M_1, M_2, M_3, respectively.

The smallest value is

$$108,$$

so AIC selects

$$\boxed{M_2}.$$

For BIC,

$$125, \qquad 118, \qquad 123$$

are obtained.

The smallest value is

$$118,$$

so BIC also selects

$$\boxed{M_2}.$$

(Module 3.7, slides 8–14.)

(c) Training error versus validation

Increasing model complexity generally gives the model more freedom to fit the available training data.

Consequently, a more complex model can have a smaller training error while fitting noise or other features specific to that data set.

This can lead to poor prediction on unseen data, i.e. overfitting.

Therefore,

$$\boxed{\text{model selection should consider validation/generalisation, not training error alone.}}$$

AIC and BIC explicitly introduce complexity penalties to address this trade-off.

(Module 3.7, slides 8–17.)

(d) Validation checks

Two valid checks are:

1. residual analysis, including checking whether the residuals behave consistently with the assumed noise model;
2. cross-validation using data that were not used for parameter estimation.

Another valid check is comparison of the measured and predicted frequency responses.

Thus one possible answer is

$$\boxed{\text{residual analysis and cross-validation.}}$$

(Module 3.7, slides 15–22.)

(e) Lack of excitation

If the input contains almost no variation over the frequency band of interest, the experiment does not sufficiently excite the system dynamics in that band.

Consequently, the measured input-output data contain little information with which to distinguish or accurately estimate the corresponding parameters.

Therefore,

$$\boxed{\text{poor excitation can lead to weak identifiability and unreliable parameter estimates.}}$$

A suitable experiment should provide sufficient persistent excitation over the frequencies relevant to the intended model.

(Module 3.2, slides 3–20; Module 3.7, slides 3–6.)

Where to Study Next

- **Module 3.1, slides 4–22:** least-squares criteria, normal equations, regressors, uniqueness, and dynamic-model parameter estimation.
- **Module 3.2, slides 3–20:** information matrices, persistent excitation, identifiability, and experiment design.

- **Module 3.3, slides 3–15:** recursive least squares, gain, covariance, and recursive updates.
- **Module 3.4, slides 3–18:** exponential forgetting, tracking, memory, and variance trade-offs.
- **Module 3.5, slides 3–10:** random variables, stationarity, white noise, and covariance functions.
- **Module 3.6, slides 3–22:** LS bias, white versus colored noise, consistency, efficiency, and maximum likelihood.
- **Module 3.7, slides 3–22:** experiment design, model structures, AIC/BIC, residual analysis, cross-validation, and validation.
- **Module 3.8, slides 3–20:** physical system identification, pole/zero interpretation, inverse response, and parsimonious model selection.

Mock MAP 2 – Control of Cyber-Physical Systems

Module 3 **Time: 30 min** **Total: 20 points**

Four problems, worth 5 points each. Budget approximately 7 minutes per problem.

Problem 1 — Batch least squares (5p)

Consider the model

$$y(k) = a\,x(k) + e(k),$$

and the measured data

k	1	2	3	4
$x(k)$	1	2	3	4
$y(k)$	1.1	1.9	3.2	4.0

(a) (1.0p) Write the least-squares criterion $J(a)$ and derive the normal equation satisfied by $\hat{a}$.

(b) (1.5p) Compute the least-squares estimate $\hat{a}$.

(c) (1.5p) Compute the residual sum of squares

$$\mathrm{SSE} = \sum_k \left(y(k) - \hat{a}x(k)\right)^2$$

and the estimate

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{N-1}.$$

(d) (1.0p) Using

$$\mathrm{Var}(\hat{a}) = \hat{\sigma}^2 \left(\sum_k x^2(k)\right)^{-1},$$

compute the estimated variance of $\hat{a}$.

Problem 2 — Dynamic-model least squares (5p)

Consider the ARX model

$$y(k) = a\,y(k-1) + b\,u(k-1) + e(k).$$

Suppose that the following quantities have been computed from an experiment:

$$\sum_k y^2(k-1) = 10, \qquad \sum_k u^2(k-1) = 20,$$

$$\sum_k y(k-1)u(k-1) = 4,$$

$$\sum_k y(k)y(k-1) = 7, \qquad \sum_k y(k)u(k-1) = 9.$$

(a) (1.5p) Define the regression vector $\phi(k-1)$ and parameter vector θ, and write the model in the form

$$y(k) = \phi^\top(k-1)\theta + e(k).$$

(b) (1.5p) Construct explicitly the matrices

$$\Phi^\top\Phi \qquad \text{and} \qquad \Phi^\top Y$$

using the supplied sums.

(c) (1.5p) Determine the least-squares estimate

$$\hat{\theta} = \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix}.$$

(d) (0.5p) Explain why the invertibility of $\Phi^\top\Phi$ matters for obtaining a unique least-squares estimate.

Problem 3 — Recursive least squares (5p)

Consider one recursive least-squares update without forgetting,

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \phi^\top(k-1)\hat{\theta}(k-1)\right],$$

with

$$K(k) = \frac{P(k-1)\phi(k-1)}{1 + \phi^\top(k-1)P(k-1)\phi(k-1)},$$

and

$$P(k) = \Big(I - K(k)\phi^\top(k-1)\Big)\,P(k-1).$$

The current quantities are

$$\hat{\theta}(k-1) = \begin{bmatrix} 0.2 \\ -0.1 \end{bmatrix}, \qquad P(k-1) = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},$$

$$\phi(k-1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y(k) = 0.8.$$

(a) (1.0p) Compute

$$P(k-1)\phi(k-1)$$

and

$$\phi^\top(k-1)P(k-1)\phi(k-1).$$

(b) (1.0p) Compute the gain $K(k)$.

(c) (1.0p) Compute the prediction error

$$\varepsilon(k) = y(k) - \phi^\top(k-1)\hat{\theta}(k-1).$$

(d) (1.0p) Compute the updated parameter estimate $\hat{\theta}(k)$.

(e) (1.0p) Compute the updated covariance matrix $P(k)$. State whether the uncertainty in the direction of the current regressor has increased or decreased.

Problem 4 — Forgetting factor and stochastic processes (5p)

An RLS estimator with exponential forgetting is used to identify a slowly time-varying parameter. The estimator uses

$$\lambda = 0.96.$$

(a) (1.0p) Using the effective-memory approximation

$$N_{\text{eff}} \approx \frac{1}{1-\lambda},$$

estimate the number of samples effectively retained by the estimator.

(b) (1.5p) Suppose the true parameter changes abruptly at $k = 500$. Explain qualitatively what happens if λ is changed from 0.96 to 0.99. Discuss both tracking speed and estimation variance.

(c) (1.5p) A zero-mean noise sequence $e(k)$ satisfies

$$E[e(k)e(k-\tau)] = \begin{cases} \sigma^2, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases}$$

State whether the process is white and explain what property of its covariance function leads to your conclusion.

(d) (1.0p) Explain why the assumption of white noise is relevant to the usual least-squares identification setting.

Solutions

Solution 1 — Batch least squares

(a)

The least-squares criterion is

$$J(a) = \sum_{k=1}^{4} [y(k) - ax(k)]^2 .$$

Differentiate with respect to a:

$$\frac{dJ}{da} = -2 \sum_{k=1}^{4} x(k) [y(k) - ax(k)] .$$

At the minimum,

$$\frac{dJ}{da} = 0,$$

hence

$$\sum_k x(k)y(k) = \hat{a} \sum_k x^2(k).$$

Therefore,

$$\boxed{\hat{a} = \frac{\sum_k x(k)y(k)}{\sum_k x^2(k)}}.$$

This is the scalar normal equation for batch least squares.

(b)

First compute

$$\sum_k x(k)y(k) = 1(1.1) + 2(1.9) + 3(3.2) + 4(4.0).$$

Thus

$$\sum_k x(k)y(k) = 1.1 + 3.8 + 9.6 + 16 = 30.5.$$

Also,

$$\sum_k x^2(k) = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30.$$

Therefore,

$$\hat{a} = \frac{30.5}{30} = \boxed{1.0167}.$$

(c)

The predicted outputs are

$$\hat{y}(k) = \hat{a}x(k).$$

Hence the residuals are

$$\begin{aligned}
\varepsilon(1) &= 1.1 - 1.0167(1) = -0.0167, \\
\varepsilon(2) &= 1.9 - 1.0167(2) = -0.1333, \\
\varepsilon(3) &= 3.2 - 1.0167(3) = 0.1500, \\
\varepsilon(4) &= 4.0 - 1.0167(4) = -0.0667.
\end{aligned}$$

Therefore,

$$\text{SSE} = (-0.0167)^2 + (-0.1333)^2 + (0.1500)^2 + (-0.0667)^2.$$

Numerically,

$$\text{SSE} \approx 0.0483.$$

With $N = 4$,

$$\hat{\sigma}^2 = \frac{0.0483}{4-1} = \boxed{0.0161}.$$

(d)

Using

$$\text{Var}(\hat{a}) = \hat{\sigma}^2 \left(\sum_k x^2(k) \right)^{-1},$$

we obtain

$$\text{Var}(\hat{a}) = \frac{0.0161}{30} \approx 5.37 \times 10^{-4}.$$

Therefore,

$$\boxed{\text{Var}(\hat{a}) \approx 5.37 \times 10^{-4}}.$$

Check: the residuals are relatively small compared with the measured outputs, and the estimate is close to $a = 1$.

Where to study: Module 3.1, least-squares criteria, normal equations, regressors, uniqueness, and dynamic-model parameter estimation.

Solution 2 — Dynamic-model least squares

(a)

The model is

$$y(k) = a\, y(k-1) + b\, u(k-1) + e(k).$$

Define

$$\phi(k-1) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}, \qquad \theta = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Then

$$\phi^\top(k-1)\theta = \begin{bmatrix} y(k-1) & u(k-1) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix},$$

so

$$\boxed{y(k) = \phi^\top(k-1)\theta + e(k)}.$$

(b)

The information matrix is

$$\Phi^\top \Phi = \sum_k \phi(k-1)\phi^\top(k-1).$$

Therefore,

$$\Phi^\top \Phi = \begin{bmatrix} \sum y^2(k-1) & \sum y(k-1)u(k-1) \\ \sum y(k-1)u(k-1) & \sum u^2(k-1) \end{bmatrix}.$$

Using the supplied values,

$$\boxed{\Phi^\top \Phi = \begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix}}.$$

Similarly,

$$\Phi^\top Y = \sum_k \phi(k-1)y(k),$$

so

$$\boxed{\Phi^\top Y = \begin{bmatrix} 7 \\ 9 \end{bmatrix}}.$$

(c)

The least-squares estimate satisfies

$$(\Phi^\top \Phi)\hat{\theta} = \Phi^\top Y.$$

Thus,

$$\begin{bmatrix} 10 & 4 \\ 4 & 20 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \end{bmatrix}.$$

The determinant is

$$10(20) - 4(4) = 200 - 16 = 184.$$

Therefore,

$$(\Phi^\top \Phi)^{-1} = \frac{1}{184} \begin{bmatrix} 20 & -4 \\ -4 & 10 \end{bmatrix}.$$

Hence,

$$\hat{\theta} = \frac{1}{184} \begin{bmatrix} 20 & -4 \\ -4 & 10 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix}.$$

For the first component,

$$\hat{a} = \frac{140 - 36}{184} = \frac{104}{184} = \boxed{0.5652}.$$

For the second component,

$$\hat{b} = \frac{-28 + 90}{184} = \frac{62}{184} = \boxed{0.3370}.$$

Therefore,

$$\boxed{\hat{\theta} = \begin{bmatrix} 0.5652 \\ 0.3370 \end{bmatrix}}.$$

(d)

A unique least-squares solution requires

$$\Phi^\top \Phi$$

to be invertible. Here,

$$\det(\Phi^\top \Phi) = 184 \neq 0,$$

so the estimate is unique.

Equivalently, the regressors must contain sufficient independent information to distinguish the parameters.

Where to study: Module 3.1, particularly dynamic-model parameter estimation, regressors, normal equations, and uniqueness.

Solution 3 — Recursive least squares

(a)

We have

$$P(k-1) = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \phi(k-1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Therefore,

$$P(k-1)\phi(k-1) = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Next,

$$\phi^\top(k-1)P(k-1)\phi(k-1) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 12.$$

Thus,

$$\boxed{P\phi = \begin{bmatrix} 4 \\ 4 \end{bmatrix}, \qquad \phi^\top P\phi = 12}.$$

(b)

The denominator is

$$1 + \phi^\top P\phi = 1 + 12 = 13.$$

Therefore,

$$K(k) = \frac{1}{13} \begin{bmatrix} 4 \\ 4 \end{bmatrix},$$

giving

$$\boxed{K(k) = \begin{bmatrix} 4/13 \\ 4/13 \end{bmatrix}}.$$

(c)

First calculate the predicted output:

$$\phi^\top \hat{\theta} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 0.2 \\ -0.1 \end{bmatrix}.$$

Hence,

$$\phi^\top \hat{\theta} = 0.2 - 0.2 = 0.$$

The prediction error is therefore

$$\varepsilon(k) = 0.8 - 0 = \boxed{0.8}.$$

(d)

The update is

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k).$$

Therefore,

$$K(k)\varepsilon(k) = \begin{bmatrix} 4/13 \\ 4/13 \end{bmatrix} (0.8) = \begin{bmatrix} 3.2/13 \\ 3.2/13 \end{bmatrix}.$$

Thus,

$$\hat{\theta}(k) = \begin{bmatrix} 0.2 \\ -0.1 \end{bmatrix} + \begin{bmatrix} 0.24615 \\ 0.24615 \end{bmatrix}.$$

Hence,

$$\boxed{\hat{\theta}(k) = \begin{bmatrix} 0.44615 \\ 0.14615 \end{bmatrix}}.$$

(e)

Using

$$P(k) = (I - K\phi^\top)P(k-1),$$

first compute

$$K\phi^\top = \begin{bmatrix}4/13\\4/13\end{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix} = \begin{bmatrix}4/13 & 8/13\\4/13 & 8/13\end{bmatrix}.$$

Therefore,

$$I - K\phi^\top = \begin{bmatrix}9/13 & -8/13\\-4/13 & 5/13\end{bmatrix}.$$

Multiplying by $P(k-1)$,

$$P(k) = \begin{bmatrix}9/13 & -8/13\\-4/13 & 5/13\end{bmatrix}\begin{bmatrix}4 & 0\\0 & 2\end{bmatrix}.$$

Thus,

$$\boxed{P(k) = \begin{bmatrix}36/13 & -16/13\\-16/13 & 10/13\end{bmatrix}}$$

or numerically,

$$\boxed{P(k) \approx \begin{bmatrix}2.7692 & -1.2308\\-1.2308 & 0.7692\end{bmatrix}}.$$

The covariance in the direction informed by the current regressor decreases. This is the expected behavior: a new measurement supplies information and therefore reduces parameter uncertainty.

Where to study: Module 3.3, recursive least squares, gain, covariance, and recursive parameter updates.

Solution 4 — Forgetting factor and stochastic processes

(a)

The effective-memory approximation is

$$N_{\text{eff}} \approx \frac{1}{1-\lambda}.$$

For

$$\lambda = 0.96,$$

we obtain

$$N_{\text{eff}} \approx \frac{1}{1-0.96} = \frac{1}{0.04} = \boxed{25}.$$

Thus the estimator effectively emphasizes approximately the most recent 25 samples.

(b)

If

$$\lambda = 0.96,$$

then

$$N_{\text{eff}} \approx 25.$$

If

$$\lambda = 0.99,$$

then

$$N_{\text{eff}} \approx \frac{1}{0.01} = 100.$$

Therefore, increasing λ from 0.96 to 0.99 gives the estimator a substantially longer memory.

After the parameter changes at $k = 500$, the estimator with $\lambda = 0.96$ gives greater weight to recent data and therefore tracks the parameter change faster.

The estimator with $\lambda = 0.99$ retains more information from data before $k = 500$. Consequently, it adapts more slowly to the change.

The trade-off is that stronger forgetting generally allows faster tracking of parameter changes but produces a noisier parameter estimate, whereas weaker forgetting gives a smoother estimate but slower tracking.

Thus,

$$\boxed{\lambda \downarrow \Rightarrow \text{shorter memory, faster tracking, generally larger variance}}$$

and

$$\boxed{\lambda \uparrow \Rightarrow \text{longer memory, slower tracking, generally smaller variance}}.$$

(c)

The covariance is

$$R_e(\tau) = E[e(k)e(k-\tau)].$$

It satisfies

$$R_e(0) = \sigma^2$$

and

$$R_e(\tau) = 0, \qquad \tau \neq 0.$$

Therefore, samples at different time lags are uncorrelated.

Hence the process is white:

$$\boxed{e(k) \text{ is a white-noise process.}}$$

The key property is that its autocovariance is zero for every non-zero lag.

(d)

Least-squares identification treats the unexplained part of the measured output as an error term. The usual statistical interpretation is particularly clean when that error is white and uncorrelated across time.

If the disturbance is instead coloured, past disturbances can be correlated with current regressors or with each other, which can invalidate the usual least-squares assumptions and may produce biased parameter estimates.

Therefore, checking the noise assumptions is important when interpreting least-squares estimates statistically.

Where to study: Module 3.4 for forgetting factors, memory, tracking and variance trade-offs; Module 3.5 for random processes, variance, stationarity, white noise and covariance functions.

Mock MAP 2 – Control of Cyber-Physical Systems

Module 3 **Time: 30 min** **Total: 20 points**

Four problems, worth 5 points each. Budget approximately 7 minutes per problem.

Problem 1 — Least-squares estimation and model interpretation (5p)

Consider the model

$$y(k) = \theta_1 + \theta_2 u(k) + e(k),$$

with measurements

k	1	2	3	4	5
$u(k)$	0	1	2	3	4
$y(k)$	1.2	2.1	2.9	4.2	5.1

(a) (1.0p) Define the parameter vector θ and regression vector $\phi(k)$ such that

$$y(k) = \phi^\top(k)\theta + e(k).$$

(b) (1.5p) Construct $\Phi^\top\Phi$ and $\Phi^\top Y$.

(c) (1.5p) Compute the least-squares estimate $\hat{\theta}$.

(d) (1.0p) Write the identified model $\hat{y}(k) = \hat{\theta}_1 + \hat{\theta}_2 u(k)$ and interpret $\hat{\theta}_1$ and $\hat{\theta}_2$.

Problem 2 — Least-squares properties and residuals (5p)

A scalar parameter is estimated from

$$y(k) = a\,x(k) + e(k)$$

using the data

$$x = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}, \qquad y = \begin{bmatrix} 2.1 & 3.9 & 6.2 & 7.8 \end{bmatrix}.$$

(a) (1.5p) Determine the least-squares estimate $\hat{a}$.

(b) (1.5p) Compute the four residuals

$$\hat{e}(k) = y(k) - \hat{a}x(k)$$

and verify numerically that the least-squares orthogonality condition

$$\sum_k x(k)\hat{e}(k) = 0$$

is satisfied up to numerical rounding.

(c) (1.0p) Explain why the residuals themselves do not generally have to be zero even though the least-squares criterion is minimized.

(d) (1.0p) State what would happen to the uniqueness of the scalar estimate if all measured values satisfied $x(k) = 0$.

Problem 3 — Recursive least squares with forgetting (5p)

Consider the recursive least-squares update

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k),$$

where

$$\varepsilon(k) = y(k) - \phi^\top(k-1)\hat{\theta}(k-1),$$

and

$$K(k) = \frac{P(k-1)\phi(k-1)}{\lambda + \phi^\top(k-1)P(k-1)\phi(k-1)}.$$

The covariance update is

$$P(k) = \frac{1}{\lambda}\left[I - K(k)\phi^\top(k-1)\right]P(k-1).$$

Let

$$\hat{\theta}(k-1) = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix}, \qquad P(k-1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

$$\phi(k-1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad y(k) = 1.0, \qquad \lambda = 0.9.$$

(a) (1.0p) Compute the prediction error $\varepsilon(k)$.

(b) (1.0p) Compute the gain $K(k)$.

(c) (1.5p) Compute the updated parameter estimate $\hat{\theta}(k)$.

(d) (1.5p) Compute $P(k)$.

Problem 4 — Identification, excitation and model validation (5p)

Suppose two candidate models are identified from the same input-output experiment:

$$M_1: \qquad y(k) = a_1 y(k-1) + b_1 u(k-1) + e(k),$$

and

$$M_2: \qquad y(k) = a_1 y(k-1) + a_2 y(k-2) + b_1 u(k-1) + b_2 u(k-2) + e(k).$$

The following results are obtained:

	M_1	M_2
Training SSE	4.8	3.1
Validation SSE	5.0	5.7

(a) (1.0p) Which model has the smaller training error? Which model has the smaller validation error?

(b) (1.5p) Explain why a model can have a smaller training error but a larger validation error.

(c) (1.0p) The input signal used during the experiment is approximately constant:

$$u(k) \approx 2.$$

Explain why this can create difficulties for parameter identification.

(d) (1.5p) Give two checks that could be performed on the validation residuals before accepting an identified model.

Solutions

Solution 1 — Least-squares estimation and model interpretation

(a)

We write

$$y(k) = \begin{bmatrix} 1 & u(k) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + e(k).$$

Therefore,

$$\boxed{\phi(k) = \begin{bmatrix} 1 \\ u(k) \end{bmatrix}, \qquad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}}.$$

(b)

The regression matrix is

$$\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

Hence,

$$\Phi^\top \Phi = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}.$$

Also,

$$\Phi^\top Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1.2 \\ 2.1 \\ 2.9 \\ 4.2 \\ 5.1 \end{bmatrix}.$$

The first component is

$$1.2 + 2.1 + 2.9 + 4.2 + 5.1 = 15.5.$$

The second component is

$$0(1.2) + 1(2.1) + 2(2.9) + 3(4.2) + 4(5.1) = 40.0.$$

Therefore,

$$\boxed{\Phi^\top Y = \begin{bmatrix} 15.5 \\ 40 \end{bmatrix}}.$$

(c)

The normal equations are

$$\begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} 15.5 \\ 40 \end{bmatrix}.$$

The determinant is

$$5(30) - 10(10) = 50.$$

Thus,

$$(\Phi^\top \Phi)^{-1} = \frac{1}{50} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix}.$$

Consequently,

$$\hat{\theta} = \frac{1}{50}\begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix}\begin{bmatrix} 15.5 \\ 40 \end{bmatrix}.$$

Therefore,

$$\hat{\theta}_1 = \frac{465 - 400}{50} = \boxed{1.30},$$

and

$$\hat{\theta}_2 = \frac{-155 + 200}{50} = \boxed{0.90}.$$

Hence,

$$\boxed{\hat{\theta} = \begin{bmatrix} 1.30 \\ 0.90 \end{bmatrix}}.$$

(d)

The identified model is

$$\boxed{\hat{y}(k) = 1.30 + 0.90u(k)}.$$

The parameter $\hat{\theta}_1 = 1.30$ represents the estimated output offset when $u(k) = 0$.

The parameter $\hat{\theta}_2 = 0.90$ represents the estimated change in output associated with a unit change in the input.

Check:

$$\hat{y} = \begin{bmatrix} 1.3 & 2.2 & 3.1 & 4.0 & 4.9 \end{bmatrix},$$

which is close to the measured data.

Where to study: Module 3, batch least squares, regression formulation, normal equations and interpretation of identified parameters.

Solution 2 — Least-squares properties and residuals

(a)

For the scalar model,

$$\hat{a} = \frac{\sum_k x(k)y(k)}{\sum_k x^2(k)}.$$

Compute the numerator:

$$\sum_k x(k)y(k) = 1(2.1) + 2(3.9) + 3(6.2) + 4(7.8).$$

Therefore,

$$\sum_k x(k)y(k) = 2.1 + 7.8 + 18.6 + 31.2 = 59.7.$$

The denominator is

$$\sum_k x^2(k) = 1 + 4 + 9 + 16 = 30.$$

Thus,

$$\boxed{\hat{a} = \frac{59.7}{30} = 1.99}.$$

(b)

The residuals are

$$\hat{e}(k) = y(k) - 1.99x(k).$$

Therefore,

$$\hat{e}(1) = 2.1 - 1.99 = 0.11,$$
$$\hat{e}(2) = 3.9 - 3.98 = -0.08,$$
$$\hat{e}(3) = 6.2 - 5.97 = 0.23,$$
$$\hat{e}(4) = 7.8 - 7.96 = -0.16.$$

Now calculate

$$\sum_k x(k)\hat{e}(k) = 1(0.11) + 2(-0.08) + 3(0.23) + 4(-0.16).$$

Hence,

$$\sum_k x(k)\hat{e}(k) = 0.11 - 0.16 + 0.69 - 0.64 = 0.$$

Therefore,

$$\boxed{\sum_k x(k)\hat{e}(k) = 0}.$$

This is the least-squares orthogonality condition.

(c)

Least squares minimizes

$$J(a) = \sum_k e^2(k),$$

but it does not generally have enough free parameters to reproduce every measurement exactly.

Therefore,

$$\boxed{\text{minimum squared error} \neq \text{zero squared error}}$$

in general.

The residuals are zero only if the measured data lie exactly on the model structure being estimated.

(d)

If

$$x(k) = 0 \qquad \forall k,$$

then

$$\sum_k x^2(k) = 0.$$

The estimate

$$\hat{a} = \frac{\sum_k x(k)y(k)}{\sum_k x^2(k)}$$

would therefore be undefined.

There is no information in the experiment about a.

Hence,

$$\boxed{a \text{ cannot be uniquely identified from these data.}}$$

Where to study: Module 3, least-squares optimality, orthogonality of residuals, uniqueness and informative experiments.

Solution 3 — Recursive least squares with forgetting

(a)

First compute

$$\phi^\top(k-1)\hat{\theta}(k-1) = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}0.5\\-0.2\end{bmatrix}.$$

Thus,

$$\phi^\top\hat{\theta} = 0.5 - 0.2 = 0.3.$$

Therefore,

$$\varepsilon(k) = 1.0 - 0.3 = \boxed{0.7}.$$

(b)

First,

$$P(k-1)\phi(k-1) = \begin{bmatrix}2 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}.$$

Also,

$$\phi^\top P\phi = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = 3.$$

The denominator is

$$\lambda + \phi^\top P\phi = 0.9 + 3 = 3.9.$$

Therefore,

$$K(k) = \frac{1}{3.9}\begin{bmatrix}2\\1\end{bmatrix},$$

giving

$$\boxed{K(k) = \begin{bmatrix}0.51282\\0.25641\end{bmatrix}}.$$

(c)

The parameter update is

$$\hat{\theta}(k) = \begin{bmatrix}0.5\\-0.2\end{bmatrix} + \begin{bmatrix}0.51282\\0.25641\end{bmatrix}(0.7).$$

Therefore,

$$\hat{\theta}(k) = \begin{bmatrix}0.5 + 0.35897\\-0.2 + 0.17949\end{bmatrix}.$$

Hence,

$$\boxed{\hat{\theta}(k) = \begin{bmatrix}0.85897\\-0.02051\end{bmatrix}}.$$

(d)

First compute

$$K\phi^\top = \begin{bmatrix}0.51282\\0.25641\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}0.51282 & 0.51282\\0.25641 & 0.25641\end{bmatrix}.$$

Thus,

$$I - K\phi^\top = \begin{bmatrix}0.48718 & -0.51282\\-0.25641 & 0.74359\end{bmatrix}.$$

Multiplying by $P(k-1)$,

$$\left(I - K\phi^\top\right) P(k-1) = \begin{bmatrix} 0.97436 & -0.51282 \\ -0.51282 & 0.74359 \end{bmatrix}.$$

Finally, divide by $\lambda = 0.9$:

$$P(k) = \frac{1}{0.9} \begin{bmatrix} 0.97436 & -0.51282 \\ -0.51282 & 0.74359 \end{bmatrix}.$$

Therefore,

$$\boxed{P(k) \approx \begin{bmatrix} 1.08262 & -0.56980 \\ -0.56980 & 0.82621 \end{bmatrix}}.$$

Check: the covariance update contains the factor $1/\lambda > 1$. This reflects the forgetting mechanism: before the new measurement is incorporated, older information is effectively discounted.

Where to study: Module 3, recursive least squares, forgetting factor, gain, prediction error and covariance update.

Solution 4 — Identification, excitation and model validation

(a)

The training SSE values are

$$\mathrm{SSE}_{\mathrm{train},1} = 4.8, \qquad \mathrm{SSE}_{\mathrm{train},2} = 3.1.$$

Therefore, M_2 has the smaller training error.

For validation,

$$\mathrm{SSE}_{\mathrm{val},1} = 5.0, \qquad \mathrm{SSE}_{\mathrm{val},2} = 5.7.$$

Therefore, M_1 has the smaller validation error.

Thus,

$$\boxed{M_2 : \text{ smaller training SSE}}$$

while

$$\boxed{M_1 : \text{ smaller validation SSE.}}$$

(b)

M_2 has more parameters and therefore greater flexibility. It can adapt more closely to the particular data used for estimation.

Consequently, it can reduce the training error without necessarily representing the underlying system better on new data.

This is the usual distinction between fitting the identification data and generalizing to independent validation data.

$$\boxed{\text{small training error alone is not sufficient evidence of a good model.}}$$

(c)

If

$$u(k) \approx 2$$

for essentially all k, then the input provides very little variation.

Consequently, different parameter combinations may produce very similar predicted outputs. The experiment therefore contains insufficient information to reliably distinguish the effects of different input terms.

In matrix terms, the regression matrix can become poorly conditioned or lose rank.

Hence,

$$\boxed{\text{poor excitation} \Rightarrow \text{poor parameter identifiability.}}$$

(d)

Two appropriate validation checks are:

1. Check whether the validation residuals have approximately zero mean and no systematic structure.
2. Check whether the residuals are approximately uncorrelated with past input signals and relevant regressors.

Other appropriate checks include comparing prediction performance on independent data and inspecting residual autocorrelation.

The purpose is to determine whether the residuals behave consistently with the assumed unexplained-noise component rather than containing systematic information that the model has failed to capture.

Where to study: Module 3, experiment design and excitation, model identification, validation and residual analysis.

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Mock MAP 2 – Control of Cyber-Physical Systems

Module 3 **Time: 30 min** **Total: 20 points**

Four problems, worth 5 points each. Budget approximately 7 minutes per problem.

Problem 1 — Least-squares parameter estimation (5p)

Consider the model

$$y(k) = a\,y(k-1) + b\,u(k-1) + e(k).$$

From an experiment, the following quantities were computed:

$$\sum_k y^2(k-1) = 40, \qquad \sum_k u^2(k-1) = 25,$$

$$\sum_k y(k-1)u(k-1) = 10,$$

$$\sum_k y(k)y(k-1) = 14, \qquad \sum_k y(k)u(k-1) = 11.$$

(a) (1.0p) Define the regression vector $\phi(k-1)$ and parameter vector θ such that

$$y(k) = \phi^\top(k-1)\theta + e(k).$$

(b) (1.5p) Construct $\Phi^\top\Phi$ and $\Phi^\top Y$.

(c) (2.0p) Compute the least-squares estimate

$$\hat{\theta} = \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix}.$$

(d) (0.5p) Verify that the estimate is unique.

Problem 2 — Excitation and identifiability (5p)

Consider the regression model

$$y(k) = \theta_1 u(k) + \theta_2 u(k-1) + e(k).$$

Two experiments are performed.
In Experiment A,

$$u(k) = 1, \qquad \forall k.$$

In Experiment B, the input alternates according to

$$u(k) = \begin{cases} 1, & k \text{ even}, \\ -1, & k \text{ odd}. \end{cases}$$

(a) (1.5p) For Experiment A, write the regression vector $\phi(k)$ and explain whether θ_1 and θ_2 can be uniquely identified.

(b) (1.5p) Show that the information matrix

$$\Phi^\top \Phi = \sum_k \phi(k)\phi^\top(k)$$

is singular for Experiment A.

(c) (1.0p) Explain why Experiment B provides more information for distinguishing θ_1 from θ_2.

(d) (1.0p) State, in words, the role of persistent excitation in parameter identification.

Problem 3 — Recursive least squares with forgetting (5p)

Consider RLS with exponential forgetting:

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k),$$

where

$$\varepsilon(k) = y(k) - \phi^\top(k-1)\hat{\theta}(k-1),$$

$$K(k) = \frac{P(k-1)\phi(k-1)}{\lambda + \phi^\top(k-1)P(k-1)\phi(k-1)},$$

and

$$P(k) = \frac{1}{\lambda}\Big(I - K(k)\phi^\top(k-1)\Big)P(k-1).$$

Let

$$\lambda = 0.95, \qquad \hat{\theta}(k-1) = \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix},$$

$$P(k-1) = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \phi(k-1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad y(k) = 1.5.$$

(a) (1.0p) Compute the prediction error $\varepsilon(k)$.

(b) (1.0p) Compute the gain $K(k)$.

(c) (1.5p) Compute the updated estimate $\hat{\theta}(k)$.

(d) (1.5p) Compute the updated covariance matrix $P(k)$.

Problem 4 — White noise, bias and model validation (5p)

Suppose the disturbance in an identified model is

$$v(k) = e(k) + 0.8e(k-1),$$

where $e(k)$ is a zero-mean white-noise sequence with variance

$$E[e^2(k)] = \sigma^2.$$

(a) (1.5p) Determine whether $v(k)$ is white. Justify your answer by computing $E[v(k)v(k-1)]$.

(b) (1.0p) Explain why using $v(k)$ as if it were white noise can affect the statistical interpretation of a least-squares estimate.

(c) (1.0p) Two candidate models produce the following validation residual sums:

$$\mathrm{SSE}_{V,1} = 6.2, \qquad \mathrm{SSE}_{V,2} = 4.7.$$

What does this information tell us about the two models on the validation data?

(d) (1.5p) Give two residual-based checks that should be performed before concluding that an identified model adequately describes the system.

Solutions

Solution 1 — Least-squares parameter estimation

(a)

Define

$$\phi(k-1) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}, \qquad \theta = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Then

$$\phi^\top(k-1)\theta = a\,y(k-1) + b\,u(k-1),$$

so

$$\boxed{y(k) = \phi^\top(k-1)\theta + e(k)}.$$

(b)

The information matrix is

$$\Phi^\top\Phi = \sum_k \phi(k-1)\phi^\top(k-1).$$

Therefore,

$$\Phi^\top\Phi = \begin{bmatrix} \sum y^2(k-1) & \sum y(k-1)u(k-1) \\ \sum y(k-1)u(k-1) & \sum u^2(k-1) \end{bmatrix}.$$

Using the given values,

$$\boxed{\Phi^\top\Phi = \begin{bmatrix} 40 & 10 \\ 10 & 25 \end{bmatrix}}.$$

Similarly,

$$\Phi^\top Y = \begin{bmatrix} \sum y(k)y(k-1) \\ \sum y(k)u(k-1) \end{bmatrix},$$

giving

$$\boxed{\Phi^\top Y = \begin{bmatrix} 14 \\ 11 \end{bmatrix}}.$$

(c)

The normal equations are

$$\begin{bmatrix} 40 & 10 \\ 10 & 25 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 14 \\ 11 \end{bmatrix}.$$

The determinant is

$$40(25) - 10(10) = 1000 - 100 = 900.$$

Hence,

$$(\Phi^\top\Phi)^{-1} = \frac{1}{900} \begin{bmatrix} 25 & -10 \\ -10 & 40 \end{bmatrix}.$$

Therefore,

$$\hat{\theta} = \frac{1}{900} \begin{bmatrix} 25 & -10 \\ -10 & 40 \end{bmatrix} \begin{bmatrix} 14 \\ 11 \end{bmatrix}.$$

For $\hat{a}$,

$$\hat{a} = \frac{350 - 110}{900} = \frac{240}{900} = \boxed{0.2667}.$$

For $\hat{b}$,

$$\hat{b} = \frac{-140 + 440}{900} = \frac{300}{900} = \boxed{0.3333}.$$

Thus,

$$\boxed{\hat{\theta} = \begin{bmatrix} 0.2667 \\ 0.3333 \end{bmatrix}}.$$

(d)

A unique least-squares solution requires

$$\det(\Phi^\top \Phi) \neq 0.$$

Since

$$\det(\Phi^\top \Phi) = 900 \neq 0,$$

the matrix is invertible and the estimate is unique:

$$\boxed{\text{unique least-squares estimate.}}$$

Course connection: This directly follows the dynamic-model least-squares structure in Module 3.1, including regressors, normal equations and uniqueness. The Problem Series treats the same structure in S3P4/S3P9.

Solution 2 — Excitation and identifiability

(a)

The regression vector is

$$\phi(k) = \begin{bmatrix} u(k) \\ u(k-1) \end{bmatrix}.$$

For Experiment A,

$$u(k) = 1,$$

so

$$\phi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

for every k.

Consequently, the two regressors are identical. The model becomes

$$y(k) = (\theta_1 + \theta_2) + e(k).$$

Only the sum

$$\theta_1 + \theta_2$$

can be determined from the data. The individual parameters cannot be uniquely identified.

Therefore,

$$\boxed{\theta_1, \theta_2 \text{ are not separately identifiable.}}$$

(b)

For N measurements,

$$\Phi^\top \Phi = \sum_{k=1}^{N} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

Hence,

$$\Phi^\top \Phi = N \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Its determinant is

$$\det(\Phi^\top \Phi) = N^2 - N^2 = 0.$$

Therefore,

$$\boxed{\Phi^\top \Phi \text{ is singular.}}$$

This is precisely why the two parameters cannot be uniquely recovered.

(c)

For Experiment B,

$$u(k) = (-1)^k.$$

Therefore,

$$u(k-1) = -u(k).$$

The regression vector alternates between

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

The two regressors are no longer identical. The experiment therefore provides information that distinguishes the contribution of θ_1 from that of θ_2.

For an even number N of samples,

$$\Phi^\top \Phi = N \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Important: this particular alternating signal still gives a rank-one information matrix, because the two regressors are perfectly related by a sign. Thus it improves the distinction relative to the constant-input case but does **not**, by itself, provide full identifiability of both parameters.

A truly informative experiment would require the regressors $u(k)$ and $u(k-1)$ not to remain linearly dependent.

(d)

Persistent excitation means that the input contains sufficient variation, over time, to make the relevant regression directions informative.

In identification terms,

$$\boxed{\text{adequate excitation} \Rightarrow \text{informative data} \Rightarrow \text{parameter identifiability.}}$$

Poor excitation can make $\Phi^\top \Phi$ singular or poorly conditioned, making parameter estimates non-unique or highly uncertain.

Course connection: Module 3.2 covers information matrices, persistent excitation and identifiability. The Problem Series explicitly lists these as the main objectives of Module 3.2.

Solution 3 — Recursive least squares with forgetting

(a)

First compute the predicted output:

$$\phi^\top(k-1)\hat{\theta}(k-1) = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix}.$$

Therefore,

$$\phi^\top \hat{\theta} = 0.4 - 0.1 = 0.3.$$

Hence,

$$\varepsilon(k) = 1.5 - 0.3 = \boxed{1.2}.$$

(b)

Compute

$$P(k-1)\phi(k-1) = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Next,

$$\phi^\top P\phi = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = 5.$$

Therefore,

$$\lambda + \phi^\top P\phi = 0.95 + 5 = 5.95.$$

Thus,

$$K(k) = \frac{1}{5.95} \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Hence,

$$\boxed{K(k) = \begin{bmatrix} 0.5042 \\ -0.3361 \end{bmatrix}}.$$

(c)

The update is

$$\hat{\theta}(k) = \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.5042 \\ -0.3361 \end{bmatrix} (1.2).$$

Therefore,

$$\hat{\theta}(k) = \begin{bmatrix} 0.4 + 0.6050 \\ 0.1 - 0.4034 \end{bmatrix}.$$

Thus,

$$\boxed{\hat{\theta}(k) = \begin{bmatrix} 1.0050 \\ -0.3034 \end{bmatrix}}.$$

(d)

We first calculate

$$K\phi^\top = \begin{bmatrix} 0.5042 \\ -0.3361 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix},$$

so

$$K\phi^\top = \begin{bmatrix} 0.5042 & -0.5042 \\ -0.3361 & 0.3361 \end{bmatrix}.$$

Therefore,

$$I - K\phi^\top = \begin{bmatrix} 0.4958 & 0.5042 \\ 0.3361 & 0.6639 \end{bmatrix}.$$

Multiplying by $P(k-1)$,

$$(I - K\phi^\top)P(k-1) = \begin{bmatrix} 1.4874 & 1.0084 \\ 1.0084 & 1.3277 \end{bmatrix}.$$

Finally,

$$P(k) = \frac{1}{0.95} \begin{bmatrix} 1.4874 & 1.0084 \\ 1.0084 & 1.3277 \end{bmatrix}.$$

Hence,

$$\boxed{P(k) \approx \begin{bmatrix} 1.5657 & 1.0615 \\ 1.0615 & 1.3976 \end{bmatrix}}.$$

Check: the forgetting factor appears explicitly as $1/\lambda$. Since $\lambda < 1$, old information is discounted, keeping the estimator responsive to parameter changes.

Course connection: Module 3.3 covers RLS gain, covariance and recursive updates; Module 3.4 covers exponential forgetting and the tracking/variance trade-off. The previous MAP2 uses essentially this same type of RLS update.

Solution 4 — White noise, bias and model validation

(a)

We have

$$v(k) = e(k) + 0.8e(k-1)$$

and

$$v(k-1) = e(k-1) + 0.8e(k-2).$$

Therefore,

$$E[v(k)v(k-1)] = E[(e(k) + 0.8e(k-1))(e(k-1) + 0.8e(k-2))].$$

Expanding,

$$\begin{aligned} E[v(k)v(k-1)] = {} & E[e(k)e(k-1)] + 0.8E[e(k)e(k-2)] \\ & + 0.8E[e^2(k-1)] + 0.64E[e(k-1)e(k-2)]. \end{aligned}$$

Because $e(k)$ is white,

$$E[e(k)e(k-1)] = 0,$$

$$E[e(k)e(k-2)] = 0,$$

and

$$E[e(k-1)e(k-2)] = 0.$$

However,

$$E[e^2(k-1)] = \sigma^2.$$

Therefore,

$$\boxed{E[v(k)v(k-1)] = 0.8\sigma^2 \neq 0}.$$

Thus $v(k)$ is not white.

$$\boxed{v(k) \text{ is colored noise.}}$$

(b)

Least-squares statistical properties rely on assumptions concerning the relationship between the disturbance and the regressors.

Because $v(k)$ is colored, it contains temporal correlation. In dynamic identification, such correlation can also interact with regressors containing past outputs.

Therefore, the usual white-noise interpretation of the least-squares estimate cannot simply be applied without checking the assumptions.

In particular, the estimate can exhibit bias under relevant colored-noise structures.

(c)

The validation errors are

$$\mathrm{SSE}_{V,1} = 6.2, \qquad \mathrm{SSE}_{V,2} = 4.7.$$

Therefore, on the supplied validation data,

$$\boxed{M_2 \text{ has the smaller validation SSE.}}$$

This is a statement about the observed validation performance only; it does not, by itself, establish that M_2 is the correct physical model.

(d)

Two important residual checks are:

1. Check whether the residuals are approximately uncorrelated in time. Significant residual autocorrelation indicates that systematic dynamics may remain unexplained.
2. Check whether the residuals are uncorrelated with the input and relevant past regressors. Significant correlation indicates that information contained in the regressors has not been fully captured by the model.

Other useful checks include comparing performance on independent validation data and examining whether the residual mean is approximately zero.

Course connection: Module 3.6 covers white versus colored noise and least-squares bias, while Module 3.7 covers model validation, residual analysis, experiment design and cross-validation. The Problem Series explicitly includes these topics in S3P9–S3P10. “‘

Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Justify your calculations and state any assumptions you use.

Problem 1 – Least-squares estimation from first principles (5 points)

Consider the scalar measurement model

$$y(k) = \theta + \varepsilon(k), \qquad k = 1, \ldots, N,$$

where θ is an unknown constant parameter and the measurement errors are independent, zero-mean random variables with variance σ^2.

(a) Define the least-squares cost function $J(\theta)$. (1 point)

(b) Derive the normal equation and obtain the least-squares estimate $\hat{\theta}$. (2 points)

(c) Show that the estimator is unbiased and calculate $\mathrm{Var}(\hat{\theta})$. (2 points)

Problem 2 – One-step recursive least squares (5 points)

Consider the linear regression model

$$y(k) = \boldsymbol{\varphi}^T(k)\boldsymbol{\theta} + \varepsilon(k),$$

with two unknown parameters. At time $k-1$, the recursive least-squares algorithm has

$$\hat{\boldsymbol{\theta}}(k-1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad P(k-1) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

At the new measurement,

$$\boldsymbol{\varphi}(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad y(k) = 1.5.$$

Assume a forgetting factor $\lambda = 1$.

Use the update equations

$$K(k) = \frac{P(k-1)\boldsymbol{\varphi}(k)}{\lambda + \boldsymbol{\varphi}^T(k)P(k-1)\boldsymbol{\varphi}(k)},$$

$$\hat{\boldsymbol{\theta}}(k) = \hat{\boldsymbol{\theta}}(k-1) + K(k)\left[y(k) - \boldsymbol{\varphi}^T(k)\hat{\boldsymbol{\theta}}(k-1)\right],$$

$$P(k) = \frac{1}{\lambda}\left[P(k-1) - K(k)\boldsymbol{\varphi}^T(k)P(k-1)\right].$$

(a) Calculate the prediction error before the update. (1 point)

(b) Calculate the gain vector $K(k)$. (1.5 points)

(c) Calculate $\hat{\boldsymbol{\theta}}(k)$ and $P(k)$. (2.5 points)

Problem 3 – Stochastic processes and correlation (5 points)

Let

$$v(k) = e(k) + 0.5e(k-1),$$

where $e(k)$ is a zero-mean white-noise process with variance $\sigma_e^2 = 4$.

(a) Calculate the mean and variance of $v(k)$. (2 points)

(b) Calculate the lag-one autocovariance $\mathrm{Cov}(v(k), v(k-1))$. (2 points)

(c) Is $v(k)$ white noise? Justify your answer. (1 point)

Problem 4 – Model validation and identifiability (5 points)

Two candidate models have been estimated using the same input-output dataset.

Model	Training SSE	Validation SSE
A	12	15
B	5	28

(a) Which model has the smaller training error? Which has the smaller validation error? (1 point)

(b) Explain why training error alone is insufficient for selecting a model. (2 points)

(c) Suppose the input used for estimating a two-parameter model is constant, so that every regressor vector is $\boldsymbol{\varphi}(k) = [1 \;\; 1]^T$. Explain why the two parameters cannot be uniquely identified from these data. (2 points)

Solutions

Problem 1 – Least-squares estimation

(a) Cost function

The residual at sample k is $y(k) - \theta$. Thus,

$$J(\theta) = \sum_{k=1}^{N} [y(k) - \theta]^2.$$

(b) Normal equation and estimate

Differentiate the cost:

$$\frac{dJ}{d\theta} = -2 \sum_{k=1}^{N} [y(k) - \theta].$$

At the minimum, set the derivative to zero:

$$-2 \sum_{k=1}^{N} [y(k) - \hat{\theta}] = 0.$$

Therefore,

$$\sum_{k=1}^{N} y(k) - N\hat{\theta} = 0,$$

and hence

$$\boxed{\hat{\theta} = \frac{1}{N} \sum_{k=1}^{N} y(k)}.$$

The second derivative is $d^2J/d\theta^2 = 2N > 0$, so this is the unique minimum for $N > 0$.

(c) Bias and variance

Using $y(k) = \theta + \varepsilon(k)$,

$$\hat{\theta} = \theta + \frac{1}{N} \sum_{k=1}^{N} \varepsilon(k).$$

Since $E[\varepsilon(k)] = 0$,

$$E[\hat{\theta}] = \theta + \frac{1}{N} \sum_{k=1}^{N} E[\varepsilon(k)] = \theta.$$

Thus, $\hat{\theta}$ is unbiased.

Independence gives zero covariance between distinct errors:

$$\begin{aligned} \mathrm{Var}(\hat{\theta}) &= \mathrm{Var}\left(\frac{1}{N} \sum_{k=1}^{N} \varepsilon(k) \right) \\ &= \frac{1}{N^2} \sum_{k=1}^{N} \mathrm{Var}(\varepsilon(k)) \\ &= \frac{1}{N^2} N\sigma^2. \end{aligned}$$

Therefore,

$$\boxed{\mathrm{Var}(\hat{\theta}) = \frac{\sigma^2}{N}}.$$

Problem 2 – Recursive least squares

(a) Prediction error

The predicted measurement is

$$\hat{y}(k|k-1) = \boldsymbol{\varphi}^T(k)\hat{\boldsymbol{\theta}}(k-1) = 1(1) + 1(-1) = 0.$$

Therefore,

$$\boxed{\epsilon(k) = y(k) - \hat{y}(k|k-1) = 1.5}.$$

(b) Gain vector

First calculate

$$P(k-1)\boldsymbol{\varphi}(k) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The denominator is

$$\lambda + \boldsymbol{\varphi}^T(k)P(k-1)\boldsymbol{\varphi}(k) = 1 + \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 5.$$

Thus,

$$\boxed{K(k) = \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix}}.$$

(c) Parameter and covariance updates

The parameter update is

$$\begin{aligned} \hat{\boldsymbol{\theta}}(k) &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix} (1.5) \\ &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0.6 \\ 0.6 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{\hat{\boldsymbol{\theta}}(k) = \begin{bmatrix} 1.6 \\ -0.4 \end{bmatrix}}.$$

For the covariance,

$$\boldsymbol{\varphi}^T(k)P(k-1) = \begin{bmatrix} 2 & 2 \end{bmatrix},$$

so

$$\begin{aligned} P(k) &= P(k-1) - K(k)\boldsymbol{\varphi}^T(k)P(k-1) \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 0.4 \\ 0.4 \end{bmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 0.8 & 0.8 \\ 0.8 & 0.8 \end{bmatrix}. \end{aligned}$$

Hence,

$$\boxed{P(k) = \begin{bmatrix} 1.2 & -0.8 \\ -0.8 & 1.2 \end{bmatrix}}.$$

Problem 3 – Stochastic processes

(a) Mean and variance

Because $e(k)$ is zero-mean,

$$E[v(k)] = E[e(k)] + 0.5E[e(k-1)] = 0.$$

White noise has zero covariance at nonzero lags, so $e(k)$ and $e(k-1)$ are uncorrelated. Consequently,

$$\begin{aligned}\mathrm{Var}(v(k)) &= \mathrm{Var}(e(k) + 0.5e(k-1)) \\ &= \mathrm{Var}(e(k)) + 0.25\,\mathrm{Var}(e(k-1)) \\ &= 4 + 0.25(4).\end{aligned}$$

Therefore,

$$\boxed{E[v(k)] = 0, \qquad \mathrm{Var}(v(k)) = 5}.$$

(b) Lag-one autocovariance

Write

$$v(k-1) = e(k-1) + 0.5e(k-2).$$

Since both processes have zero mean,

$$\begin{aligned}\mathrm{Cov}(v(k), v(k-1)) = E[&(e(k) + 0.5e(k-1)) \\ &(e(k-1) + 0.5e(k-2))].\end{aligned}$$

Expanding,

$$\begin{aligned}&E[e(k)e(k-1)] + 0.5E[e(k)e(k-2)] \\ &\quad + 0.5E[e(k-1)^2] + 0.25E[e(k-1)e(k-2)].\end{aligned}$$

All terms involving distinct time indices are zero. The remaining term is

$$0.5E[e(k-1)^2] = 0.5(4) = 2.$$

Thus,

$$\boxed{\mathrm{Cov}(v(k), v(k-1)) = 2}.$$

(c) Whiteness

A white-noise process must have zero autocovariance at every nonzero lag. Here the lag-one autocovariance is $2 \neq 0$. Therefore,

$$\boxed{v(k) \text{ is not white noise.}}$$

Problem 4 – Model validation and identifiability

(a) Error comparison

Model B has the smaller training SSE:

$$5 < 12.$$

Model A has the smaller validation SSE:

$$15 < 28.$$

Thus, B fits the training data more closely, while A performs better on the supplied validation data.

(b) Why training error is insufficient

A model with more parameters or excessive flexibility can fit the training observations closely, including their noise, without generalizing well to unseen data. The validation SSE measures performance on data not used for estimating the parameters. Here, Model B's training SSE is lower, but its validation SSE is higher, which is consistent with possible overfitting. The table alone does not establish the precise cause of the difference.

(c) Identifiability

For every sample,

$$y(k) = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}\theta_1 \\ \theta_2\end{bmatrix} + \varepsilon(k) = \theta_1 + \theta_2 + \varepsilon(k).$$

The measurements depend only on the sum $\theta_1 + \theta_2$. Any parameter pair with the same sum produces the same predicted output.

The information matrix is

$$\Phi^T\Phi = \sum_{k=1}^{N} \begin{bmatrix}1\\1\end{bmatrix} \begin{bmatrix}1 & 1\end{bmatrix} = N \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}.$$

Its determinant is

$$\det(\Phi^T\Phi) = N^2(1-1) = 0.$$

Thus, the information matrix is singular, and the two parameters cannot be uniquely identified separately from these data:

$$\boxed{\text{Only } \theta_1 + \theta_2 \text{ is identifiable.}}$$

A richer input or regressor sequence is needed to distinguish the parameters.

LaTeX

‘‘‘

Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Justify your reasoning and show the essential calculation steps.

Problem 1 – Batch least-squares estimation (5 points)

Consider the model

$$y(k) = \theta u(k) + e(k), \qquad k = 1, \dots, N,$$

where θ is an unknown scalar parameter and $e(k)$ is a zero-mean white-noise sequence with variance σ^2.

The available measurements are

k	1	2	3	4
$u(k)$	1	1	2	2
$y(k)$	1.2	0.8	2.2	1.8

(a) Write the least-squares cost function $J(\theta)$. (1 point)

(b) Derive the normal equation and calculate $\hat{\theta}$. (2 points)

(c) Calculate the residual sum of squares

$$J(\hat{\theta}) = \sum_{k=1}^{4} [y(k) - \hat{\theta}u(k)]^2.$$

(2 points)

Problem 2 – Recursive least squares and forgetting (5 points)

A scalar recursive least-squares estimator uses

$$K(k) = \frac{P(k-1)u(k)}{\lambda + u^2(k)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - u(k)\hat{\theta}(k-1)],$$
$$P(k) = \frac{1}{\lambda}\left[P(k-1) - K(k)u(k)P(k-1)\right].$$

At the current sample, let

$$\hat{\theta}(k-1) = 0.5, \quad P(k-1) = 2, \quad u(k) = 1, \quad y(k) = 2, \quad \lambda = 0.5.$$

(a) Calculate the prediction error. (1 point)

(b) Calculate $K(k)$ and the updated estimate $\hat{\theta}(k)$. (2 points)

(c) Calculate $P(k)$. Briefly explain the role of the forgetting factor $\lambda < 1$. (2 points)

Problem 3 – Noise correlation and least-squares estimation (5 points)

Consider the measurement model

$$y(k) = \theta u(k) + v(k),$$

where $u(k)$ is a known input and $v(k)$ is a zero-mean disturbance.

(a) State the usual condition on the relationship between the regressor and the disturbance that supports unbiased least-squares estimation. (1.5 points)

(b) Suppose $v(k)$ is correlated with $u(k)$. Explain why the least-squares estimate may be biased. (2 points)

(c) Explain why increasing the number of measurements does not, by itself, remove bias caused by this correlation. (1.5 points)

Problem 4 – Excitation and parameter identifiability (5 points)

Consider the two-parameter model

$$y(k) = \theta_1 + \theta_2 u(k) + e(k).$$

(a) Write the regressor vector $\boldsymbol{\varphi}(k)$ and the corresponding regression equation. (1 point)

(b) Suppose $u(k) = 3$ for every sample. Show that the parameters cannot be uniquely identified. (2 points)

(c) Now suppose the input takes at least two distinct values. Explain the condition under which the two parameters can be uniquely identified from the regression data. (2 points)

Solutions

Problem 1 – Batch least-squares estimation

(a) Cost function

The residual is $y(k) - \theta u(k)$, so

$$\boxed{J(\theta) = \sum_{k=1}^{4} [y(k) - \theta u(k)]^2.}$$

(b) Normal equation and estimate

Differentiate:

$$\frac{dJ}{d\theta} = -2 \sum_{k=1}^{4} u(k)[y(k) - \theta u(k)].$$

Setting the derivative to zero gives

$$\sum_{k=1}^{4} u(k)y(k) - \hat{\theta} \sum_{k=1}^{4} u^2(k) = 0.$$

Therefore,

$$\hat{\theta} = \frac{\sum_{k=1}^{4} u(k)y(k)}{\sum_{k=1}^{4} u^2(k)}.$$

Calculate the sums:

$$\begin{aligned} \sum u(k)y(k) &= 1(1.2) + 1(0.8) + 2(2.2) + 2(1.8) \\ &= 1.2 + 0.8 + 4.4 + 3.6 = 10, \\ \sum u^2(k) &= 1 + 1 + 4 + 4 = 10. \end{aligned}$$

Hence,

$$\boxed{\hat{\theta} = 1.}$$

(c) Residual sum of squares

Using $\hat{\theta} = 1$:

k	1	2	3	4
$y(k) - \hat{\theta}u(k)$	0.2	−0.2	0.2	−0.2

Thus,

$$\begin{aligned} J(\hat{\theta}) &= (0.2)^2 + (-0.2)^2 + (0.2)^2 + (-0.2)^2 \\ &= 4(0.04). \end{aligned}$$

Therefore,

$$\boxed{J(\hat{\theta}) = 0.16.}$$

Problem 2 – Recursive least squares

(a) Prediction error

The predicted output is

$$\hat{y}(k|k-1) = u(k)\hat{\theta}(k-1) = 1(0.5) = 0.5.$$

Therefore,

$$\boxed{\epsilon(k) = 2 - 0.5 = 1.5.}$$

(b) Gain and updated estimate

The gain is

$$K(k) = \frac{2(1)}{0.5 + 1^2(2)} = \frac{2}{2.5} = 0.8.$$

Thus,

$$\boxed{K(k) = 0.8.}$$

The parameter update is

$$\hat{\theta}(k) = 0.5 + 0.8(1.5) = 0.5 + 1.2.$$

Hence,

$$\boxed{\hat{\theta}(k) = 1.7.}$$

(c) Covariance and forgetting factor

Using the covariance update,

$$P(k) = \frac{1}{0.5}[2 - 0.8(1)(2)] = 2(2 - 1.6) = 2(0.4).$$

Therefore,

$$\boxed{P(k) = 0.8.}$$

A forgetting factor $\lambda < 1$ gives less weight to older data relative to new measurements. This can help the estimator adapt when the underlying parameters change over time. The trade-off is that forgetting can increase estimate variability.

Problem 3 – Noise correlation and least-squares estimation

(a) Required condition

A standard sufficient condition is that the disturbance has zero conditional mean given the regressors:

$$\boxed{E[v(k) \mid u(1), \ldots, u(N)] = 0.}$$

In particular, the disturbance should be uncorrelated with the regressors under the usual zero-mean assumptions.

(b) Why correlation may cause bias

For the scalar model, the least-squares estimate is

$$\hat{\theta} = \frac{\sum_{k=1}^{N} u(k)y(k)}{\sum_{k=1}^{N} u^2(k)}.$$

Substituting $y(k) = \theta u(k) + v(k)$ gives

$$\hat{\theta} = \theta + \frac{\sum_{k=1}^{N} u(k)v(k)}{\sum_{k=1}^{N} u^2(k)}.$$

The estimation error is therefore

$$\hat{\theta} - \theta = \frac{\sum_{k=1}^{N} u(k)v(k)}{\sum_{k=1}^{N} u^2(k)}.$$

If the input and disturbance are correlated, the numerator need not have zero expectation. The estimator may consequently be biased.

(c) Effect of more measurements

More measurements can reduce random estimation variability, but they do not automatically eliminate systematic correlation between the regressor and disturbance. If the correlation persists as the dataset grows, the estimator may converge to a value different from the true parameter. Additional data alone are therefore not a guarantee of consistency.

Problem 4 – Excitation and identifiability

(a) Regressor vector

The model can be written as

$$y(k) = \boldsymbol{\varphi}^T(k)\boldsymbol{\theta} + e(k),$$

where

$$\boxed{\boldsymbol{\varphi}(k) = \begin{bmatrix} 1 \\ u(k) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.}$$

(b) Constant input $u(k) = 3$

For every sample,

$$\boldsymbol{\varphi}(k) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

The model becomes

$$y(k) = \theta_1 + 3\theta_2 + e(k).$$

The observations depend only on the combination $\theta_1 + 3\theta_2$. Different parameter pairs can have the same value of this combination.

The information matrix is

$$\begin{aligned} \Phi^T\Phi &= \sum_{k=1}^{N} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} \\ &= N \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}. \end{aligned}$$

Its determinant is

$$\det(\Phi^T\Phi) = N^2(9 - 9) = 0.$$

It is singular, so the two parameters cannot be uniquely identified.

(c) Distinct input values

The information matrix is

$$\Phi^T\Phi = \begin{bmatrix} N & \sum_{k=1}^{N} u(k) \\ \sum_{k=1}^{N} u(k) & \sum_{k=1}^{N} u^2(k) \end{bmatrix}.$$

Its determinant is

$$\det(\Phi^T\Phi) = N \sum_{k=1}^{N} u^2(k) - \left(\sum_{k=1}^{N} u(k) \right)^2.$$

For this two-parameter model, unique least-squares identification requires

$$\boxed{\operatorname{rank}(\Phi) = 2}$$

or, equivalently,

$$\boxed{N \sum_{k=1}^{N} u^2(k) - \left(\sum_{k=1}^{N} u(k) \right)^2 > 0.}$$

This holds when the observed input values are not all identical. Having at least two distinct input values makes the constant and input-dependent regressors linearly independent. “‘

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Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Show the main derivation steps and justify your answers.

Problem 1 – Least-squares estimation and parameter uncertainty (5 points)

Consider the regression model

$$y(k) = \theta\varphi(k) + e(k), \qquad k = 1, \dots, N,$$

where θ is unknown and $e(k)$ is zero-mean white noise with variance σ^2.

The available data satisfy

$$\sum_{k=1}^{N} \varphi^2(k) = 20, \qquad \sum_{k=1}^{N} \varphi(k)y(k) = 30, \qquad N = 10.$$

(a) Derive the least-squares estimate $\hat{\theta}$. (2 points)

(b) Calculate its numerical value. (1 point)

(c) Assuming the regressors are deterministic, calculate $\mathrm{Var}(\hat{\theta})$ in terms of σ^2. (2 points)

Problem 2 – Recursive least squares with a new measurement (5 points)

A scalar RLS estimator is initialized with

$$\hat{\theta}(k-1) = 1, \qquad P(k-1) = 4.$$

The new observation is

$$\varphi(k) = 2, \qquad y(k) = 3,$$

and the forgetting factor is $\lambda = 1$.

Use

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^2(k)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \varphi(k)\hat{\theta}(k-1)\right],$$

$$P(k) = \frac{1}{\lambda}\left[P(k-1) - K(k)\varphi(k)P(k-1)\right].$$

(a) Calculate the prior prediction and prediction error. (1 point)

(b) Calculate the gain $K(k)$ and updated estimate $\hat{\theta}(k)$. (2 points)

(c) Calculate $P(k)$ and explain briefly what its decrease represents in this update. (2 points)

Problem 3 – Autocorrelation of a moving-average process (5 points)

Let

$$v(k) = e(k) + ae(k-1),$$

where $e(k)$ is zero-mean white noise with variance σ_e^2 and a is a constant.

(a) Derive the variance $R_v(0) = E[v^2(k)]$. (2 points)

(b) Derive the lag-one autocorrelation $R_v(1) = E[v(k)v(k-1)]$. (2 points)

(c) For $\sigma_e^2 = 2$ and $a = 0.5$, calculate $R_v(0)$ and $R_v(1)$. Is $v(k)$ white noise? (1 point)

Problem 4 – Identifying a first-order dynamic model (5 points)

Consider the model

$$y(k) = ay(k-1) + bu(k-1) + e(k).$$

Assume $y(k-1)$ and $u(k-1)$ are available measurements.

(a) Express the model as a linear regression in the unknown parameter vector $\boldsymbol{\theta} = [a \;\; b]^T$. (1 point)

(b) Write the batch least-squares estimate using the regression matrix Φ and output vector $\mathbf{y}$. State the condition required for the inverse to exist. (2 points)

(c) Explain why a dataset for which $u(k-1) = y(k-1)$ at every sample cannot uniquely identify a and b separately. (2 points)

Solutions

Problem 1 – Least-squares estimation and uncertainty

(a) Derivation

The cost function is

$$J(\theta) = \sum_{k=1}^{N} [y(k) - \theta\varphi(k)]^2.$$

Differentiate with respect to θ:

$$\frac{dJ}{d\theta} = -2\sum_{k=1}^{N} \varphi(k)[y(k) - \theta\varphi(k)].$$

Setting the derivative to zero yields

$$\sum_{k=1}^{N} \varphi(k)y(k) - \hat{\theta}\sum_{k=1}^{N} \varphi^2(k) = 0.$$

Therefore,

$$\boxed{\hat{\theta} = \frac{\sum_{k=1}^{N} \varphi(k)y(k)}{\sum_{k=1}^{N} \varphi^2(k)}}.$$

(b) Numerical value

Substituting the given sums,

$$\hat{\theta} = \frac{30}{20}.$$

Hence,

$$\boxed{\hat{\theta} = 1.5}.$$

(c) Estimator variance

With deterministic regressors,

$$\hat{\theta} - \theta = \frac{\sum_{k=1}^{N} \varphi(k)e(k)}{\sum_{k=1}^{N} \varphi^2(k)}.$$

Because the errors are white with variance σ^2,

$$\begin{aligned} \mathrm{Var}(\hat{\theta}) &= \frac{\mathrm{Var}\left(\sum_{k=1}^{N} \varphi(k)e(k)\right)}{\left(\sum_{k=1}^{N} \varphi^2(k)\right)^2} \\ &= \frac{\sigma^2 \sum_{k=1}^{N} \varphi^2(k)}{\left(\sum_{k=1}^{N} \varphi^2(k)\right)^2}. \end{aligned}$$

Thus,

$$\mathrm{Var}(\hat{\theta}) = \frac{\sigma^2}{\sum_{k=1}^{N} \varphi^2(k)} = \boxed{\frac{\sigma^2}{20}}.$$

Problem 2 – Recursive least squares

(a) Prediction and error

The prior predicted output is

$$\hat{y}(k|k-1) = \varphi(k)\hat{\theta}(k-1) = 2(1) = 2.$$

Therefore, the prediction error is

$$\boxed{\epsilon(k) = y(k) - \hat{y}(k|k-1) = 3 - 2 = 1}.$$

(b) Gain and updated estimate

The gain is

$$\begin{aligned} K(k) &= \frac{4(2)}{1 + 2^2(4)} \\ &= \frac{8}{17}. \end{aligned}$$

Hence,

$$\boxed{K(k) = \frac{8}{17} \approx 0.4706}.$$

The parameter update is

$$\begin{aligned} \hat{\theta}(k) &= 1 + \frac{8}{17}(1) \\ &= \boxed{\frac{25}{17} \approx 1.4706}. \end{aligned}$$

(c) Covariance update

Using the covariance recursion,

$$\begin{aligned} P(k) &= 4 - \frac{8}{17}(2)(4) \\ &= 4 - \frac{64}{17} \\ &= \frac{68 - 64}{17}. \end{aligned}$$

Therefore,

$$\boxed{P(k) = \frac{4}{17} \approx 0.2353}.$$

The new measurement supplies information about the parameter, reducing the estimator's covariance measure from 4 to approximately 0.2353.

Problem 3 – Moving-average process

(a) Variance

Since $e(k)$ and $e(k-1)$ are uncorrelated,

$$\begin{aligned} R_v(0) &= E[(e(k) + ae(k-1))^2] \\ &= E[e^2(k)] + 2aE[e(k)e(k-1)] + a^2E[e^2(k-1)] \\ &= \sigma_e^2 + a^2\sigma_e^2. \end{aligned}$$

Thus,

$$\boxed{R_v(0) = (1 + a^2)\sigma_e^2}.$$

(b) Lag-one autocorrelation

We have

$$v(k-1) = e(k-1) + ae(k-2).$$

Therefore,

$$\begin{aligned} R_v(1) = E[&(e(k) + ae(k-1)) \\ &(e(k-1) + ae(k-2))] \\ = E[&e(k)e(k-1)] + aE[e(k)e(k-2)] \\ &+ aE[e^2(k-1)] + a^2E[e(k-1)e(k-2)]. \end{aligned}$$

All cross-time expectations vanish except $E[e^2(k-1)] = \sigma_e^2$. Consequently,

$$\boxed{R_v(1) = a\sigma_e^2}.$$

(c) Numerical values and whiteness

For $\sigma_e^2 = 2$ and $a = 0.5$,

$$R_v(0) = (1 + 0.5^2)2 = 1.25(2) = \boxed{2.5},$$

and

$$R_v(1) = 0.5(2) = \boxed{1}.$$

Since $R_v(1) \neq 0$, the process is not white noise.

Problem 4 – Dynamic-model identification

(a) Regression form

Rearranging the model,

$$y(k) = \begin{bmatrix} y(k-1) & u(k-1) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + e(k).$$

Therefore,

$$\boxed{\boldsymbol{\varphi}(k) = \begin{bmatrix} y(k-1) \\ u(k-1) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}.}$$

(b) Batch least-squares estimate

Stacking all N observations gives

$$\mathbf{y} = \Phi\boldsymbol{\theta} + \mathbf{e},$$

where

$$\Phi = \begin{bmatrix} y(0) & u(0) \\ y(1) & u(1) \\ \vdots & \vdots \\ y(N-1) & u(N-1) \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y(1) \\ y(2) \\ \vdots \\ y(N) \end{bmatrix}.$$

The least-squares estimate is

$$\boxed{\hat{\boldsymbol{\theta}} = (\Phi^T\Phi)^{-1}\Phi^T\mathbf{y}}.$$

The inverse exists if $\Phi^T\Phi$ is nonsingular, equivalently if Φ has full column rank:

$$\boxed{\text{rank}(\Phi) = 2}.$$

(c) Lack of unique identification

If $u(k-1) = y(k-1)$ for every sample, then each row of the regression matrix has the form

$$\begin{bmatrix} r(k) & r(k) \end{bmatrix}.$$

The model becomes

$$y(k) = [a + b]y(k-1) + e(k).$$

Only the sum $a + b$ affects the predicted output. Moreover, the two columns of Φ are identical, so

$$\text{rank}(\Phi) \leq 1 < 2.$$

Consequently, $\Phi^T\Phi$ is singular and a and b cannot be uniquely identified separately:

$$\boxed{\text{Only the combination } a + b \text{ is identifiable.}}$$

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Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Show the essential derivations and justify your conclusions.

Problem 1 – Least-squares estimation with two parameters (5 points)

Consider the regression model

$$y(k) = \theta_1 + \theta_2 u(k) + e(k).$$

The measured data are

k	1	2	3	4
$u(k)$	-1	0	1	2
$y(k)$	1	2	2	5

(a) Write the regression matrix Φ, the output vector $\mathbf{y}$, and the parameter vector $\boldsymbol{\theta}$. (1 point)

(b) Calculate $\Phi^T\Phi$ and $\Phi^T\mathbf{y}$. (2 points)

(c) Calculate the least-squares estimate

$$\hat{\boldsymbol{\theta}} = (\Phi^T\Phi)^{-1}\Phi^T\mathbf{y}.$$

(2 points)

Problem 2 – Recursive least squares with forgetting (5 points)

A scalar RLS estimator has the current estimates

$$\hat{\theta}(k-1) = 2, \qquad P(k-1) = 3.$$

A new measurement arrives:

$$\varphi(k) = 1, \qquad y(k) = 1.$$

The forgetting factor is $\lambda = 0.8$.

Use

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^2(k)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \varphi(k)\hat{\theta}(k-1)\right],$$

$$P(k) = \frac{1}{\lambda}\left[P(k-1) - K(k)\varphi(k)P(k-1)\right].$$

(a) Calculate the prediction error. (1 point)

(b) Calculate $K(k)$ and $\hat{\theta}(k)$. (2 points)

(c) Calculate $P(k)$ and explain how the forgetting factor affects the estimator. (2 points)

Problem 3 – Statistical properties of measurement noise (5 points)

Let the measurement noise be

$$v(k) = e(k) + 0.5e(k-1),$$

where $e(k)$ is zero-mean white noise with variance $\sigma_e^2 = 4$.

(a) Calculate $E[v(k)]$ and $\mathrm{Var}(v(k))$. (2 points)

(b) Calculate the autocovariance

$$R_v(1) = E[v(k)v(k-1)].$$

(2 points)

(c) Is $v(k)$ white noise? Explain. (1 point)

Problem 4 – Persistent excitation and identifiability (5 points)

Consider the model

$$y(k) = \theta_1 u(k) + \theta_2 u(k-1) + e(k).$$

(a) Write the regressor vector and the corresponding regression equation. (1 point)

(b) Suppose $u(k) = 1$ for every sample. Show that the two parameters cannot be uniquely identified. (2 points)

(c) State the matrix-rank condition required for unique least-squares identification of both parameters. Explain the role of the input sequence. (2 points)

Solutions

Problem 1 – Least-squares estimation with two parameters

(a) Regression representation

The model is

$$y(k) = \begin{bmatrix} 1 & u(k) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + e(k).$$

Thus,

$$\Phi = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 5 \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

(b) Calculate the normal-equation terms

First,

$$\Phi^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \Phi^T\Phi &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}. \end{aligned}$$

Similarly,

$$\begin{aligned} \Phi^T\mathbf{y} &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 1+2+2+5 \\ -1+0+2+10 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ 11 \end{bmatrix}. \end{aligned}$$

(c) Least-squares estimate

The normal equations are

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 11 \end{bmatrix}.$$

The determinant is

$$\det(\Phi^T\Phi) = 4(6) - 2(2) = 20.$$

Hence,

$$(\Phi^T\Phi)^{-1} = \frac{1}{20} \begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix}.$$

Then

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \frac{1}{20} \begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 10 \\ 11 \end{bmatrix} \\ &= \frac{1}{20} \begin{bmatrix} 60 - 22 \\ -20 + 44 \end{bmatrix} \\ &= \begin{bmatrix} 1.9 \\ 1.2 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{\hat{\theta}_1 = 1.9, \qquad \hat{\theta}_2 = 1.2.}$$

Problem 2 – Recursive least squares with forgetting

(a) Prediction error

The predicted output is

$$\hat{y}(k|k-1) = \varphi(k)\hat{\theta}(k-1) = 1(2) = 2.$$

Therefore,

$$\boxed{\epsilon(k) = y(k) - \hat{y}(k|k-1) = 1 - 2 = -1.}$$

(b) Gain and parameter update

The gain is

$$\begin{aligned} K(k) &= \frac{3(1)}{0.8 + 1^2(3)} \\ &= \frac{3}{3.8} = \frac{15}{19}. \end{aligned}$$

Thus,

$$\boxed{K(k) \approx 0.7895.}$$

The estimate update gives

$$\begin{aligned} \hat{\theta}(k) &= 2 + \frac{15}{19}(-1) \\ &= \frac{38 - 15}{19}. \end{aligned}$$

Hence,

$$\boxed{\hat{\theta}(k) = \frac{23}{19} \approx 1.2105.}$$

(c) Covariance and forgetting

The covariance update is

$$\begin{aligned} P(k) &= \frac{1}{0.8}\left[3 - \frac{15}{19}(1)(3)\right] \\ &= 1.25\left[3 - \frac{45}{19}\right] \\ &= 1.25\left(\frac{12}{19}\right) \\ &= \frac{15}{19}. \end{aligned}$$

Therefore,

$$\boxed{P(k) \approx 0.7895.}$$

A forgetting factor below one reduces the influence of older measurements relative to newer ones. It helps the estimator track changing parameters, although it can increase sensitivity to measurement noise.

Problem 3 – Statistical properties of measurement noise

(a) Mean and variance

The mean is

$$E[v(k)] = E[e(k)] + 0.5E[e(k-1)] = 0.$$

Since distinct samples of white noise are uncorrelated,

$$\begin{aligned}\mathrm{Var}(v(k)) &= E[(e(k)+0.5e(k-1))^2] \\ &= E[e^2(k)] + 0.25E[e^2(k-1)] \\ &\quad + E[e(k)e(k-1)] \\ &= 4 + 0.25(4) + 0 \\ &= 5.\end{aligned}$$

Thus,

$$\boxed{E[v(k)] = 0, \qquad \mathrm{Var}(v(k)) = 5.}$$

(b) Lag-one autocovariance

Write

$$v(k-1) = e(k-1) + 0.5e(k-2).$$

Then

$$\begin{aligned}R_v(1) = E[&(e(k)+0.5e(k-1)) \\ &(e(k-1)+0.5e(k-2))].\end{aligned}$$

Expanding,

$$\begin{aligned}R_v(1) = E[e(k)e(k-1)] + 0.5E[e(k)e(k-2)] \\ + 0.5E[e^2(k-1)] + 0.25E[e(k-1)e(k-2)].\end{aligned}$$

The cross-time terms are zero, leaving

$$R_v(1) = 0.5\sigma_e^2 = 0.5(4).$$

Therefore,

$$\boxed{R_v(1) = 2.}$$

(c) Whiteness

A white-noise process has zero autocovariance at every nonzero lag. Since

$$R_v(1) = 2 \neq 0,$$

the process $v(k)$ is not white noise.

Problem 4 – Persistent excitation and identifiability

(a) Regression equation

The model is linear in the unknown parameters:

$$y(k) = \begin{bmatrix} u(k) & u(k-1) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + e(k).$$

Therefore,

$$\boxed{\boldsymbol{\varphi}(k) = \begin{bmatrix} u(k) \\ u(k-1) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.}$$

(b) Constant input

If $u(k) = 1$ for every sample, then $u(k-1) = 1$ as well. Hence,

$$y(k) = \theta_1 + \theta_2 + e(k).$$

The data depend only on $\theta_1 + \theta_2$, not on each parameter separately.

The regression matrix has identical columns:

$$\Phi = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix}.$$

Thus,

$$\operatorname{rank}(\Phi) = 1 < 2.$$

The parameters cannot be uniquely identified.

(c) Rank condition and input design

Unique least-squares identification requires

$$\boxed{\operatorname{rank}(\Phi) = 2,}$$

or equivalently that $\Phi^T\Phi$ be nonsingular.

The input must provide sufficiently varied regressors so that the columns corresponding to $u(k)$ and $u(k-1)$ are linearly independent over the observed data. A constant input fails this condition because the two regressor columns are identical. More informative input variation can make the parameters distinguishable, provided the resulting regression matrix has full column rank. ‘‘‘

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Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Justify your reasoning and show the essential calculation steps.

Problem 1 – Least-squares estimation and residual analysis (5 points)

Consider the scalar model

$$y(k) = \theta u(k) + e(k),$$

where θ is unknown. The following measurements are available:

k	1	2	3	4
$u(k)$	1	2	-1	2
$y(k)$	2	3	-1	5

(a) Derive the least-squares estimate of θ from the cost function. (2 points)

(b) Calculate the numerical estimate $\hat{\theta}$. (1 point)

(c) Calculate the residuals and the residual sum of squares

$$J(\hat{\theta}) = \sum_{k=1}^{4} [y(k) - \hat{\theta} u(k)]^2.$$

(2 points)

Problem 2 – Recursive least squares and covariance update (5 points)

A scalar RLS estimator has

$$\hat{\theta}(k-1) = 1, \qquad P(k-1) = 2.$$

The new measurement is

$$\varphi(k) = 2, \qquad y(k) = 3,$$

with forgetting factor $\lambda = 1$.

Use

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^2(k)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \varphi(k)\hat{\theta}(k-1)],$$

$$P(k) = P(k-1) - K(k)\varphi(k)P(k-1).$$

(a) Calculate the prediction error. (1 point)

(b) Calculate the gain and updated parameter estimate. (2 points)

(c) Calculate the updated covariance $P(k)$. (2 points)

Problem 3 – Properties of a stochastic process (5 points)

Let

$$v(k) = e(k) + ae(k-1),$$

where $e(k)$ is zero-mean white noise with variance σ_e^2.

(a) Derive the mean and variance of $v(k)$. (2 points)

(b) Derive the lag-one autocovariance

$$R_v(1) = E[v(k)v(k-1)].$$

(2 points)

(c) For $a = -0.5$ and $\sigma_e^2 = 4$, calculate $R_v(0)$ and $R_v(1)$. Is $v(k)$ white? (1 point)

Problem 4 – Model structure and parameter identifiability (5 points)

Consider the regression model

$$y(k) = \theta_1 + \theta_2 u(k) + e(k).$$

(a) Write the regression equation in matrix form for N measurements. (1 point)

(b) Suppose $u(k) = c$ for every sample, where c is a constant. Show that the two parameters cannot be uniquely identified. (2 points)

(c) State the rank condition for unique least-squares identification and explain how input variation affects it. (2 points)

Solutions

Problem 1 – Least-squares estimation

(a) Derivation

The least-squares cost is

$$J(\theta) = \sum_{k=1}^{4} [y(k) - \theta u(k)]^2.$$

Differentiating,

$$\frac{dJ}{d\theta} = -2 \sum_{k=1}^{4} u(k)[y(k) - \theta u(k)].$$

Set the derivative to zero:

$$\sum_{k=1}^{4} u(k)y(k) - \hat{\theta} \sum_{k=1}^{4} u^2(k) = 0.$$

Therefore,

$$\boxed{\hat{\theta} = \frac{\sum_{k=1}^{4} u(k)y(k)}{\sum_{k=1}^{4} u^2(k)}}.$$

(b) Numerical estimate

Calculate the two sums:

$$\begin{aligned} \sum_{k=1}^{4} u(k)y(k) &= 1(2) + 2(3) + (-1)(-1) + 2(5) \\ &= 2 + 6 + 1 + 10 = 19, \\ \sum_{k=1}^{4} u^2(k) &= 1 + 4 + 1 + 4 = 10. \end{aligned}$$

Hence,

$$\boxed{\hat{\theta} = \frac{19}{10} = 1.9}.$$

(c) Residuals and cost

Define the residual as

$$\hat{e}(k) = y(k) - \hat{\theta} u(k).$$

Using $\hat{\theta} = 1.9$,

$$\begin{aligned} \hat{e}(1) &= 2 - 1.9(1) = 0.1, \\ \hat{e}(2) &= 3 - 1.9(2) = -0.8, \\ \hat{e}(3) &= -1 - 1.9(-1) = 0.9, \\ \hat{e}(4) &= 5 - 1.9(2) = 1.2. \end{aligned}$$

Thus,

$$\begin{aligned} J(\hat{\theta}) &= (0.1)^2 + (-0.8)^2 + (0.9)^2 + (1.2)^2 \\ &= 0.01 + 0.64 + 0.81 + 1.44. \end{aligned}$$

Therefore,

$$\boxed{J(\hat{\theta}) = 2.90}.$$

Problem 2 – Recursive least squares

(a) Prediction error

The prior prediction is

$$\hat{y}(k|k-1) = \varphi(k)\hat{\theta}(k-1) = 2(1) = 2.$$

Therefore,

$$\boxed{\epsilon(k) = 3 - 2 = 1}.$$

(b) Gain and parameter update

The gain is

$$\begin{aligned} K(k) &= \frac{2(2)}{1 + 2^2(2)} \\ &= \frac{4}{9}. \end{aligned}$$

Thus,

$$\boxed{K(k) = \frac{4}{9}}.$$

The updated estimate is

$$\begin{aligned} \hat{\theta}(k) &= 1 + \frac{4}{9}(1) \\ &= \boxed{\frac{13}{9} \approx 1.4444}. \end{aligned}$$

(c) Covariance update

Using the specified recursion,

$$\begin{aligned} P(k) &= 2 - \frac{4}{9}(2)(2) \\ &= 2 - \frac{16}{9} \\ &= \boxed{\frac{2}{9} \approx 0.2222}. \end{aligned}$$

The covariance decreases because the new measurement provides information about the unknown parameter.

Problem 3 – Stochastic process

(a) Mean and variance

Since $e(k)$ is zero mean,

$$E[v(k)] = E[e(k)] + aE[e(k-1)] = 0.$$

Because white noise is uncorrelated at distinct time instants,

$$\begin{aligned} R_v(0) &= E[v^2(k)] \\ &= E[(e(k) + ae(k-1))^2] \\ &= \sigma_e^2 + a^2\sigma_e^2. \end{aligned}$$

Therefore,

$$\boxed{E[v(k)] = 0, \qquad \mathrm{Var}(v(k)) = (1 + a^2)\sigma_e^2}.$$

(b) Lag-one autocovariance

Since

$$v(k-1) = e(k-1) + ae(k-2),$$

we have

$$\begin{aligned} R_v(1) = E[&(e(k) + ae(k-1)) \\ &(e(k-1) + ae(k-2))]. \end{aligned}$$

Expanding and using the whiteness of $e(k)$, all cross-time terms vanish:

$$R_v(1) = aE[e^2(k-1)].$$

Consequently,

$$\boxed{R_v(1) = a\sigma_e^2}.$$

(c) Numerical values and whiteness

For $a = -0.5$ and $\sigma_e^2 = 4$,

$$R_v(0) = (1 + 0.25)4 = \boxed{5},$$

and

$$R_v(1) = (-0.5)(4) = \boxed{-2}.$$

Since $R_v(1) \neq 0$, the process is not white noise.

Problem 4 – Model structure and identifiability

(a) Matrix form

For each sample,

$$y(k) = \begin{bmatrix} 1 & u(k) \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + e(k).$$

Stacking the measurements,

$$\boxed{\mathbf{y} = \Phi\boldsymbol{\theta} + \mathbf{e}},$$

where

$$\Phi = \begin{bmatrix} 1 & u(1) \\ 1 & u(2) \\ \vdots & \vdots \\ 1 & u(N) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

(b) Constant input

If $u(k) = c$ for every sample, the model reduces to

$$y(k) = \theta_1 + c\theta_2 + e(k).$$

Only the combination $\theta_1 + c\theta_2$ affects the predicted output.

The regression matrix is

$$\Phi = \begin{bmatrix} 1 & c \\ 1 & c \\ \vdots & \vdots \\ 1 & c \end{bmatrix}.$$

Its second column is c times its first column, so

$$\operatorname{rank}(\Phi) = 1 < 2.$$

Therefore, $\Phi^T\Phi$ is singular and the two parameters cannot be uniquely identified.

(c) Rank condition and input variation

Unique least-squares identification requires

$$\boxed{\operatorname{rank}(\Phi) = 2}.$$

Equivalently, $\Phi^T\Phi$ must be nonsingular.

For this model, the input must not be constant across all measurements. If it takes at least two distinct values, the constant regressor and the input regressor are linearly independent, making unique identification possible under this rank condition. “‘

Mock MAP 2 – Control of Cyber-Physical Systems

Scope: Module 3 **Duration:** 30 minutes **Total:** 20 points

Instructions: Answer all four problems. Each problem is worth 5 points. Show the essential derivations and justify your conclusions.

Problem 1 – Least-squares identification of an ARX model (5 points)

Consider the model

$$y(k) = -a\,y(k-1) + b\,u(k-1) + e(k).$$

From an experiment, the following quantities were computed:

$$\sum_{i=1}^{N} y^2(i-1) = 20, \qquad \sum_{i=1}^{N} u^2(i-1) = 40,$$

$$\sum_{i=1}^{N} y(i)y(i-1) = 2, \qquad \sum_{i=1}^{N} y(i-1)u(i-1) = 10,$$

$$\sum_{i=1}^{N} y(i)u(i-1) = 24.$$

(a) Write the model as a linear regression in the unknown parameters a and b. (1 point)

(b) Write the normal equations using the supplied sums. (2 points)

(c) Compute the least-squares estimates $\hat{a}$ and $\hat{b}$. (2 points)

Problem 2 – Persistent excitation and information matrix (5 points)

Consider the regression model

$$y(k) = \theta_1 u(k) + \theta_2 u(k-1) + e(k).$$

For an experiment, the regressors are

$$\boldsymbol{\varphi}(1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\varphi}(2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{\varphi}(3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

(a) Compute the information matrix

$$M = \sum_{k=1}^{3} \boldsymbol{\varphi}(k)\boldsymbol{\varphi}^T(k).$$

(2 points)

(b) Determine whether the two parameters are uniquely identifiable. (1 point)

(c) Explain what would happen if all three regressors were instead

$$\boldsymbol{\varphi}(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

(2 points)

Problem 3 – Least-squares bias with colored noise (5 points)

Consider the model

$$y(k) = a\,y(k-1) + u(k-1) + v(k),$$

where

$$v(k) = e(k) + c\,e(k-1),$$

and $e(k)$ is zero-mean white noise with unit variance.

(a) For $c \neq 0$, determine whether $v(k)$ is white noise. Justify your answer by calculating its lag-one autocovariance. (2 points)

(b) Explain why the least-squares estimate of a can be biased when the regressor contains $y(k-1)$. (2 points)

(c) What happens to this bias if $c = 0$? (1 point)

Problem 4 – Maximum likelihood and sensor fusion (5 points)

Two independent sensors measure the same unknown constant parameter θ:

$$y_1 = \theta + e_1, \qquad y_2 = \theta + e_2,$$

where

$$e_1, e_2 \sim \mathcal{N}(0, 1)$$

are independent.

Suppose the measurements are

$$y_1 = 4, \qquad y_2 = 6.$$

(a) Write the likelihood function $p(y_1, y_2 \mid \theta)$, up to a multiplicative constant independent of θ. (1 point)

(b) Determine the maximum-likelihood estimate $\hat{\theta}_{\mathrm{ML}}$. (2 points)

(c) Calculate the variance of the estimation error of the ML estimator and compare it with using only sensor 1. (2 points)

Solutions

Problem 1 – Least-squares identification of an ARX model

(a) Regression form

Starting from

$$y(k) = -a\,y(k-1) + b\,u(k-1) + e(k),$$

define

$$\boldsymbol{\theta} = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad \boldsymbol{\varphi}(k) = \begin{bmatrix} -y(k-1) \\ u(k-1) \end{bmatrix}.$$

Then

$$\boxed{y(k) = \boldsymbol{\varphi}^T(k)\boldsymbol{\theta} + e(k).}$$

(b) Normal equations

The normal equations are

$$\left(\sum_k \boldsymbol{\varphi}(k)\boldsymbol{\varphi}^T(k)\right)\hat{\boldsymbol{\theta}} = \sum_k \boldsymbol{\varphi}(k)y(k).$$

Using the supplied sums,

$$\sum_k \boldsymbol{\varphi}(k)\boldsymbol{\varphi}^T(k) = \begin{bmatrix} 20 & -10 \\ -10 & 40 \end{bmatrix},$$

because

$$\sum_k [-y(k-1)]u(k-1) = -10.$$

Also,

$$\sum_k \boldsymbol{\varphi}(k)y(k) = \begin{bmatrix} -\sum_k y(k)y(k-1) \\ \sum_k u(k-1)y(k) \end{bmatrix} = \begin{bmatrix} -2 \\ 24 \end{bmatrix}.$$

Therefore,

$$\boxed{\begin{bmatrix} 20 & -10 \\ -10 & 40 \end{bmatrix}\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} -2 \\ 24 \end{bmatrix}.}$$

(c) Parameter estimates

The determinant is

$$20(40) - (-10)^2 = 700.$$

Hence,

$$\left(\Phi^T\Phi\right)^{-1} = \frac{1}{700}\begin{bmatrix} 40 & 10 \\ 10 & 20 \end{bmatrix}.$$

Thus,

$$\begin{aligned} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} &= \frac{1}{700}\begin{bmatrix} 40 & 10 \\ 10 & 20 \end{bmatrix}\begin{bmatrix} -2 \\ 24 \end{bmatrix} \\ &= \frac{1}{700}\begin{bmatrix} -80 + 240 \\ -20 + 480 \end{bmatrix} \\ &= \begin{bmatrix} 160/700 \\ 460/700 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\boxed{\hat{a} = \frac{8}{35} \approx 0.229, \qquad \hat{b} = \frac{23}{35} \approx 0.657.}$$

Problem 2 – Persistent excitation and information matrix

(a) Information matrix

The first contribution is

$$\boldsymbol{\varphi}(1)\boldsymbol{\varphi}^T(1) = \begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}.$$

The second is

$$\boldsymbol{\varphi}(2)\boldsymbol{\varphi}^T(2) = \begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}.$$

The third is

$$\boldsymbol{\varphi}(3)\boldsymbol{\varphi}^T(3) = \begin{bmatrix}0\\1\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix} = \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix}.$$

Therefore,

$$\begin{aligned} M &= \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} + \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix} \\ &= \boxed{\begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}}. \end{aligned}$$

(b) Identifiability

The determinant is

$$\det(M) = 2(2) - 1(1) = 3 > 0.$$

Thus M is nonsingular and has full rank:

$$\boxed{\text{rank}(M) = 2.}$$

Therefore, the two parameters are uniquely identifiable from these regressors.

(c) Constant regressor

If

$$\boldsymbol{\varphi}(k) = \begin{bmatrix}1\\1\end{bmatrix}$$

for all three samples, then

$$M = 3\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}3 & 3\\3 & 3\end{bmatrix}.$$

Its determinant is

$$\det(M) = 9 - 9 = 0.$$

Therefore,

$$\boxed{\text{rank}(M) = 1 < 2.}$$

The parameters cannot be uniquely identified. Only the combination $\theta_1 + \theta_2$ is observable through the regression.

Problem 3 – Least-squares bias with colored noise

(a) Whiteness of $v(k)$

We have

$$v(k) = e(k) + ce(k-1),$$

and

$$v(k-1) = e(k-1) + ce(k-2).$$

Since $e(k)$ is white noise,

$$\begin{aligned} R_v(1) &= E[v(k)v(k-1)] \\ &= E[(e(k) + ce(k-1))(e(k-1) + ce(k-2))]. \end{aligned}$$

All cross-time terms are zero except

$$cE[e^2(k-1)] = c,$$

because $e(k)$ has unit variance. Hence,

$$\boxed{R_v(1) = c.}$$

For $c \neq 0$,

$$R_v(1) \neq 0,$$

so $v(k)$ is not white noise.

(b) Source of least-squares bias

The regressor for estimating a is

$$\varphi(k) = y(k-1).$$

But the previous output contains previous noise:

$$y(k-1) = ay(k-2) + u(k-2) + v(k-1).$$

Thus, because

$$v(k) = e(k) + ce(k-1),$$

the disturbance $v(k)$ contains $e(k-1)$, which also contributes to $y(k-1)$ through $v(k-1)$.

Consequently,

$$E[y(k-1)v(k)]$$

need not be zero. The regressor and disturbance are therefore correlated, violating the usual condition required for unbiased/consistent least-squares estimation.

Hence,

$$\boxed{\text{the LS estimate of } a \text{ can be biased.}}$$

(c) Case $c = 0$

If $c = 0$,

$$v(k) = e(k),$$

so the disturbance is white. In the standard setting, the current disturbance is then uncorrelated with the past regressor $y(k-1)$.

Therefore, the particular colored-noise source of bias disappears:

$$\boxed{c = 0 \;\Rightarrow\; \text{no bias from this noise-correlation mechanism.}}$$

Problem 4 – Maximum likelihood and sensor fusion

(a) Likelihood

Because the two measurement errors are independent Gaussian variables,

$$p(y_1, y_2 \mid \theta) = p(y_1 \mid \theta)p(y_2 \mid \theta).$$

Ignoring constants independent of θ,

$$\boxed{p(y_1, y_2 \mid \theta) \propto \exp\left[-\frac{1}{2}\left((y_1-\theta)^2 + (y_2-\theta)^2\right)\right].}$$

(b) Maximum-likelihood estimate

Maximizing the likelihood is equivalent to minimizing

$$(y_1-\theta)^2 + (y_2-\theta)^2.$$

Differentiate:

$$-2(y_1 - \theta) - 2(y_2 - \theta) = 0.$$

Therefore,

$$2\theta = y_1 + y_2,$$

so

$$\hat{\theta}_{\mathrm{ML}} = \frac{y_1 + y_2}{2}.$$

With $y_1 = 4$ and $y_2 = 6$,

$$\boxed{\hat{\theta}_{\mathrm{ML}} = 5.}$$

(c) Estimation variance

The estimator is

$$\hat{\theta}_{\mathrm{ML}} = \frac{(\theta + e_1) + (\theta + e_2)}{2} = \theta + \frac{e_1 + e_2}{2}.$$

Therefore,

$$\begin{aligned} \mathrm{Var}(\hat{\theta}_{\mathrm{ML}} - \theta) &= \frac{1}{4}\,\mathrm{Var}(e_1 + e_2) \\ &= \frac{1}{4}(1 + 1) \\ &= \boxed{\frac{1}{2}}. \end{aligned}$$

If only sensor 1 is used,

$$\hat{\theta}_1 = y_1 = \theta + e_1,$$

so

$$\boxed{\mathrm{Var}(\hat{\theta}_1 - \theta) = 1.}$$

Thus, using both independent sensors reduces the estimation variance by a factor of two:

$$\boxed{\mathrm{Var}_{\text{two sensors}} = \frac{1}{2}\,\mathrm{Var}_{\text{one sensor}}\,.}$$

Mock MAP 2 – Module 3

Control of Cyber-Physical Systems **30 min**

Total: 20 points **4 problems, 5 points each**

Instructions: Give well-motivated answers. Show the main calculation steps. A calculator may be used.

Problem 1 – Least-Squares Identification [5 points]

Consider the static model

$$y(k) = \theta x(k) + e(k),$$

and the measurements

k	1	2	3	4
$x(k)$	1	2	4	5
$y(k)$	1.2	1.9	4.2	5.1

(a) Write the least-squares cost function and derive the normal equation for $\hat{\theta}$. [2 pts]

(b) Compute $\hat{\theta}$ numerically. [2 pts]

(c) Compute the sum of squared residuals

$$J(\hat{\theta}) = \sum_{k=1}^{4} \Big(y(k) - \hat{\theta} x(k) \Big)^2 .$$

[1 pt]

Problem 2 – Information Matrix and Identifiability [5 points]

A two-parameter model is described by

$$y(k) = \theta_1 \phi_1(k) + \theta_2 \phi_2(k) + e(k),$$

with regressors

$$\varphi(1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \varphi(2) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \varphi(3) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

(a) Construct the regression matrix Φ and calculate $\Phi^\top \Phi$. [2 pts]

(b) Determine whether the two parameters are uniquely identifiable from these measurements. Justify your answer using $\Phi^\top \Phi$. [2 pts]

(c) Explain what would happen if all measurements had instead been collected with

$$\varphi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \forall k.$$

[1 pt]

Problem 3 – Recursive Least Squares [5 points]

Consider a scalar parameter estimated using recursive least squares with forgetting factor $\lambda = 1$:

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k) \Big[y(k) - \varphi(k) \hat{\theta}(k-1) \Big],$$

where

$$K(k) = \frac{P(k-1)\varphi(k)}{1 + \varphi^2(k) P(k-1)},$$

and

$$P(k) = \big[1 - K(k)\varphi(k) \big] P(k-1).$$

At iteration k,

$$\hat{\theta}(k-1) = 0.8, \qquad P(k-1) = 4, \qquad \varphi(k) = 2, \qquad y(k) = 1.8.$$

(a) Calculate the prediction error

$$\varepsilon(k) = y(k) - \varphi(k) \hat{\theta}(k-1).$$

[1 pt]

(b) Calculate the gain $K(k)$. [1 pt]

(c) Calculate the updated estimate $\hat{\theta}(k)$. [2 pts]

(d) Calculate the updated covariance $P(k)$. [1 pt]

Problem 4 – Noise, Residuals and Model Validation [5 points]

Suppose a model has been identified from input-output data and produces the residual

$$r(k) = y(k) - \hat{y}(k).$$

A sequence of residuals is obtained:

$$\{r(k)\} = \{0.10,\ -0.08,\ 0.12,\ -0.11,\ 0.09,\ -0.10\}.$$

Consider the following statements:

(a) If the residuals have approximately zero mean and small variance, this alone proves that the identified model is correct. [1 pt]

(b) If residuals are strongly correlated with previous residuals, this is evidence that the model has not captured all the relevant dynamics. [1 pt]

(c) If the residuals are correlated with the input signal, this may indicate that relevant system dynamics have not been captured. [1 pt]

(d) Explain why residual analysis is useful as a model-validation tool even when the parameter estimates themselves appear reasonable. [2 pts]

For each statement in (a)–(c), answer **True** or **False** and briefly justify your answer.

Solutions

Solution 1

The model is

$$y(k) = \theta x(k) + e(k).$$

The least-squares cost is

$$J(\theta) = \sum_{k=1}^{4} \left[y(k) - \theta x(k)\right]^2 .$$

Differentiating with respect to θ,

$$\frac{dJ}{d\theta} = -2\sum_{k=1}^{4} x(k)\left[y(k) - \theta x(k)\right].$$

At the minimum,

$$\frac{dJ}{d\theta} = 0,$$

hence

$$\sum_{k=1}^{4} x(k)y(k) = \hat{\theta}\sum_{k=1}^{4} x^2(k).$$

Therefore,

$$\boxed{\hat{\theta} = \frac{\sum_{k=1}^{4} x(k)y(k)}{\sum_{k=1}^{4} x^2(k)}}.$$

The required sums are

$$\sum x(k)y(k) = 1(1.2) + 2(1.9) + 4(4.2) + 5(5.1) = 39.1,$$

and

$$\sum x^2(k) = 1^2 + 2^2 + 4^2 + 5^2 = 46.$$

Thus

$$\boxed{\hat{\theta} = \frac{39.1}{46} \approx 0.850}.$$

The residuals are

$$r(k) = y(k) - \hat{\theta}x(k).$$

Hence

$$\begin{aligned} r(1) &= 1.2 - 0.850(1) = 0.350, \\ r(2) &= 1.9 - 0.850(2) = 0.200, \\ r(3) &= 4.2 - 0.850(4) = 0.800, \\ r(4) &= 5.1 - 0.850(5) = 0.850. \end{aligned}$$

Therefore,

$$J(\hat{\theta}) = 0.350^2 + 0.200^2 + 0.800^2 + 0.850^2 \approx 1.665.$$

$$\boxed{J(\hat{\theta}) \approx 1.67}$$

Solution 2

The regression matrix is

$$\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore,

$$\Phi^\top \Phi = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}.$$

Thus,

$$\boxed{\Phi^\top \Phi = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}}.$$

Its determinant is

$$\det(\Phi^\top \Phi) = 3(5) - 3(3) = 6 \neq 0.$$

Therefore $\Phi^\top \Phi$ is nonsingular, so the least-squares estimate is unique.

$$\boxed{\text{The two parameters are uniquely identifiable.}}$$

If instead

$$\varphi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \forall k,$$

then all rows of Φ are identical:

$$\Phi = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ \vdots & \vdots \end{bmatrix}.$$

Consequently, the two columns of Φ are linearly dependent and

$$\operatorname{rank}(\Phi) = 1 < 2.$$

Hence

$$\Phi^\top \Phi$$

is singular and the two parameters cannot be uniquely separated.

$$\boxed{\text{The parameters are not individually identifiable.}}$$

Solution 3

The prediction error is

$$\varepsilon(k) = y(k) - \varphi(k)\hat{\theta}(k-1).$$

Substituting the values,

$$\varepsilon(k) = 1.8 - 2(0.8) = 0.2.$$

Therefore,

$$\boxed{\varepsilon(k) = 0.2}.$$

The gain is

$$K(k) = \frac{P(k-1)\varphi(k)}{1 + \varphi^2(k)P(k-1)}.$$

Thus,

$$K(k) = \frac{4(2)}{1 + 2^2(4)} = \frac{8}{17} \approx 0.4706.$$

Hence,

$$\boxed{K(k) \approx 0.4706}.$$

The updated estimate is

$$\hat{\theta}(k) = 0.8 + \frac{8}{17}(0.2).$$

Therefore,

$$\hat{\theta}(k) = 0.8 + \frac{1.6}{17} \approx 0.8941.$$

Thus,

$$\boxed{\hat{\theta}(k) \approx 0.894}.$$

Finally,

$$P(k) = [1 - K(k)\varphi(k)]P(k-1).$$

Therefore,

$$P(k) = \left[1 - \frac{8}{17}(2)\right] 4 = \left(1 - \frac{16}{17}\right) 4 = \frac{4}{17}.$$

Hence,

$$\boxed{P(k) \approx 0.2353}.$$

Solution 4

(a) False.

A small residual variance and approximately zero residual mean are useful properties, but they do not by themselves prove that the model is correct. A model may still fail to capture dynamics while producing residuals with reasonable mean and variance.

$$\boxed{\text{False}}$$

(b) True.

If residuals are correlated with their own past values, the residual is not behaving like an unpredictable disturbance. This suggests that information contained in the output dynamics remains unexplained by the model.

$$\boxed{\text{True}}$$

(c) True.

If the residual is correlated with the input, then part of the system's response to the input has not been adequately explained by the identified model.

$$\boxed{\text{True}}$$

(d)

Residual analysis checks whether the unexplained part of the measured output still contains systematic information. In particular, correlations with past residuals or with the input can reveal missing dynamics even when the estimated parameters appear numerically reasonable.

Therefore, residual analysis provides a complementary validation step: parameter estimation tells us which model parameters fit the data, while residual analysis helps determine whether the resulting model has actually captured the relevant system behavior.

A good model should leave residuals without systematic predictable structure.

Mock MAP 2 – Module 3

Control of Cyber-Physical Systems **30 min**

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Problem 1 – Least-Squares Identification [5 points]

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k	1	2	3	4
$x(k)$	1	2	4	5
$y(k)$	1.2	1.9	4.2	5.1

(a) Write the least-squares cost function and derive the normal equation for $\hat{\theta}$. [2 pts]

(b) Compute $\hat{\theta}$ numerically. [2 pts]

(c) Compute the sum of squared residuals

$$J(\hat{\theta}) = \sum_{k=1}^{4} \Big(y(k) - \hat{\theta} x(k)\Big)^2 .$$

[1 pt]

Problem 2 – Information Matrix and Identifiability [5 points]

A two-parameter model is described by

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(a) Construct the regression matrix Φ and calculate $\Phi^\top \Phi$. [2 pts]

(b) Determine whether the two parameters are uniquely identifiable from these measurements. Justify your answer using $\Phi^\top \Phi$. [2 pts]

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$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \varphi(k)\hat{\theta}(k-1)\right],$$

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$$K(k) = \frac{P(k-1)\varphi(k)}{1 + \varphi^2(k)P(k-1)},$$

and

$$P(k) = \left[1 - K(k)\varphi(k)\right] P(k-1).$$

At iteration k,

$$\hat{\theta}(k-1) = 0.8, \qquad P(k-1) = 4, \qquad \varphi(k) = 2, \qquad y(k) = 1.8.$$

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[1 pt]

(b) Calculate the gain $K(k)$. [1 pt]

(c) Calculate the updated estimate $\hat{\theta}(k)$. [2 pts]

(d) Calculate the updated covariance $P(k)$. [1 pt]

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A sequence of residuals is obtained:

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Differentiating with respect to θ,

$$\frac{dJ}{d\theta} = -2\sum_{k=1}^{4} x(k)\left[y(k) - \theta x(k)\right].$$

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$$\frac{dJ}{d\theta} = 0,$$

hence

$$\sum_{k=1}^{4} x(k)y(k) = \hat{\theta}\sum_{k=1}^{4} x^2(k).$$

Therefore,

$$\boxed{\hat{\theta} = \frac{\sum_{k=1}^{4} x(k)y(k)}{\sum_{k=1}^{4} x^2(k)}}.$$

The required sums are

$$\sum x(k)y(k) = 1(1.2) + 2(1.9) + 4(4.2) + 5(5.1) = 39.1,$$

and

$$\sum x^2(k) = 1^2 + 2^2 + 4^2 + 5^2 = 46.$$

Thus

$$\boxed{\hat{\theta} = \frac{39.1}{46} \approx 0.850}.$$

The residuals are

$$r(k) = y(k) - \hat{\theta}x(k).$$

Hence

$$\begin{aligned} r(1) &= 1.2 - 0.850(1) = 0.350, \\ r(2) &= 1.9 - 0.850(2) = 0.200, \\ r(3) &= 4.2 - 0.850(4) = 0.800, \\ r(4) &= 5.1 - 0.850(5) = 0.850. \end{aligned}$$

Therefore,

$$J(\hat{\theta}) = 0.350^2 + 0.200^2 + 0.800^2 + 0.850^2 \approx 1.665.$$

$$\boxed{J(\hat{\theta}) \approx 1.67}$$

Solution 2

The regression matrix is

$$\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore,

$$\Phi^\top \Phi = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}.$$

Thus,

$$\boxed{\Phi^\top \Phi = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}}.$$

Its determinant is

$$\det(\Phi^\top \Phi) = 3(5) - 3(3) = 6 \neq 0.$$

Therefore $\Phi^\top \Phi$ is nonsingular, so the least-squares estimate is unique.

$$\boxed{\text{The two parameters are uniquely identifiable.}}$$

If instead

$$\varphi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \forall k,$$

then all rows of Φ are identical:

$$\Phi = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ \vdots & \vdots \end{bmatrix}.$$

Consequently, the two columns of Φ are linearly dependent and

$$\operatorname{rank}(\Phi) = 1 < 2.$$

Hence

$$\Phi^\top \Phi$$

is singular and the two parameters cannot be uniquely separated.

$$\boxed{\text{The parameters are not individually identifiable.}}$$

Solution 3

The prediction error is

$$\varepsilon(k) = y(k) - \varphi(k)\hat{\theta}(k-1).$$

Substituting the values,

$$\varepsilon(k) = 1.8 - 2(0.8) = 0.2.$$

Therefore,

$$\boxed{\varepsilon(k) = 0.2}.$$

The gain is

$$K(k) = \frac{P(k-1)\varphi(k)}{1 + \varphi^2(k)P(k-1)}.$$

Thus,

$$K(k) = \frac{4(2)}{1 + 2^2(4)} = \frac{8}{17} \approx 0.4706.$$

Hence,

$$\boxed{K(k) \approx 0.4706}.$$

The updated estimate is

$$\hat{\theta}(k) = 0.8 + \frac{8}{17}(0.2).$$

Therefore,

$$\hat{\theta}(k) = 0.8 + \frac{1.6}{17} \approx 0.8941.$$

Thus,

$$\boxed{\hat{\theta}(k) \approx 0.894}.$$

Finally,

$$P(k) = [1 - K(k)\varphi(k)]P(k-1).$$

Therefore,

$$P(k) = \left[1 - \frac{8}{17}(2)\right] 4 = \left(1 - \frac{16}{17}\right) 4 = \frac{4}{17}.$$

Hence,

$$\boxed{P(k) \approx 0.2353}.$$

Solution 4

(a) False.

A small residual variance and approximately zero residual mean are useful properties, but they do not by themselves prove that the model is correct. A model may still fail to capture dynamics while producing residuals with reasonable mean and variance.

$$\boxed{\text{False}}$$

(b) True.

If residuals are correlated with their own past values, the residual is not behaving like an unpredictable disturbance. This suggests that information contained in the output dynamics remains unexplained by the model.

$$\boxed{\text{True}}$$

(c) True.

If the residual is correlated with the input, then part of the system's response to the input has not been adequately explained by the identified model.

$$\boxed{\text{True}}$$

(d)

Residual analysis checks whether the unexplained part of the measured output still contains systematic information. In particular, correlations with past residuals or with the input can reveal missing dynamics even when the estimated parameters appear numerically reasonable.

Therefore, residual analysis provides a complementary validation step: parameter estimation tells us which model parameters fit the data, while residual analysis helps determine whether the resulting model has actually captured the relevant system behavior.

A good model should leave residuals without systematic predictable structure.

Mock MAP 2 – Module 3

Control of Cyber-Physical Systems **30 min**

Total: 20 points **4 problems, 5 points each**

Instructions: Give well-motivated answers and show the relevant calculation steps. A simple calculator may be used.

Problem 1 – Batch Least Squares for a Dynamic Model [5 points]

Consider the model

$$y(k) = a\,y(k-1) + u(k-1) + e(k).$$

The following measurements are available:

k	0	1	2	3	4
$y(k)$	1.0	1.8	2.1	2.9	3.2
$u(k)$	1.0	1.0	1.0	1.0	–

(a) Write the least-squares cost function for estimating a. [1 pt]

(b) Derive the equation satisfied by $\hat{a}$. [2 pts]

(c) Compute $\hat{a}$ numerically. [2 pts]

Problem 2 – Identifiability and Persistent Excitation [5 points]

Consider the two-parameter model

$$y(k) = \theta_1 u(k) + \theta_2 u(k-1) + e(k),$$

with the following input sequence:

$$u(0) = 0, \qquad u(1) = 1, \qquad u(2) = 0, \qquad u(3) = -1.$$

The available regression vectors are therefore

$$\varphi(1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \varphi(2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \varphi(3) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

(a) Calculate the information matrix

$$R = \sum_{k=1}^{3} \varphi(k)\varphi^{\top}(k).$$

[2 pts]

(b) Determine whether the parameters θ_1 and θ_2 are uniquely identifiable. Justify your answer. [1 pt]

(c) Explain what would happen to identifiability if the input were constant, $u(k) = 1$, for all k. [2 pts]

Problem 3 – RLS and Forgetting Factor [5 points]

A scalar parameter is estimated using recursive least squares with forgetting factor $\lambda = 0.9$:

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^2(k)P(k-1)},$$

$$\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\left[y(k) - \varphi(k)\hat{\theta}(k-1)\right].$$

At a given iteration,

$$\hat{\theta}(k-1) = 0.5, \qquad P(k-1) = 2, \qquad \varphi(k) = 1, \qquad y(k) = 1.5.$$

(a) Calculate the prediction error. [1 pt]

(b) Calculate the RLS gain $K(k)$. [1.5 pts]

(c) Calculate the updated estimate $\hat{\theta}(k)$. [1.5 pts]

(d) Explain qualitatively what decreasing λ does to the relative importance of recent versus old data. [1 pt]

Problem 4 – Maximum Likelihood Sensor Fusion [5 points]

An unknown constant parameter θ is measured by two independent Gaussian sensors:

$$y_1 = \theta + e_1, \qquad y_2 = \theta + e_2,$$

where

$$e_1 \sim \mathcal{N}(0, \sigma_1^2), \qquad e_2 \sim \mathcal{N}(0, \sigma_2^2),$$

with

$$\sigma_1^2 = 1, \qquad \sigma_2^2 = 4.$$

The measured values are

$$y_1 = 10, \qquad y_2 = 8.$$

(a) Write the likelihood function, up to a multiplicative constant independent of θ. [1 pt]

(b) Derive the maximum-likelihood estimate $\hat{\theta}_{\mathrm{ML}}$. [2 pts]

(c) Calculate $\hat{\theta}_{\mathrm{ML}}$ numerically. [1 pt]

(d) Explain why the more accurate sensor receives a larger weight. [1 pt]

Solutions

Solution 1

The model is

$$y(k) = a\,y(k-1) + u(k-1) + e(k).$$

Therefore,

$$e(k) = y(k) - a\,y(k-1) - u(k-1).$$

The least-squares criterion is

$$J(a) = \sum_{k=1}^{4} \left[y(k) - a\,y(k-1) - u(k-1)\right]^2.$$

Hence,

$$\boxed{J(a) = \sum_{k=1}^{4} \left[y(k) - a\,y(k-1) - u(k-1)\right]^2}.$$

Differentiating,

$$\frac{dJ}{da} = -2\sum_{k=1}^{4} y(k-1)\left[y(k) - a\,y(k-1) - u(k-1)\right].$$

At the minimum,

$$\sum_{k=1}^{4} y(k-1)\left[y(k) - \hat{a}\,y(k-1) - u(k-1)\right] = 0.$$

Thus,

$$\hat{a}\sum_{k=1}^{4} y^2(k-1) = \sum_{k=1}^{4} y(k)y(k-1) - \sum_{k=1}^{4} u(k-1)y(k-1),$$

and therefore

$$\boxed{\hat{a} = \frac{\sum y(k)y(k-1) - \sum u(k-1)y(k-1)}{\sum y^2(k-1)}}.$$

Using the data,

$$\sum_{k=1}^{4} y(k)y(k-1) = 1.8(1.0) + 2.1(1.8) + 2.9(2.1) + 3.2(2.9),$$

so

$$\sum y(k)y(k-1) = 1.8 + 3.78 + 6.09 + 9.28 = 20.95.$$

Also,

$$\sum_{k=1}^{4} u(k-1)y(k-1) = 1(1.0) + 1(1.8) + 1(2.1) + 1(2.9) = 7.8.$$

Finally,

$$\begin{aligned}\sum_{k=1}^{4} y^2(k-1) &= 1^2 + 1.8^2 + 2.1^2 + 2.9^2 \\ &= 1 + 3.24 + 4.41 + 8.41 = 17.06.\end{aligned}$$

Therefore,

$$\hat{a} = \frac{20.95 - 7.8}{17.06} = \frac{13.15}{17.06} \approx 0.771.$$

Hence,

$$\boxed{\hat{a} \approx 0.771}.$$

This follows the batch least-squares formulation for dynamic models in Module 3.1; the problem structure is closely aligned with S3P3. (Module 3.1, slides 4–22; Problem Series S3P3.)

Solution 2

The information matrix is

$$R = \sum_{k=1}^{3} \varphi(k)\varphi^{\top}(k).$$

The individual contributions are

$$\varphi(1)\varphi^{\top}(1) = \begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix},$$

$$\varphi(2)\varphi^{\top}(2) = \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix},$$

and

$$\varphi(3)\varphi^{\top}(3) = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}.$$

Therefore,

$$R = \begin{bmatrix}2 & 0\\0 & 1\end{bmatrix}.$$

Thus,

$$\boxed{R = \begin{bmatrix}2 & 0\\0 & 1\end{bmatrix}}.$$

Its determinant is

$$\det(R) = 2 \neq 0.$$

Hence R is nonsingular and the two parameters are uniquely identifiable:

$$\boxed{\text{The parameters are identifiable.}}$$

If instead

$$u(k) = 1, \qquad \forall k,$$

then

$$\varphi(k) = \begin{bmatrix}1\\1\end{bmatrix}$$

for every k. Consequently,

$$R = N\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix},$$

which has rank one.

Therefore,

$$\boxed{\det(R) = 0}$$

and the two parameters cannot be uniquely identified.

The information matrix, excitation, and identifiability are treated in Module 3.2. (Module 3.2, slides 3–20; Problem Series S3P10.)

Solution 3

The prediction error is

$$\varepsilon(k) = y(k) - \varphi(k)\hat{\theta}(k-1).$$

Therefore,

$$\varepsilon(k) = 1.5 - (1)(0.5) = 1.$$

Hence,

$$\boxed{\varepsilon(k) = 1}.$$

The gain is

$$K(k) = \frac{P(k-1)\varphi(k)}{\lambda + \varphi^2(k)P(k-1)}.$$

Substituting,

$$K(k) = \frac{2(1)}{0.9 + 1^2(2)} = \frac{2}{2.9} \approx 0.6897.$$

Thus,

$$\boxed{K(k) \approx 0.6897}.$$

The parameter update is

$$\hat{\theta}(k) = 0.5 + (0.6897)(1).$$

Therefore,

$$\boxed{\hat{\theta}(k) \approx 1.1897}.$$

A smaller forgetting factor gives more relative weight to recent data and makes the estimator more responsive to changes in the parameters. The trade-off is that stronger forgetting generally increases estimation variability.

Thus,

$$\boxed{\lambda \downarrow \implies \text{more emphasis on recent data and faster tracking.}}$$

The RLS update and forgetting-factor trade-off are treated in Modules 3.3–3.4. (Module 3.3, slides 3–15; Module 3.4, slides 3–18; Problem Series S3P3.)

Solution 4

Because the sensor errors are independent Gaussian variables,

$$p(y_1, y_2 \mid \theta) = p(y_1 \mid \theta)p(y_2 \mid \theta).$$

Ignoring multiplicative constants that do not depend on θ,

$$L(\theta) \propto \exp\left[-\frac{(y_1-\theta)^2}{2\sigma_1^2} - \frac{(y_2-\theta)^2}{2\sigma_2^2}\right].$$

Hence,

$$\boxed{L(\theta) \propto \exp\left[-\frac{(y_1-\theta)^2}{2} - \frac{(y_2-\theta)^2}{8}\right]}.$$

It is easier to maximize the log-likelihood:

$$\ell(\theta) = -\frac{(y_1-\theta)^2}{2\sigma_1^2} - \frac{(y_2-\theta)^2}{2\sigma_2^2} + \text{constant}.$$

Differentiating,

$$\frac{d\ell}{d\theta} = \frac{y_1 - \theta}{\sigma_1^2} + \frac{y_2 - \theta}{\sigma_2^2}.$$

Setting the derivative to zero,

$$\frac{y_1 - \hat{\theta}}{\sigma_1^2} + \frac{y_2 - \hat{\theta}}{\sigma_2^2} = 0.$$

Therefore,

$$\hat{\theta}\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right) = \frac{y_1}{\sigma_1^2} + \frac{y_2}{\sigma_2^2},$$

so

$$\boxed{\hat{\theta}_{\mathrm{ML}} = \frac{y_1/\sigma_1^2 + y_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2}}.$$

Substituting the values,

$$\hat{\theta}_{\mathrm{ML}} = \frac{10/1 + 8/4}{1/1 + 1/4} = \frac{10 + 2}{1.25} = 9.6.$$

Hence,

$$\boxed{\hat{\theta}_{\mathrm{ML}} = 9.6}.$$

The weights are inversely proportional to the sensor variances:

$$w_1 = \frac{1}{\sigma_1^2} = 1, \qquad w_2 = \frac{1}{\sigma_2^2} = 0.25.$$

Thus, the sensor with smaller variance contributes more strongly because it provides more precise information about θ.

This is the sensor-fusion maximum-likelihood structure treated in Module 3.6 and directly reflected in Problem Series S3P11. (Module 3.6, slides 3–22; Problem Series S3P11.)

MOCK MAP30 #2 — Module 3 — Version A

Time: 30 minutes **Total: 20 points**

Instructions. Show the main steps of your reasoning. A correct numerical answer without the relevant setup may receive only partial credit. You may use a simple calculator and the formula sheet.

Problem 1 — Least squares with intercept [4 points]

Consider

$$y(k) = a\,x(k) + b + e(k)$$

with measurements

$$(x, y) = \{(1, 1.2), (2, 2.1), (3, 2.8), (4, 4.1)\}.$$

(a) Write the regressor $\phi(k)$, parameter vector θ, and regression matrix Φ. [1 pt]

(b) Compute $\Phi^\top\Phi$ and $\Phi^\top\bar{y}$, and solve the normal equations

$$\Phi^\top\Phi\hat{\theta} = \Phi^\top\bar{y}.$$

[2 pts]

(c) Compute the residuals and SSE. [1 pt]

Problem 2 — Information, covariance and excitation [4 points]

A two-parameter model has regressor

$$\phi(k) = \begin{bmatrix} u(k-1) \\ u(k-2) \end{bmatrix}.$$

(a) For the input

$$u(k) = 1,$$

determine whether the two parameters are identifiable. Justify using $\Phi^\top \Phi$. [1 pt]

(b) For the input sequence

$$u(k) = 1, 1, -1, -1, 1, 1, -1, -1, \ldots$$

determine whether the two parameters are identifiable. [1 pt]

(c) Suppose that for an identified experiment

$$\hat{\sigma}^2 = 0.01, \qquad (\Phi^\top \Phi)^{-1} = \begin{bmatrix} 0.20 & -0.05 \\ -0.05 & 0.10 \end{bmatrix}.$$

Compute the covariance matrix of $\hat{\theta}$ and the standard errors of both parameters. [1 pt]

(d) Interpret the negative off-diagonal covariance. [1 pt]

Problem 3 — RLS with forgetting [4 points]

At time $t-1$,

$$\hat{\theta}(t-1) = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix}, \qquad P(t-1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

and

$$\phi(t-1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y(t) = 0.9.$$

Use forgetting factor

$$\lambda = 0.9.$$

Compute:

(a) The prediction error

$$\varepsilon(t) = y(t) - \phi^\top(t-1)\hat{\theta}(t-1).$$

[1 pt]

(b) The RLS gain

$$K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)}.$$

[1 pt]

(c) The updated estimate

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\varepsilon(t).$$

[1 pt]

(d) Explain qualitatively why forgetting can cause $P(t)$ to increase when the regressor does not excite a parameter direction. [1 pt]

Problem 4 — Noise, stationarity and ML [4 points]

(a) Let

$$v(k) = e(k) + 0.5e(k-1),$$

where $e(k)$ is white noise with variance σ^2. Compute

$$R_v(0), \qquad R_v(1), \qquad R_v(\ell), \quad |\ell| > 1.$$

[1.5 pts]

(b) Consider

$$y(k) = a\, y(k-1) + e(k).$$

State the condition on a for a stationary AR(1) process. [0.5 pt]

(c) Three measurements of an unknown constant are

$$y = \{1, 2, 10\}.$$

Compute the LS estimate. [0.5 pt]

(d) If instead the measurement noise is Laplacian,

$$p(e) \propto e^{-|e|/b},$$

state the ML criterion and compute the ML estimate. [1.5 pts]

Problem 5 — Synthesis [4 points]

A student estimates

$$y(k) = a\, u(k-1) + v(k)$$

using LS and obtains residuals that are strongly correlated in time.
Answer briefly:

(a) Is LS necessarily unbiased? Explain the relevant condition. [1 pt]

(b) Why can the usual covariance formula become over-optimistic? [1 pt]

(c) Give one suitable method to address the problem. [1 pt]

(d) State the condition required for a unique LS estimate. [1 pt]

Solutions — MOCK MAP30 #2 — Version A

Solution 1

$$\phi(k) = \begin{bmatrix} x(k) \\ 1 \end{bmatrix}, \qquad \theta = \begin{bmatrix} a \\ b \end{bmatrix},$$

and

$$\Phi = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}.$$

Therefore

$$\Phi^\top \Phi = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}.$$

Also,

$$\Phi^\top \bar{y} = \begin{bmatrix} 1(1.2) + 2(2.1) + 3(2.8) + 4(4.1) \\ 1.2 + 2.1 + 2.8 + 4.1 \end{bmatrix} = \begin{bmatrix} 29.8 \\ 10.2 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 29.8 \\ 10.2 \end{bmatrix}.$$

The determinant is

$$30(4) - 10^2 = 20.$$

Hence

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 29.8 \\ 10.2 \end{bmatrix},$$

giving

$$\boxed{\hat{a} = 0.94, \qquad \hat{b} = 0.30}.$$

Predictions:

$$\hat{y} = \{1.24, 2.18, 3.12, 4.06\}.$$

Residuals:

$$\hat{e} = \{-0.04, -0.08, -0.32, 0.04\}.$$

Therefore

$$\text{SSE} = (-0.04)^2 + (-0.08)^2 + (-0.32)^2 + (0.04)^2 = 0.112.$$

$$\boxed{\text{SSE} = 0.112}.$$

Course reference: Module 3.1–3.2, Appendix Exercise A.1.

Solution 2

(a) For $u(k) = 1$, every regressor is

$$\phi(k) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus

$$\Phi^\top \Phi = \begin{bmatrix} N & N \\ N & N \end{bmatrix},$$

whose determinant is zero.

Therefore

$$\boxed{\text{the parameters are not identifiable.}}$$

(b) For

$$u(k) = 1, 1, -1, -1, \ldots$$

the shifted pairs cycle through distinct directions

$$[1, 1], \quad [1, -1], \quad [-1, -1], \quad [-1, 1].$$

Over complete periods,

$$\Phi^\top \Phi \propto I,$$

so it is nonsingular.

Therefore

$$\boxed{\text{the two parameters are identifiable.}}$$

(c) The covariance matrix is

$$P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}.$$

Hence

$$P = 0.01 \begin{bmatrix} 0.20 & -0.05 \\ -0.05 & 0.10 \end{bmatrix} = \begin{bmatrix} 0.002 & -0.0005 \\ -0.0005 & 0.001 \end{bmatrix}.$$

Therefore

$$\mathrm{se}(\hat{\theta}_1) = \sqrt{0.002} \approx 0.0447,$$

and

$$\mathrm{se}(\hat{\theta}_2) = \sqrt{0.001} \approx 0.0316.$$

(d) The negative covariance means that the two parameter estimates are negatively correlated: an increase in one estimate tends to be accompanied by a decrease in the other.

Course reference: Module 3.2, Appendix Exercises A.2–A.3.

Solution 3

First,

$$\phi^\top \hat{\theta} = [1 \quad 2] \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} = 0.5 - 0.4 = 0.1.$$

Thus

$$\boxed{\varepsilon(t) = 0.9 - 0.1 = 0.8}.$$

Next,

$$P\phi = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Also,

$$\phi^\top P \phi = [1 \quad 2] \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 6.$$

Therefore

$$K = \frac{1}{0.9 + 6} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.2899 \\ 0.2899 \end{bmatrix}.$$

Thus

$$\hat{\theta}(t) = \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} + 0.8 \begin{bmatrix} 0.2899 \\ 0.2899 \end{bmatrix},$$

so

$$\boxed{\hat{\theta}(t) \approx \begin{bmatrix} 0.7319 \\ 0.0319 \end{bmatrix}}.$$

The covariance update is

$$P(t) = \frac{1}{\lambda}\left(I - K\phi^\top\right) P(t-1).$$

Because $\lambda < 1$, the old covariance is inflated by $1/\lambda$. If a direction is not excited by ϕ, there is little or no measurement-information reduction in that direction, while the division by λ still increases uncertainty.

This is the **estimator windup** phenomenon.

Course reference: Module 3.3–3.4, Appendix Exercise A.5.

Solution 4

(a) We have

$$v(k) = e(k) + 0.5e(k-1).$$

Since $e(k)$ is white,

$$R_v(0) = E[v(k)^2] = \sigma^2 + 0.25\sigma^2 = 1.25\sigma^2.$$

For lag 1,

$$R_v(1) = E[v(k)v(k-1)] = 0.5\sigma^2.$$

For $|\ell| > 1$, there are no common white-noise terms, hence

$$\boxed{R_v(\ell) = 0, \qquad |\ell| > 1.}$$

Thus this is an MA(1)-type process.

(b) For

$$y(k) = a\, y(k-1) + e(k),$$

stationarity requires

$$\boxed{|a| < 1}.$$

(c) Under Gaussian noise, LS is the sample mean:

$$\hat{\theta}_{\text{LS}} = \frac{1+2+10}{3} = \boxed{\frac{13}{3} \approx 4.33}.$$

(d) For Laplacian noise,

$$p(e) \propto e^{-|e|/b}.$$

Therefore

$$\log L(\theta) = \text{constant} - \frac{1}{b}\sum_t |y(t) - \theta|.$$

Maximizing likelihood is therefore equivalent to minimizing

$$\sum_t |y(t) - \theta|.$$

The minimizer is the median:

$$\boxed{\hat{\theta}_{\mathrm{ML}} = 2}.$$

Hence

$$\boxed{\hat{\theta}_{\mathrm{LS}} \approx 4.33, \qquad \hat{\theta}_{\mathrm{ML}} = 2.}$$

The difference illustrates that

$$\boxed{\text{ML} = \text{LS only under the Gaussian-noise assumption.}}$$

Course reference: Module 3.5–3.6, Appendix Exercises A.6 and A.8.

Solution 5

(a) LS is unbiased when the consistency condition

$$\boxed{E[\phi(k)v(k)] = 0}$$

holds.

If the residual noise is correlated with the regressor, this condition fails and LS can be biased.

(b) The standard covariance formula assumes white residuals. If residuals are correlated, treating them as independent provides too much information and therefore can underestimate the actual uncertainty.

Hence the reported covariance can be

$$\boxed{\text{over-optimistic}}.$$

(c) A suitable solution is to explicitly model the noise, for example using

$$\boxed{\text{Extended Least Squares (ELS)}}$$

or an appropriate ARMAX/ML formulation.

(d) A unique LS estimate requires

$$\boxed{\Phi^{\top}\Phi \text{ invertible}}$$

or equivalently

$$\boxed{\mathrm{rank}(\Phi) = p}.$$

Course reference: Module 3.1–3.6, Appendix Exercises A.7–A.9.

MOCK MAP30 #2 — Module 3 — Version B

Time: 30 minutes **Total: 20 points**

Problem 1 — LS with intercept [4 points]

Fit

$$y(k) = ax(k) + b + e(k)$$

to

$$(x, y) = \{(1, 1.1), (2, 2.2), (3, 2.8), (4, 4.0)\}.$$

(a) Define $\phi(k)$ and θ. [1 pt]

(b) Compute $\Phi^\top \Phi$ and $\Phi^\top \bar{y}$. [1 pt]

(c) Solve for $\hat{a}, \hat{b}$. [1.5 pts]

(d) Compute the SSE. [0.5 pt]

Problem 2 — Identifiability and covariance [4 points]

Consider

$$\phi(k) = \begin{bmatrix} 1 \\ u(k) \end{bmatrix}.$$

(a) Explain why an input with $u(k) = 3$ for all k cannot separately identify the two parameters. [1 pt]

(b) Explain why varying $u(k)$ makes $\Phi^\top \Phi$ potentially invertible. [1 pt]

(c) Suppose

$$\hat{\sigma}^2 = 0.04, \qquad (\Phi^\top \Phi)^{-1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.25 \end{bmatrix}.$$

Compute P and both standard errors. [1 pt]

(d) What experiment would you design to reduce correlation between slope and intercept? [1 pt]

Problem 3 — RLS from scratch [4 points]

For

$$y(k) = ax(k) + e(k)$$

use the data

$$(1, 1.0), \quad (2, 2.1), \quad (3, 2.8).$$

Initialize

$$\hat{a}(0) = 0, \qquad P(0) = 10^6.$$

Run RLS without forgetting.

(a) Write the three scalar RLS equations. [1 pt]

(b) Compute the update for $k = 1$. [1 pt]

(c) Compute the final estimate after all three samples. [1 pt]

(d) Verify that it agrees with batch LS. [1 pt]

Problem 4 — Forgetting and windup [4 points]

(a) Estimate the effective memory

$$N_{\text{eff}} \approx \frac{1}{1-\lambda}$$

for

$$\lambda = 0.95, \quad 0.98, \quad 0.99.$$

[1.5 pts]

(b) If $h = 0.1$ s, convert these to approximate seconds of memory. [1 pt]

(c) Explain what happens to $P(t)$ under forgetting if $\phi(t) = 0$ for many consecutive samples. [1 pt]

(d) Give one way of preventing this windup. [0.5 pt]

Problem 5 — Coloured noise and ML [4 points]

(a) Explain why LS can become biased when the noise is coloured. [1 pt]

(b) Consider

$$v(k) = e(k) + ce(k-1).$$

For white $e(k)$, state the non-zero covariance lags of $v(k)$. [1 pt]

(c) For measurements $\{2, 2, 9\}$ of a constant, compute LS under Gaussian noise. [0.5 pt]

(d) If the noise is Laplacian, compute ML and explain why it differs from LS. [1.5 pts]

Solutions — MOCK MAP30 #2 — Version B

Solution 1

$$\phi(k) = \begin{bmatrix} x(k) \\ 1 \end{bmatrix}, \qquad \theta = \begin{bmatrix} a \\ b \end{bmatrix}.$$

Thus

$$\Phi = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}, \qquad \Phi^\top \Phi = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}.$$

Furthermore,

$$\Phi^\top \bar{y} = \begin{bmatrix} 1(1.1) + 2(2.2) + 3(2.8) + 4(4.0) \\ 1.1 + 2.2 + 2.8 + 4.0 \end{bmatrix} = \begin{bmatrix} 28.9 \\ 10.1 \end{bmatrix}.$$

Therefore

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 28.9 \\ 10.1 \end{bmatrix},$$

giving

$$\boxed{\hat{a} = 0.95, \qquad \hat{b} = 0.225}.$$

Predictions:

$$\hat{y} = \{1.175, 2.125, 3.075, 4.025\}.$$

Residuals:

$$\hat{e} = \{-0.075, 0.075, -0.275, -0.025\}.$$

Hence

$$\text{SSE} = 0.005625 + 0.005625 + 0.075625 + 0.000625 = \boxed{0.0875}.$$

Solution 2

(a) If $u(k) = 3$,

$$\phi(k) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

for every sample. Thus every row of Φ is identical up to repetition, so the two columns are linearly dependent.

Therefore

$$\boxed{\det(\Phi^\top \Phi) = 0}$$

and the parameters cannot be separately identified.

(b) If $u(k)$ varies, the second column is no longer a constant multiple of the first column. With sufficiently rich variation, the two columns become linearly independent, giving

$$\boxed{\Phi^\top \Phi \text{ invertible}}.$$

(c)

$$P = 0.04 \begin{bmatrix} 0.5 & 0 \\ 0 & 0.25 \end{bmatrix} = \begin{bmatrix} 0.02 & 0 \\ 0 & 0.01 \end{bmatrix}.$$

Thus

$$\boxed{\mathrm{se}(\hat{\theta}_1) = \sqrt{0.02} \approx 0.1414}$$

and

$$\boxed{\mathrm{se}(\hat{\theta}_2) = \sqrt{0.01} = 0.1}.$$

(d) Centre the regressor:

$$\tilde{u}(k) = u(k) - \bar{u}.$$

Then

$$\sum_k \tilde{u}(k) = 0,$$

which makes the cross term in $\Phi^\top \Phi$ vanish and therefore reduces the correlation between slope and intercept.

Solution 3

For the scalar model,

$$K(k) = \frac{P(k-1)x(k)}{1 + x^2(k)P(k-1)},$$
$$\hat{a}(k) = \hat{a}(k-1) + K(k)[y(k) - x(k)\hat{a}(k-1)],$$
$$P(k) = [1 - K(k)x(k)]P(k-1).$$

With $P(0) = 10^6$, the first update gives approximately

$$K(1) \approx 1, \qquad \hat{a}(1) \approx 1, \qquad P(1) \approx 1.$$

Continuing with $x = 2$ and $x = 3$ gives

$$\hat{a}(3) = \frac{1(1.0) + 2(2.1) + 3(2.8)}{1^2 + 2^2 + 3^2} = \frac{13.6}{14} \approx 0.9714.$$

Thus

$$\boxed{\hat{a}_{\mathrm{RLS}} \approx 0.9714}.$$

Batch LS gives exactly the same expression:

$$\boxed{\hat{a}_{\mathrm{RLS}} = \hat{a}_{\mathrm{LS}}}$$

up to the negligible effect of the finite initialization $P(0) = 10^6$.

Course reference: Module 3.3, Appendix Exercise A.4.

Solution 4

$$N_{\text{eff}} \approx \frac{1}{1-\lambda}.$$

For $\lambda = 0.95$:

$$N_{\text{eff}} = 20.$$

For $\lambda = 0.98$:

$$N_{\text{eff}} = 50.$$

For $\lambda = 0.99$:

$$N_{\text{eff}} = 100.$$

Hence

$$\boxed{20,\ 50,\ 100 \text{ samples}}.$$

With $h = 0.1$ s:

$$\boxed{2,\ 5,\ 10 \text{ s}}.$$

If

$$\phi(t) = 0,$$

the measurement contains no information about the parameter direction. The covariance update contains the division by λ:

$$P(t) = \frac{P(t-1)}{\lambda}.$$

Since $\lambda < 1$,

$$\frac{1}{\lambda} > 1,$$

so

$$\boxed{P(t) \text{ grows exponentially}}.$$

This is estimator windup.

A solution is to maintain persistent excitation, or use directional/ variable forgetting.

Course reference: Module 3.4, Appendix Exercise A.5.

Solution 5

(a) LS requires

$$E[\phi(k)v(k)] = 0.$$

With coloured noise, past noise terms can appear in the regressor. Then $\phi(k)$ and $v(k)$ can be correlated, so

$$\boxed{E[\phi(k)v(k)] \neq 0}$$

and LS becomes biased.

(b) For

$$v(k) = e(k) + ce(k-1),$$
$$R_v(0) = (1+c^2)\sigma^2,$$
$$R_v(1) = c\sigma^2,$$

and

$$\boxed{R_v(\ell) = 0, \quad |\ell| > 1}.$$

(c) Under Gaussian noise,

$$\hat{\theta}_{\text{LS}} = \frac{2+2+9}{3} = \boxed{\frac{13}{3} \approx 4.33}.$$

(d) For Laplacian noise,

$$\log L(\theta) = \text{constant} - \frac{1}{b}\sum_t |y(t) - \theta|.$$

Thus ML minimizes the ℓ_1 loss:

$$\min_\theta \sum_t |y(t) - \theta|.$$

The minimizer is the median:

$$\boxed{\hat{\theta}_{\text{ML}} = 2}.$$

Therefore

$$\boxed{\hat{\theta}_{\text{LS}} \approx 4.33 \neq \hat{\theta}_{\text{ML}} = 2.}$$

The reason is that LS/ML equivalence depends on Gaussian noise; Laplacian noise leads to an ℓ_1 rather than quadratic criterion.

Course reference: Module 3.5–3.6, Appendix Exercises A.6–A.8.

MOCK MAP30 #2 — Module 3 — Version C

Time: 30 minutes **Total: 20 points**

Problem 1 — Normal equation and uncertainty [4 points]

For

$$y(k) = a\,x(k) + b + e(k)$$

the following information is available:

$$N = 5, \qquad \sum x(k) = 5, \qquad \sum x^2(k) = 9,$$
$$\sum y(k) = 6, \qquad \sum x(k)y(k) = 8.$$

(a) Construct $\Phi^\top\Phi$ and $\Phi^\top\bar{y}$. [1 pt]

(b) Solve for $\hat{a}, \hat{b}$. [1.5 pts]

(c) Explain what would happen if $\det(\Phi^\top\Phi) = 0$. [0.5 pt]

(d) State the covariance formula using an estimate $\hat{\sigma}^2$ of the residual variance. [1 pt]

Problem 2 — Persistent excitation [4 points]

For

$$\phi^\top(k) = [u(k-1) \;\; u(k-2)],$$

consider:

Case A: $u(k) = 1$,
Case B: $u(k) = (-1)^k$,
Case C: $u(k) = 1, 1, -1, -1, 1, 1, -1, -1, \ldots$

For each case, state whether both parameters can be identified and give the key reason.

Problem 3 — RLS and batch LS [4 points]

A scalar RLS estimator has

$$\hat{a}(t-1) = 0.8, \qquad P(t-1) = 2,$$
$$x(t) = 2, \qquad y(t) = 1.9.$$

(a) Compute the prediction error. [1 pt]

(b) Compute $K(t)$ without forgetting. [1 pt]

(c) Compute $\hat{a}(t)$ and $P(t)$. [1 pt]

(d) Explain why $K(t)$ tends to decrease as information accumulates. [1 pt]

Problem 4 — Windup and coloured noise [4 points]

(a) Explain why $\lambda < 1$ is useful for tracking a changing plant. [1 pt]

(b) Explain why the same forgetting factor can be dangerous without excitation. [1 pt]

(c) State the consistency condition that is violated when the regressor is correlated with the noise. [1 pt]

(d) Name one identification method that can address coloured noise. [1 pt]

Problem 5 — ML versus LS [4 points]

An unknown constant θ is measured as

$$y = \{1, 3, 4, 20\}.$$

(a) Compute the Gaussian-noise LS/ML estimate. [1 pt]

(b) For Laplacian noise, state the ML criterion. [1 pt]

(c) Compute the Laplacian ML estimate. [1 pt]

(d) Explain which estimate is less affected by the outlier 20 and why. [1 pt]

Solutions — MOCK MAP30 #2 — Version C

Solution 1

For a model with intercept,

$$\Phi^\top \Phi = \begin{bmatrix} \sum x^2 & \sum x \\ \sum x & N \end{bmatrix}.$$

Therefore

$$\Phi^\top \Phi = \begin{bmatrix} 9 & 5 \\ 5 & 5 \end{bmatrix}, \qquad \Phi^\top \bar{y} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 9 & 5 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}.$$

The determinant is

$$9(5) - 25 = 20.$$

Hence

$$\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 5 & -5 \\ -5 & 9 \end{bmatrix} \begin{bmatrix} 8 \\ 6 \end{bmatrix}.$$

Thus

$$\boxed{\hat{a} = 0.5, \qquad \hat{b} = 0.7}.$$

If

$$\det(\Phi^\top \Phi) = 0,$$

then $\Phi^\top \Phi$ is singular, so the parameters are not uniquely identifiable from the experiment.
The covariance estimate is

$$\boxed{P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}.}$$

Solution 2

Case A For $u(k) = 1$,

$$\phi(k) = [1 \;\; 1]^\top$$

for every sample. There is only one excited direction.

$$\boxed{\text{Not identifiable.}}$$

Case B For $u(k) = (-1)^k$, the regressor alternates between

$$[1 \;\; -1]^\top$$

and

$$[-1 \;\; 1]^\top.$$

These are the same one-dimensional direction with opposite signs.
Therefore

$$\boxed{\text{Not identifiable.}}$$

The important lesson is that a fast input is not necessarily a rich input.

Case C The period-four sequence generates several independent regressor directions:

$$[1,1], \quad [-1,1], \quad [-1,-1], \quad [1,-1].$$

Therefore

$$\Phi^\top \Phi$$

is nonsingular over a suitable number of samples.

Hence

$$\boxed{\text{identifiable.}}$$

Course reference: Module 3.2, Appendix Exercise A.2.

Solution 3

The prediction error is

$$\varepsilon(t) = y(t) - x(t)\hat{a}(t-1) = 1.9 - 2(0.8) = \boxed{0.3}.$$

Without forgetting,

$$K(t) = \frac{P(t-1)x(t)}{1 + x^2(t)P(t-1)}.$$

Therefore

$$K(t) = \frac{2(2)}{1+4(2)} = \frac{4}{9} \approx 0.4444.$$

Hence

$$\hat{a}(t) = 0.8 + \frac{4}{9}(0.3) = 0.8 + 0.1333 = \boxed{0.9333}.$$

The covariance is

$$P(t) = [1 - K(t)x(t)]P(t-1),$$

so

$$P(t) = \left(1 - \frac{4}{9}(2)\right) 2 = \left(1 - \frac{8}{9}\right) 2 = \boxed{\frac{2}{9} \approx 0.2222}.$$

The covariance decreases because the new sample adds information:

$$\boxed{P(t) < P(t-1)}.$$

Consequently the gain tends to decrease as information accumulates.

Course reference: Module 3.3, Exercise 3.

Solution 4

(a) With $\lambda < 1$, older measurements receive exponentially smaller weight:

$$\lambda^{t-k}.$$

Thus recent data dominate and the estimator can adapt to changes in the plant.

(b) The covariance update contains

$$\frac{1}{\lambda}.$$

If a direction is not excited, the new measurement does not provide information in that direction, but forgetting still inflates its uncertainty.

Therefore

$$\boxed{P \text{ can grow: estimator windup.}}$$

(c) The required consistency condition is

$$\boxed{E[\phi(k)v(k)] = 0}.$$

If the regressor and noise are correlated, this condition fails and LS can be biased.

(d) One appropriate method is

$$\boxed{\text{Extended Least Squares (ELS)}}.$$

Other suitable formulations include ARMAX or an appropriate ML approach.

Course reference: Module 3.4, Appendix Exercises A.5 and A.7.

Solution 5

Under Gaussian noise, ML and LS coincide. Thus

$$\hat{\theta}_{\text{LS}} = \frac{1+3+4+20}{4} = \boxed{7}.$$

For Laplacian noise,

$$p(e) \propto e^{-|e|/b},$$

so

$$\log L(\theta) = \text{constant} - \frac{1}{b}\sum_t |y(t) - \theta|.$$

Therefore ML minimizes

$$\sum_t |y(t) - \theta|,$$

whose minimizer is the median.

The ordered data are

$$1,\ 3,\ 4,\ 20.$$

Hence

$$\hat{\theta}_{\text{ML}} = \frac{3+4}{2} = \boxed{3.5}.$$

Therefore

$$\boxed{\hat{\theta}_{\text{LS}} = 7, \qquad \hat{\theta}_{\text{ML}} = 3.5.}$$

The Laplacian ML estimate is much less affected by the outlier 20 because the ℓ_1 loss grows linearly with the error, whereas the LS quadratic loss grows as the square of the error.

Course reference: Module 3.6, Appendix Exercise A.8.

30-second MAP30 memory check

Before the real MAP, be able to write these from memory:

$$\boxed{\Phi^\top \Phi\, \hat{\theta} = \Phi^\top \bar{y}}$$

$$\boxed{\hat{\theta} = (\Phi^\top \Phi)^{-1} \Phi^\top \bar{y}}$$

$$\boxed{\Phi^\top \Phi \text{ invertible } \iff \text{ unique LS estimate}}$$

$$\boxed{P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}}$$

$$\boxed{K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)}}$$

$$\boxed{\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\varepsilon(t)}$$

$$\boxed{P(t) = \frac{[I - K(t)\phi^\top(t-1)]P(t-1)}{\lambda}}$$

$$\boxed{N_{\text{eff}} \approx \frac{1}{1-\lambda}}$$

$$\boxed{E[\phi v] = 0 \quad \text{is required for LS consistency}}$$

$$\boxed{\text{Gaussian noise} \Rightarrow \text{ML} = \text{LS}}$$

$$\boxed{\text{Laplacian noise} \Rightarrow \text{ML minimizes } \sum |e|}$$

$$\boxed{\text{MA(1): } R_v(0) = (1+c^2)\sigma^2, \quad R_v(1) = c\sigma^2, \quad R_v(\ell) = 0 \ (|\ell| > 1)}$$

Highest-priority procedures:

$$\boxed{\text{LS setup} \to \Phi^\top \Phi \to \text{normal equation} \to \hat{\theta} \to \text{residuals/SSE} \to \text{covariance}}$$

$$\boxed{\text{RLS} \to \varepsilon \to K \to \hat{\theta}^+ \to P^+}$$

$$\boxed{\text{Identifiability} \to \Phi^\top \Phi \text{ nonsingular} \to \text{all parameter directions excited}}$$

Course coverage: Module 3.1–3.6; Lecture 3, Exercises 1–4 and Appendix A.1–A.9.

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MOCK MAP30 #2 — Final Exam-Style

Module 3 **Time: 30 minutes** **Total: 20 points**

Instructions. Show the essential reasoning. Numerical answers should be simplified. The formula sheet is available, but no notes or AI tools are allowed in the real MAP.

Problem 1 — Least Squares and quality of the estimate [5 points]

Consider

$$y(k) = a\,x(k) + e(k)$$

with measurements

$$(x, y) = \{(1, 1.1), (2, 2.0), (3, 2.9), (4, 4.2)\}.$$

(a) Write the regressor form and the normal equation. [1 pt]

(b) Compute $\hat{a}$. [1 pt]

(c) Compute the residuals and

$$\text{SSE} = \sum_k \hat{e}^2(k).$$

[1 pt]

(d) Compute

$$\hat{\sigma}^2 = \frac{\text{SSE}}{N - p}$$

and the standard error $\text{se}(\hat{a})$. [1.5 pts]

(e) If four additional measurements with $x = 5, 6, 7, 8$ are collected with the same noise level, does the standard error increase or decrease? Explain. [0.5 pt]

Course reference: Module 3.1–3.2, Lecture 3, slides 9–11.

Problem 2 — Identifiability and persistent excitation [4 points]

Consider

$$y(k) = b_0 u(k-1) + b_1 u(k-2) + e(k),$$

with

$$\phi^\top(k) = \begin{bmatrix} u(k-1) & u(k-2) \end{bmatrix}.$$

For each input, determine whether b_0 and b_1 can be uniquely identified.

(a)

$$u(k) = 1.$$

[1 pt]

(b)

$$u(k) = (-1)^k.$$

[1 pt]

(c)

$$u(k) = 1, 1, -1, -1, 1, 1, -1, -1, \ldots$$

[1 pt]

(d) State the general condition on $\Phi^\top\Phi$ required for a unique LS estimate. [1 pt]

Course reference: Module 3.2, Lecture 3, slides 35–37.

Problem 3 — One RLS update with forgetting [4 points]

At time $t-1$,

$$\hat{\theta}(t-1) = \begin{bmatrix} 0.6 \\ -0.1 \end{bmatrix}, \qquad P(t-1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

The new measurement is

$$y(t) = 1.0, \qquad \phi(t-1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

and

$$\lambda = 0.9.$$

(a) Compute the prediction error

$$\varepsilon(t) = y(t) - \phi^\top(t-1)\hat{\theta}(t-1).$$

[1 pt]

(b) Compute the gain

$$K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)}.$$

[1 pt]

(c) Compute the updated parameter estimate

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\varepsilon(t).$$

[1 pt]

(d) State what forgetting does to $P(t)$ and explain the estimator-windup problem when there is poor excitation. [1 pt]

Course reference: Module 3.3–3.4, Lecture 3, slides 7–8.

Problem 4 — Coloured noise and stationarity [3 points]

(a) Let

$$v(k) = e(k) + 0.5e(k-1),$$

where $e(k)$ is zero-mean white noise with variance σ^2. Compute

$$r_v(0), \qquad r_v(1), \qquad r_v(\tau),\ |\tau| \geq 2.$$

[1.5 pts]

(b) For

$$x(k) = 0.8x(k-1) + e(k),$$

with white noise variance $\sigma^2 = 1$, compute the stationary variance $r_x(0)$. [0.75 pt]

(c) Is the AR(1) process stationary? Justify. [0.75 pt]

Course reference: Module 3.5, Lecture 3, slides 20 and 43–45.

Problem 5 — Bias and ML versus LS [4 points]

(a) An LS model is written as

$$y(k) = \phi^\top(k)\theta + v(k).$$

State the condition required for asymptotic LS consistency. [1 pt]

(b) Explain why coloured noise can violate this condition and produce a biased estimate. [1 pt]

(c) An unknown constant is measured as

$$y = \{1, 2, 10\}.$$

Compute the LS estimate under i.i.d. Gaussian noise. [0.5 pt]

(d) Suppose instead the measurement noise is Laplacian:

$$p(e) \propto \exp\left(-\frac{|e|}{b}\right).$$

State the ML objective and compute the ML estimate. [1 pt]

(e) Which estimate is more robust to the outlier 10? [0.5 pt]

Course reference: Module 3.4 and 3.6, Lecture 3, slides 21–25 and 45–48.

Solutions — MOCK MAP30 #2

Solution 1

The model is

$$y(k) = \phi(k)\theta + e(k),$$

with

$$\phi(k) = x(k), \qquad \theta = a.$$

Therefore

$$\Phi = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

The normal equation is

$$\Phi^\top \Phi\, \hat{a} = \Phi^\top \bar{y},$$

or

$$\left(\sum_k x^2(k) \right) \hat{a} = \sum_k x(k) y(k).$$

Compute

$$\sum_k x(k)y(k) = 1(1.1) + 2(2.0) + 3(2.9) + 4(4.2) = 30.6,$$

and

$$\sum_k x^2(k) = 1 + 4 + 9 + 16 = 30.$$

Hence

$$\boxed{\hat{a} = \frac{30.6}{30} = 1.02}.$$

The predictions are

$$\hat{y}(k) = 1.02x(k),$$

so

k	1	2	3	4
$\hat{y}(k)$	1.02	2.04	3.06	4.08
$\hat{e}(k)$	0.08	−0.04	−0.16	0.12

Thus

$$\text{SSE} = 0.08^2 + (-0.04)^2 + (-0.16)^2 + 0.12^2 = \boxed{0.048}.$$

There are

$$N = 4, \qquad p = 1,$$

therefore

$$\hat{\sigma}^2 = \frac{0.048}{4 - 1} = \boxed{0.016}.$$

Since

$$P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1},$$

we obtain

$$\text{Var}(\hat{a}) = \frac{0.016}{30} = 5.333 \times 10^{-4}.$$

Therefore

$$\boxed{\text{se}(\hat{a}) = \sqrt{5.333 \times 10^{-4}} \approx 0.0231.}$$

With additional points at $x = 5, \ldots, 8$,

$$\Phi^\top \Phi = \sum x^2(k)$$

increases. Therefore the information increases and the standard error decreases.

$$\boxed{\text{More informative data} \Rightarrow \text{smaller parameter uncertainty.}}$$

Course reference: Module 3.1–3.2, Lecture 3, slides 9–11.

Solution 2

(a) Constant input If

$$u(k) = 1,$$

then every row of Φ is

$$[1 \;\; 1].$$

Therefore

$$\Phi^\top \Phi = \begin{bmatrix} N & N \\ N & N \end{bmatrix},$$

so

$$\det(\Phi^\top \Phi) = 0.$$

Hence

$$\boxed{\text{not identifiable.}}$$

(b) Period-2 input For

$$u(k) = (-1)^k,$$

the regressor alternates between

$$[1 \;\; -1]$$

and

$$[-1 \;\; 1].$$

These are the same direction with opposite signs. Thus

$$\Phi^\top \Phi = \begin{bmatrix} N & -N \\ -N & N \end{bmatrix},$$

whose determinant is zero.

Hence

$$\boxed{\text{not identifiable.}}$$

The important lesson is

$$\boxed{\text{fast input} \not\Rightarrow \text{persistently exciting input.}}$$

(c) Period-4 input The regressor cycles through

$$[1 \;\; 1], \quad [-1 \;\; 1], \quad [-1 \;\; -1], \quad [1 \;\; -1].$$

These span both parameter directions. For a complete number of periods,

$$\Phi^\top \Phi \propto I,$$

and therefore

$$\det(\Phi^\top \Phi) \neq 0.$$

Hence

$$\boxed{\text{identifiable.}}$$

(d) A unique LS estimate requires

$$\boxed{\Phi^\top \Phi \text{ to be invertible}}$$

or equivalently

$$\boxed{\text{rank}(\Phi) = p.}$$

Course reference: Module 3.2, Lecture 3, slides 35–37.

Solution 3

First compute the predicted output:

$$\phi^\top \hat{\theta} = [1 \;\; 2] \begin{bmatrix} 0.6 \\ -0.1 \end{bmatrix} = 0.6 - 0.2 = 0.4.$$

Therefore

$$\boxed{\varepsilon(t) = 1.0 - 0.4 = 0.6}.$$

Next,

$$P(t-1)\phi(t-1) = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Also,

$$\phi^\top P \phi = [1 \;\; 2] \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 6.$$

Hence

$$K(t) = \frac{1}{0.9 + 6} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} 0.2899 \\ 0.2899 \end{bmatrix}}.$$

The updated estimate is

$$\hat{\theta}(t) = \begin{bmatrix} 0.6 \\ -0.1 \end{bmatrix} + 0.6 \begin{bmatrix} 0.2899 \\ 0.2899 \end{bmatrix},$$

giving

$$\boxed{\hat{\theta}(t) \approx \begin{bmatrix} 0.7739 \\ 0.0739 \end{bmatrix}}.$$

With forgetting,

$$P(t) = \frac{[I - K(t)\phi^\top(t-1)]P(t-1)}{\lambda}.$$

The division by

$$\lambda < 1$$

inflates uncertainty. This allows RLS to keep learning from recent data instead of eventually becoming insensitive to new measurements.

However, if a parameter direction is poorly excited, the data do not reduce uncertainty in that direction while forgetting continues to inflate $P(t)$. Consequently,

$$\boxed{P(t) \text{ can grow in unexcited directions.}}$$

A later measurement can then cause an excessively large parameter jump: this is estimator windup.

Course reference: Module 3.3–3.4, Lecture 3, slides 7–8.

Solution 4

Given

$$v(k) = e(k) + 0.5e(k-1),$$

with white noise

$$E[e(k)e(j)] = \sigma^2\delta(k-j),$$

we obtain

$$r_v(0) = E[v^2(k)] = (1+0.5^2)\sigma^2 = \boxed{1.25\sigma^2}.$$

For lag one,

$$r_v(1) = E[v(k+1)v(k)] = 0.5\sigma^2.$$

Hence

$$\boxed{r_v(1) = 0.5\sigma^2}.$$

For $|\tau| \geq 2$, there are no common white-noise terms:

$$\boxed{r_v(\tau) = 0, \qquad |\tau| \geq 2.}$$

Thus the MA(1) process only has non-zero covariance at lag zero and lag one.
For

$$x(k) = 0.8x(k-1) + e(k),$$

the stationary variance satisfies

$$r_x(0) = 0.8^2 r_x(0) + \sigma^2.$$

Therefore

$$r_x(0)(1-0.64) = 1,$$

and

$$\boxed{r_x(0) = \frac{1}{0.36} \approx 2.778.}$$

The process is stationary because

$$\boxed{|0.8| < 1}.$$

Its covariance is

$$r_x(\tau) = 0.8^{|\tau|} r_x(0),$$

which decays geometrically.
Course reference: Module 3.5, Lecture 3, slides 43–45.

Solution 5

For

$$y = \phi^\top\theta + v,$$

the asymptotic LS expression is

$$\hat{\theta}_N = \theta_0 + \left[\frac{1}{N}\sum \phi\phi^\top\right]^{-1}\left[\frac{1}{N}\sum \phi v\right].$$

Therefore consistency requires

$$\boxed{E[\phi v] = 0}.$$

With coloured noise, past noise can enter the regressor. Consequently,

$$\boxed{E[\phi v] \neq 0}$$

can occur, producing an asymptotic bias:

$$\boxed{\hat{\theta}_\infty \neq \theta_0}.$$

A standard course solution is Extended Least Squares, which augments the regressor with past residuals.

For the measurements

$$\{1, 2, 10\},$$

under Gaussian noise,

$$\hat{\theta}_{\mathrm{LS}} = \frac{1+2+10}{3} = \boxed{\frac{13}{3} \approx 4.33}.$$

For Laplacian noise,

$$p(e) \propto \exp\left(-\frac{|e|}{b}\right).$$

The log-likelihood is

$$\log L(\theta) = \text{constant} - \frac{1}{b}\sum_t |y(t) - \theta|.$$

Thus ML minimizes

$$\boxed{\sum_t |y(t) - \theta|}.$$

The minimizer is the median:

$$\boxed{\hat{\theta}_{\mathrm{ML}} = 2}.$$

Therefore

$$\boxed{\hat{\theta}_{\mathrm{LS}} \approx 4.33, \qquad \hat{\theta}_{\mathrm{ML}} = 2.}$$

The Laplacian ML estimate is more robust to the outlier 10 because the ℓ_1 loss grows linearly with the error, while LS uses a quadratic loss.

Course reference: Module 3.4 and 3.6, Lecture 3, slides 21–25 and 45–48.

MAP30 #2 — 60-second final checklist

Before the MAP, make sure you can write these without looking:

$$\boxed{\hat{\theta} = (\Phi^\top \Phi)^{-1} \Phi^\top \bar{y}}$$

$$\boxed{\Phi^\top \Phi \text{ invertible} \iff \text{unique LS estimate}}$$

$$\boxed{\hat{\sigma}^2 = \frac{\text{SSE}}{N-p}}$$

$$\boxed{P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}}$$

$$\boxed{\text{se}(\hat{\theta}_i) = \sqrt{P_{ii}}}$$

$$\boxed{K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)}}$$

$$\boxed{\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\varepsilon(t)}$$

$$\boxed{P(t) = \frac{[I - K(t)\phi^\top(t-1)]P(t-1)}{\lambda}}$$

$$\boxed{N_{\text{eff}} \approx \frac{1}{1-\lambda}}$$

$$\boxed{E[\phi v] = 0 \quad \text{for LS consistency}}$$

$$\boxed{\text{Gaussian noise} \Rightarrow ML = LS}$$

$$\boxed{\text{Laplacian noise} \Rightarrow ML = \arg\min_\theta \sum |y - \phi^\top \theta|}$$

$$\boxed{\text{MA(1): } r(0) = (1+c^2)\sigma^2, \quad r(1) = c\sigma^2, \quad r(\tau) = 0 \ (|\tau| \geq 2)}$$

$$\boxed{\text{AR(1) stationary} \iff |a| < 1}$$

The five patterns most likely to appear:

1. LS setup → normal equation → numerical estimate → residuals/SSE → uncertainty.
2. Check $\Phi^\top \Phi$ / rank to decide identifiability.
3. One RLS update:
$$\varepsilon \to K \to \hat{\theta}^+ \to P^+.$$

4. Explain coloured-noise bias through
$$E[\phi v] \neq 0.$$
5. Distinguish Gaussian ML/LS from Laplacian ML and recognize the median as the constant-parameter Laplacian ML estimate.

Most important conceptual traps:

- "Fast" input does **not** necessarily mean persistent excitation.
- More samples help only if they provide information in the relevant parameter directions.
- With forgetting, P can **increase** without excitation.
- Coloured noise can make LS **biased**; more data do not remove this asymptotic bias.
- ML equals LS only under the relevant Gaussian-noise assumptions.

Course reference: Lecture 3, slides 25–28 and Appendix A.1–A.9. "' "'latex

MOCK MAP30 #2 — Module 3 — Version D

Time: 30 minutes **Total: 20 points**

Instructions. Show the essential reasoning. Assume all noise variables have zero mean. Use the notation and conventions from Module 3.

Problem 1 — LS, intercept and covariance [4 points]

Consider
$$y(k) = a\,u(k) + b + e(k),$$
with the following sufficient statistics:
$$N = 5, \qquad \sum_{k=1}^{5} u(k) = 10, \qquad \sum_{k=1}^{5} u^2(k) = 30,$$
$$\sum_{k=1}^{5} y(k) = 8, \qquad \sum_{k=1}^{5} u(k)y(k) = 20.$$

(a) Write $\Phi^\top\Phi$ and $\Phi^\top\bar{y}$. [1 pt]

(b) Solve the normal equations and determine
$$\hat{\theta} = \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix}.$$
[1.5 pts]

(c) Suppose
$$\hat{\sigma}^2 = 0.20.$$
Compute the covariance matrix
$$P = \hat{\sigma}^2(\Phi^\top\Phi)^{-1}.$$
[1 pt]

(d) Which parameter has the larger standard error? [0.5 pt]

Problem 2 — Excitation and identifiability [4 points]

Consider

$$y(k) = a\,u(k-1) + b\,u(k-2) + e(k).$$

The regressor is

$$\phi(k) = \begin{bmatrix} u(k-1) \\ u(k-2) \end{bmatrix}.$$

For each experiment, determine whether the two parameters are uniquely identifiable.

(a)

$$u(k) = 2$$

for every k. [1 pt]

(b)

$$u(k) = \{1, -1, 1, -1, 1, -1, \ldots\}.$$

[1 pt]

(c)

$$u(k) = \{1, 1, -1, -1, 1, 1, -1, -1, \ldots\}.$$

[1 pt]

(d) Give the matrix condition that distinguishes the identifiable and non-identifiable cases. [1 pt]

Problem 3 — RLS update and forgetting [4 points]

At time $k-1$,

$$\hat{\theta}(k-1) = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix}, \qquad P(k-1) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

A new observation is

$$\phi(k-1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad y(k) = 0.5.$$

Use

$$\lambda = 0.8.$$

(a) Compute the prediction error

$$\varepsilon(k) = y(k) - \phi^{\top}(k-1)\hat{\theta}(k-1).$$

[1 pt]

(b) Compute the RLS gain $K(k)$. [1 pt]

(c) Compute $\hat{\theta}(k)$. [1 pt]

(d) Without performing the complete matrix calculation, explain why the covariance is affected by the forgetting factor. [1 pt]

Problem 4 — Stochastic processes [4 points]

Consider

$$v(k) = e(k) + c\,e(k-1),$$

where $e(k)$ is white noise with

$$E[e(k)] = 0, \qquad E[e^2(k)] = \sigma^2.$$

(a) Derive $r_v(0)$. [1 pt]

(b) Derive $r_v(1)$. [1 pt]

(c) What is $r_v(\ell)$ for $|\ell| > 1$? [0.5 pt]

(d) Explain why $v(k)$ is not white for $c \neq 0$. [0.5 pt]

(e) Consider

$$x(k) = a\,x(k-1) + e(k).$$

State the stationarity condition and derive its stationary variance. [1 pt]

Problem 5 — Coloured noise, ML and robustness [4 points]

An unknown constant θ is measured through

$$y(k) = \theta + v(k).$$

The observations are

$$y = \{2,\ 3,\ 3,\ 20\}.$$

(a) Under i.i.d. Gaussian noise, compute the LS/ML estimate. [1 pt]

(b) Under Laplacian noise,

$$p(v) \propto \exp\left(-\frac{|v|}{b}\right),$$

write the ML optimization problem. [1 pt]

(c) Compute the Laplacian ML estimate. [1 pt]

(d) Suppose instead that $v(k)$ is coloured and correlated with the regressor in a dynamic identification problem. State the condition that fails and explain the consequence for LS. [1 pt]

Solutions — MOCK MAP30 #2 — Version D

Solution 1

The model is

$$y(k) = \begin{bmatrix} u(k) & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + e(k).$$

Therefore

$$\Phi^\top \Phi = \begin{bmatrix} \sum u^2(k) & \sum u(k) \\ \sum u(k) & N \end{bmatrix} = \begin{bmatrix} 30 & 10 \\ 10 & 5 \end{bmatrix},$$

and

$$\Phi^\top \bar{y} = \begin{bmatrix} \sum u(k)y(k) \\ \sum y(k) \end{bmatrix} = \begin{bmatrix} 20 \\ 8 \end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix} 30 & 10 \\ 10 & 5 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 20 \\ 8 \end{bmatrix}.$$

The determinant is

$$30(5) - 10^2 = 50.$$

Thus

$$(\Phi^\top \Phi)^{-1} = \frac{1}{50} \begin{bmatrix} 5 & -10 \\ -10 & 30 \end{bmatrix}.$$

Hence

$$\hat{\theta} = \frac{1}{50} \begin{bmatrix} 5 & -10 \\ -10 & 30 \end{bmatrix} \begin{bmatrix} 20 \\ 8 \end{bmatrix},$$

so

$$\boxed{\hat{a} = 0.4, \qquad \hat{b} = 1.6.}$$

With

$$\hat{\sigma}^2 = 0.20,$$

the covariance matrix is

$$P = 0.20 \frac{1}{50} \begin{bmatrix} 5 & -10 \\ -10 & 30 \end{bmatrix}.$$

Therefore

$$\boxed{P = \begin{bmatrix} 0.02 & -0.04 \\ -0.04 & 0.12 \end{bmatrix}.}$$

The standard errors are

$$\mathrm{se}(\hat{a}) = \sqrt{0.02} \approx 0.1414,$$

and

$$\mathrm{se}(\hat{b}) = \sqrt{0.12} \approx 0.3464.$$

Hence

$$\boxed{\hat{b} \text{ has the larger standard error.}}$$

Course reference: Module 3.1–3.2; Appendix Exercise A.1/A.3.

Solution 2

(a) Constant input If

$$u(k) = 2,$$

then

$$\phi(k) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Every row of Φ is identical, so

$$\operatorname{rank}(\Phi) = 1 < 2.$$

Therefore

$$\boxed{\text{the two parameters are not identifiable.}}$$

Equivalently,

$$\Phi^\top \Phi$$

is singular.

(b) Alternating input For

$$u(k) = 1, -1, 1, -1, \ldots,$$

the regressor alternates between

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

and

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

These vectors are linearly dependent. Consequently

$$\operatorname{rank}(\Phi) = 1.$$

Thus

$$\boxed{\text{not identifiable.}}$$

This is a useful trap:

$$\boxed{\text{periodic} \not\Rightarrow \text{persistently exciting.}}$$

(c) Period-four input For

$$u(k) = 1, 1, -1, -1, \ldots,$$

the regressor cycles through four directions:

$$[1, 1]^\top, \quad [1, -1]^\top, \quad [-1, -1]^\top, \quad [-1, 1]^\top.$$

These span the complete two-dimensional parameter space.

Hence

$$\operatorname{rank}(\Phi) = 2$$

over complete periods, and

$$\boxed{\text{the parameters are identifiable.}}$$

(d) The required condition is

$$\boxed{\det(\Phi^\top \Phi) \neq 0}$$

or equivalently

$$\boxed{\operatorname{rank}(\Phi) = p.}$$

Course reference: Module 3.2; Appendix Exercise A.2.

Solution 3

The predicted output is

$$\hat{y}(k) = \phi^\top(k-1)\hat{\theta}(k-1).$$

Thus

$$\hat{y}(k) = [1 \quad -1] \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix} = 0.4 - 0.2 = 0.2.$$

Therefore

$$\boxed{\varepsilon(k) = 0.5 - 0.2 = 0.3}.$$

Next,

$$P(k-1)\phi(k-1) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Also,

$$\phi^\top P\phi = [1 \quad -1] \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 3.$$

Therefore

$$K(k) = \frac{1}{0.8+3} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \boxed{\begin{bmatrix} 0.2632 \\ -0.5263 \end{bmatrix}}.$$

The updated estimate is

$$\hat{\theta}(k) = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix} + 0.3 \begin{bmatrix} 0.2632 \\ -0.5263 \end{bmatrix}.$$

Hence

$$\boxed{\hat{\theta}(k) \approx \begin{bmatrix} 0.4789 \\ 0.0421 \end{bmatrix}.}$$

The covariance update is

$$P(k) = \frac{[I - K(k)\phi^\top(k-1)]P(k-1)}{\lambda}.$$

The factor

$$\frac{1}{\lambda} > 1$$

inflates the prior covariance. Thus forgetting deliberately reduces the effective amount of memory in the estimator.

If the regressor does not excite a parameter direction, the data provide little information in that direction while forgetting continues to inflate its covariance. This can produce estimator windup.

Course reference: Module 3.3–3.4; Appendix Exercise A.4/A.5.

Solution 4

We have

$$v(k) = e(k) + ce(k-1).$$

For zero lag,

$$r_v(0) = E[v^2(k)].$$

Expanding:

$$r_v(0) = E[(e(k) + ce(k-1))^2].$$

Therefore

$$r_v(0) = E[e^2(k)] + 2cE[e(k)e(k-1)] + c^2E[e^2(k-1)].$$

Since $e(k)$ is white,

$$E[e(k)e(k-1)] = 0.$$

Hence

$$\boxed{r_v(0) = (1 + c^2)\sigma^2.}$$

For lag one,

$$r_v(1) = E[v(k)v(k-1)].$$

Using

$$v(k-1) = e(k-1) + ce(k-2),$$

we obtain

$$r_v(1) = E[(e(k) + ce(k-1))(e(k-1) + ce(k-2))].$$

All cross terms vanish except

$$cE[e^2(k-1)].$$

Thus

$$\boxed{r_v(1) = c\sigma^2.}$$

By symmetry,

$$r_v(-1) = c\sigma^2.$$

For $|\ell| > 1$, there are no common white-noise terms:

$$\boxed{r_v(\ell) = 0.}$$

Therefore, for $c \neq 0$,

$$r_v(1) \neq 0,$$

so samples at adjacent time instants are correlated.

Hence

$$\boxed{v(k) \text{ is coloured, not white.}}$$

For the AR(1) process

$$x(k) = a\,x(k-1) + e(k),$$

stationarity requires

$$\boxed{|a| < 1}.$$

For the stationary variance,

$$r_x(0) = E[x^2(k)].$$

Using the model,

$$r_x(0) = a^2 r_x(0) + \sigma^2.$$

Thus

$$r_x(0)(1 - a^2) = \sigma^2,$$

giving

$$\boxed{r_x(0) = \frac{\sigma^2}{1 - a^2}}.$$

Course reference: Module 3.5; Appendix Exercise A.6.

Solution 5

Under i.i.d. Gaussian noise, the likelihood is maximized by minimizing the squared error:

$$\sum_k (y(k) - \theta)^2.$$

Therefore

$$\hat{\theta}_{\text{LS}} = \frac{2 + 3 + 3 + 20}{4} = \boxed{7}.$$

For Laplacian noise,

$$p(v) \propto \exp\left(-\frac{|v|}{b}\right).$$

Hence the negative log-likelihood is proportional to

$$\sum_k |y(k) - \theta|.$$

Therefore

$$\boxed{\hat{\theta}_{\text{ML}} = \arg\min_\theta \sum_k |y(k) - \theta|.}$$

The minimizer is the median.

The ordered observations are

$$2,\ 3,\ 3,\ 20.$$

Since there are four observations,

$$\hat{\theta}_{\text{ML}} = \frac{3 + 3}{2} = \boxed{3}.$$

Thus

$$\boxed{\hat{\theta}_{\text{Gaussian}} = 7, \qquad \hat{\theta}_{\text{Laplacian}} = 3.}$$

The Laplacian estimate is much less affected by the outlier 20.

For a dynamic identification problem with coloured noise, LS consistency requires

$$\boxed{E[\phi(k)v(k)] = 0.}$$

If the regressor and noise are correlated,

$$E[\phi(k)v(k)] \neq 0.$$

Then the asymptotic bias does not disappear simply by collecting more data:

$$\boxed{\hat{\theta}_\infty \neq \theta_0.}$$

Course reference: Module 3.4 and 3.6; Appendix Exercises A.7–A.8.

Module 3 — Absolute Must-Know Procedures

1. Batch LS

For

$$\bar{y} = \Phi\theta + e,$$

immediately write

$$\boxed{\hat{\theta} = (\Phi^\top \Phi)^{-1} \Phi^\top \bar{y}}.$$

If $\Phi^\top \Phi$ is singular:

$$\boxed{\text{no unique LS estimate.}}$$

For a scalar model

$$y(k) = ax(k) + e(k),$$

use

$$\boxed{\hat{a} = \frac{\sum x(k)y(k)}{\sum x^2(k)}}.$$

For an intercept,

$$y(k) = ax(k) + b + e(k),$$

use

$$\boxed{\Phi = \begin{bmatrix} x(1) & 1 \\ \vdots & \vdots \\ x(N) & 1 \end{bmatrix}}.$$

2. Parameter uncertainty

Remember

$$\boxed{P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}}$$

and

$$\boxed{\text{se}(\hat{\theta}_i) = \sqrt{P_{ii}}.}$$

Also

$$\boxed{\hat{\sigma}^2 = \frac{\text{SSE}}{N - p}.}$$

3. Identifiability

For p parameters:

$$\boxed{\text{rank}(\Phi) = p}$$

or

$$\boxed{\Phi^\top \Phi \text{ nonsingular.}}$$

Interpretation:

$$\boxed{\text{Every parameter direction must be sufficiently excited.}}$$

4. RLS

Always calculate in this order:

$$\boxed{\varepsilon \to K \to \hat{\theta}^{+} \to P^{+}.}$$

Prediction error:

$$\boxed{\varepsilon(k) = y(k) - \phi^{\top}(k-1)\hat{\theta}(k-1).}$$

Gain:

$$\boxed{K(k) = \frac{P(k-1)\phi(k-1)}{\lambda + \phi^{\top}(k-1)P(k-1)\phi(k-1)}.}$$

Estimate:

$$\boxed{\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)\varepsilon(k).}$$

Covariance:

$$\boxed{P(k) = \frac{[I - K(k)\phi^{\top}(k-1)]P(k-1)}{\lambda}.}$$

5. Forgetting factor

$$\boxed{0 < \lambda \leq 1.}$$

Approximate effective memory:

$$\boxed{N_{\text{eff}} \approx \frac{1}{1-\lambda}.}$$

Smaller λ:

$$\boxed{\text{faster adaptation but greater sensitivity/noise.}}$$

Without excitation:

$$\boxed{P(k) \uparrow \quad \Rightarrow \quad \text{possible estimator windup.}}$$

6. Coloured noise

For

$$v(k) = e(k) + ce(k-1),$$

remember

$$\boxed{r_v(0) = (1+c^2)\sigma^2}$$

$$\boxed{r_v(1) = c\sigma^2}$$

$$\boxed{r_v(\ell) = 0, \qquad |\ell| > 1.}$$

For

$$x(k) = ax(k-1) + e(k),$$

stationarity:

$$\boxed{|a| < 1}$$

and

$$\boxed{r_x(0) = \frac{\sigma^2}{1-a^2}.}$$

7. LS bias

The key condition is

$$\boxed{E[\phi(k)v(k)] = 0.}$$

If

$$E[\phi(k)v(k)] \neq 0,$$

then LS can be biased:

$$\boxed{\hat{\theta}_\infty \neq \theta_0.}$$

Do not confuse:

$$\boxed{\text{coloured noise}}$$

with

$$\boxed{\text{necessarily biased LS}}.$$

The critical issue is whether the noise is correlated with the regressor.

8. Gaussian versus Laplacian ML

Gaussian:

$$p(e) \propto e^{-e^2/(2\sigma^2)}$$

gives

$$\boxed{\text{ML} = \text{LS}}$$

and minimizes

$$\boxed{\sum e^2.}$$

Laplacian:

$$p(e) \propto e^{-|e|/b}$$

gives

$$\boxed{\text{ML} = \arg\min_\theta \sum |e|.}$$

For a constant parameter:

$$\boxed{\hat{\theta}_{\text{ML,Laplacian}} = \text{median of the observations.}}$$

9. MAP30 decision tree

If the question gives data:

$$\boxed{\text{Build } \Phi \to \Phi^\top \Phi \to \hat{\theta}.}$$

If it asks whether parameters can be estimated:

$$\boxed{\text{Check rank/invertibility.}}$$

If it gives $\hat{\theta}$, P, ϕ, y:

$$\boxed{\text{RLS update.}}$$

If it mentions $\lambda < 1$:

$$\boxed{\text{Forgetting + effective memory + possible windup.}}$$

If it mentions correlated residuals/noise:

$$\boxed{\text{Check } E[\phi v] = 0.}$$

If it asks Gaussian ML:

$$\boxed{\text{LS / quadratic loss.}}$$

If it asks Laplacian ML:

$$\boxed{\ell_1 \text{ loss / median for a constant.}}$$

10. Full-score checklist

Before submitting, check:

- Did I define Φ and θ correctly?
- Did I check whether $\Phi^\top \Phi$ is invertible?
- Did I distinguish identifiability from merely having varying input?
- Did I calculate the RLS prediction error before the gain?
- Did I use the forgetting factor in the denominator of K?
- Did I divide the covariance update by λ?
- Did I recognize estimator windup?
- Did I check $E[\phi v] = 0$ before claiming LS consistency?
- Did I distinguish white from coloured noise using covariance?
- Did I remember $|a| < 1$ for stationary AR(1)?
- Did I distinguish Gaussian ML/LS from Laplacian ML?
- If an outlier appears, did I consider the ℓ_1/median solution?

“‘latex

Mock MAP 2 – Module 3

Time: 30 minutes **Total:** 20 points

Instructions: Show the essential reasoning. Numerical answers without justification may receive only partial credit.

Learning objectives:

- Formulate linear-in-the-parameters models in regressor form.
- Solve small least-squares problems and assess parameter uncertainty.
- Determine excitation and identifiability from $\Phi^\top \Phi$.
- Perform recursive least-squares (RLS) updates with forgetting.
- Characterize stochastic processes and distinguish white and coloured noise.
- Recognize when LS is biased and when ML differs from LS.

Problem 1 – Least squares with intercept and uncertainty [4 points]

Consider

$$y(k) = a\,x(k) + b + e(k)$$

with measurements

$$(x, y) = \{(1, 1.30), (2, 2.05), (3, 3.10), (4, 3.95)\}.$$

(a) Write the regressor $\phi(k)$, parameter vector θ, and matrices Φ and $\bar{y}$.

(b) Form $\Phi^\top \Phi$ and $\Phi^\top \bar{y}$ and solve the normal equations

$$\Phi^\top \Phi\, \hat{\theta} = \Phi^\top \bar{y}.$$

(c) Compute the residuals and

$$\mathrm{SSE} = \sum_k \hat{e}^2(k).$$

(d) Estimate the residual variance using

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{N - p},$$

and compute

$$P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}.$$

Give the standard errors of $\hat{a}$ and $\hat{b}$ and interpret the sign of $\mathrm{cov}(\hat{a}, \hat{b})$.

Problem 2 – Excitation and identifiability [4 points]

Consider the FIR model

$$y(k) = b_0 u(k-1) + b_1 u(k-2) + e(k), \qquad \phi^\top(k) = \begin{bmatrix} u(k-1) & u(k-2) \end{bmatrix}.$$

Three possible experiments are proposed:

(A) $u(k) = 1$ for every k.

(B) $u(k) = (-1)^k$.

(C) $u(k) = 1, 1, -1, -1, 1, 1, -1, -1, \ldots$

(a) For each experiment, describe the possible rows of Φ and determine whether

$$\Phi^\top \Phi$$

is invertible.

(b) State which experiments identify both b_0 and b_1.

(c) Explain why experiment (B) is not sufficient even though the input changes at every sample.

(d) State the general connection between excitation, the rank of Φ, and identifiability.

Problem 3 – RLS with forgetting [4 points]

Consider the two-parameter model

$$y(t) = \phi^\top(t-1)\theta + e(t)$$

and suppose that, immediately before a new measurement,

$$\hat{\theta}(t-1) = \begin{bmatrix} 0.40 \\ 0.20 \end{bmatrix}, \qquad P(t-1) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

with

$$\phi(t-1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad y(t) = 0.50, \qquad \lambda = 0.80.$$

Use

$$K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)},$$

$$\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\left[y(t) - \phi^\top(t-1)\hat{\theta}(t-1)\right],$$

and

$$P(t) = \left[I - K(t)\phi^\top(t-1)\right]\frac{P(t-1)}{\lambda}.$$

(a) Compute the prediction error.

(b) Compute $K(t)$.

(c) Compute $\hat{\theta}(t)$.

(d) Compute $P(t)$ and explain qualitatively why forgetting can prevent the estimator from becoming permanently insensitive to new data.

Problem 4 – Stochastic processes [4 points]

Let $e(k)$ be zero-mean white noise with variance σ^2.

(a) Consider the MA(1) process

$$v(k) = e(k) + c\,e(k-1).$$

Derive

$$r_v(0), \qquad r_v(1), \qquad r_v(\tau), \quad |\tau| > 1.$$

(b) Consider the AR(1) process

$$v(k) = a\,v(k-1) + e(k).$$

State the condition for weak stationarity and derive its stationary variance $r_v(0)$.

(c) For $c = 0.5$ and $\sigma^2 = 2$, calculate the MA(1) covariance values $r_v(0)$ and $r_v(1)$.

(d) Explain in one or two sentences why correlated residuals are evidence that the white-noise assumption behind ordinary LS may be inappropriate.

Problem 5 – Bias, ML and robustness [4 points]

Consider the scalar identification model

$$y(k) = a\,\phi(k) + v(k).$$

(a) State the condition

$$E[\phi(k)v(k)] = 0$$

and explain why its violation can produce asymptotic LS bias.

(b) Suppose instead that

$$v(k) = e(k) + c\,e(k-1),$$

while

$$\phi(k) = y(k-1).$$

Explain why $\phi(k)$ can become correlated with $v(k)$.

(c) You now estimate a constant θ from

$$y = \{1, 2, 10\}.$$

Under i.i.d. Gaussian noise, compute the ML/LS estimate.

(d) Suppose instead the noise is i.i.d. Laplacian,

$$p(e) \propto \exp\left(-\frac{|e|}{b}\right).$$

Show what objective ML minimizes and compute the ML estimate. Explain which estimate is more robust to the outlier 10.

Solutions

Solution 1

(a) The model is linear in the unknown parameters:

$$y(k) = \begin{bmatrix} x(k) & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + e(k).$$

Hence

$$\phi(k) = \begin{bmatrix} x(k) \\ 1 \end{bmatrix}, \qquad \theta = \begin{bmatrix} a \\ b \end{bmatrix},$$

and

$$\Phi = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}, \qquad \bar{y} = \begin{bmatrix} 1.30 \\ 2.05 \\ 3.10 \\ 3.95 \end{bmatrix}.$$

(b) We obtain

$$\Phi^\top \Phi = \begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix}, \qquad \Phi^\top \bar{y} = \begin{bmatrix} 30.5 \\ 10.4 \end{bmatrix}.$$

Therefore

$$\begin{bmatrix} 30 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} = \begin{bmatrix} 30.5 \\ 10.4 \end{bmatrix}.$$

Since

$$\det(\Phi^\top \Phi) = 30(4) - 10^2 = 20 \neq 0,$$

the solution is unique.

Using

$$(\Phi^\top \Phi)^{-1} = \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix},$$

gives

$$\hat{\theta} = \begin{bmatrix} 0.90 \\ 0.35 \end{bmatrix}.$$

(c) The fitted values are

$$\hat{y} = \{1.25,\ 2.15,\ 3.05,\ 3.95\}.$$

Therefore

$$\hat{e} = \{0.05,\ -0.10,\ 0.05,\ 0\}.$$

Hence

$$\text{SSE} = 0.05^2 + (-0.10)^2 + 0.05^2 = 0.015.$$

(d) Here

$$N = 4, \qquad p = 2,$$

so

$$\hat{\sigma}^2 = \frac{0.015}{4 - 2} = 0.0075.$$

Thus

$$P = 0.0075 \frac{1}{20} \begin{bmatrix} 4 & -10 \\ -10 & 30 \end{bmatrix} = \begin{bmatrix} 0.0015 & -0.00375 \\ -0.00375 & 0.01125 \end{bmatrix}.$$

Therefore

$$\text{se}(\hat{a}) = \sqrt{0.0015} \approx 0.0387,$$

and

$$\mathrm{se}(\hat{b}) = \sqrt{0.01125} \approx 0.1061.$$

The negative off-diagonal entry means that the slope and intercept estimates are negatively correlated: increasing the estimated slope tends to require a decrease in the estimated intercept to preserve the fit to the same data.

Course reference: Module 3.1–3.2; Lecture 3, slides 32–38 (Exercises A.1–A.3).

Solution 2

(a) The regressor is

$$\phi^\top(k) = \begin{bmatrix} u(k-1) & u(k-2) \end{bmatrix}.$$

Experiment A: $u(k) = 1$. Every row is

$$\phi^\top(k) = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

Thus

$$\Phi^\top \Phi = N \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

whose determinant is zero.

Experiment B: $u(k) = (-1)^k$. The rows alternate between

$$[1, -1] \quad \text{and} \quad [-1, 1].$$

These are only two versions of the same direction. Consequently

$$\Phi^\top \Phi = N \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix},$$

up to the precise number of samples, and it is singular.

Experiment C: the rows cycle through

$$[1, 1], \quad [-1, 1], \quad [-1, -1], \quad [1, -1].$$

Over a complete number of periods,

$$\Phi^\top \Phi \propto \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

which is nonsingular.

(b) Only experiment C identifies both parameters.

(c) Changing at every sample is not enough. Experiment B only excites one parameter direction because

$$[-1, 1] = -[1, -1].$$

Therefore the regressors remain linearly dependent.

(d) The parameters are identifiable when

$$\Phi^\top \Phi$$

is invertible, equivalently when Φ has full column rank. Persistent excitation provides the required richness in the parameter directions.

Course reference: Module 3.2; Lecture 3, slides 12–14 and 35–37 (Exercises 2 and A.2).

Solution 3

First compute

$$P(t-1)\phi(t-1) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

The scalar denominator is

$$\lambda + \phi^\top P \phi = 0.8 + \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 0.8 + 3 = 3.8.$$

(a) The prediction is

$$\phi^\top \hat{\theta}(t-1) = [1, -1] \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix} = 0.2.$$

Hence

$$\varepsilon(t) = 0.5 - 0.2 = 0.3.$$

(b)

$$K(t) = \frac{1}{3.8} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0.26316 \\ -0.52632 \end{bmatrix}.$$

(c) Therefore

$$\hat{\theta}(t) = \begin{bmatrix} 0.4 \\ 0.2 \end{bmatrix} + 0.3 \begin{bmatrix} 0.26316 \\ -0.52632 \end{bmatrix},$$

giving

$$\boxed{\hat{\theta}(t) \approx \begin{bmatrix} 0.47895 \\ 0.04211 \end{bmatrix}}$$

(d) First,

$$\frac{P(t-1)}{\lambda} = \begin{bmatrix} 1.25 & 0 \\ 0 & 2.5 \end{bmatrix}.$$

Using the RLS covariance update,

$$P(t) = \left[I - K(t)\phi^\top(t-1) \right] \begin{bmatrix} 1.25 & 0 \\ 0 & 2.5 \end{bmatrix}.$$

Equivalently,

$$P(t) = \frac{P(t-1)}{\lambda} - \frac{P(t-1)\phi\phi^\top P(t-1)}{\lambda\left(\lambda + \phi^\top P(t-1)\phi\right)}.$$

This gives approximately

$$\boxed{P(t) \approx \begin{bmatrix} 0.92105 & 0.52632 \\ 0.52632 & 1.44737 \end{bmatrix}}$$

Forgetting divides the previous covariance by $\lambda < 1$ before incorporating the new measurement. Thus old information gradually loses weight, preventing the estimator from becoming permanently "locked" to old data.

Course reference: Module 3.3–3.4; Lecture 3, slides 15–17 and 26 (RLS and forgetting).

Solution 4

(a) For

$$v(k) = e(k) + ce(k-1),$$

with white noise $e(k)$,

$$r_v(0) = E[v^2(k)] = E[(e(k) + ce(k-1))^2].$$

Since distinct white-noise samples are uncorrelated,

$$\boxed{r_v(0) = (1 + c^2)\sigma^2}.$$

For lag 1,

$$r_v(1) = E[v(k+1)v(k)] = c\sigma^2.$$

Hence

$$\boxed{r_v(1) = r_v(-1) = c\sigma^2}.$$

For $|\tau| > 1$, there are no common white-noise samples, so

$$\boxed{r_v(\tau) = 0, \qquad |\tau| > 1}.$$

(b) For

$$v(k) = av(k-1) + e(k),$$

the weak-stationarity condition is

$$\boxed{|a| < 1}.$$

In steady state,

$$r_v(0) = a^2 r_v(0) + \sigma^2.$$

Therefore

$$r_v(0)(1 - a^2) = \sigma^2,$$

giving

$$\boxed{r_v(0) = \frac{\sigma^2}{1 - a^2}}.$$

(c) With $c = 0.5$ and $\sigma^2 = 2$,

$$r_v(0) = (1 + 0.5^2)2 = 2.5,$$

and

$$r_v(1) = 0.5(2) = 1.$$

Thus

$$\boxed{r_v(0) = 2.5, \qquad r_v(1) = 1}.$$

(d) If residuals are correlated, the noise is not white. This invalidates an important assumption behind the standard LS interpretation and can indicate that the model has failed to capture some dynamic structure. In particular, if the regressor is correlated with the noise, LS can become biased.

Course reference: Module 3.5; Lecture 3, slides 19–20 and Appendix A.6.

Solution 5

(a) For

$$y(k) = a\phi(k) + v(k),$$

the LS estimator is consistent when the relevant orthogonality condition holds:

$$\boxed{E[\phi(k)v(k)] = 0}.$$

If

$$E[\phi(k)v(k)] \neq 0,$$

the noise term contributes systematically to the parameter estimate. Consequently, the LS estimator can converge to a value different from the true parameter even as the amount of data grows.

(b) Since

$$\phi(k) = y(k-1),$$

the regressor contains the previous disturbance. In particular,

$$y(k-1) = ay(k-2) + v(k-1).$$

Meanwhile

$$v(k) = e(k) + ce(k-1).$$

Because $v(k)$ contains $e(k-1)$ and $y(k-1)$ also contains the previous disturbance, $\phi(k)$ and $v(k)$ can be correlated. Hence the orthogonality condition can fail.

(c) Under i.i.d. Gaussian noise, ML and LS are equivalent. For a constant model, LS minimizes

$$\sum_k (y(k) - \theta)^2,$$

whose minimizer is the sample mean:

$$\hat{\theta}_{\text{LS}} = \frac{1+2+10}{3} = \boxed{\frac{13}{3} \approx 4.33}.$$

(d) For Laplacian noise,

$$p(e) \propto \exp\left(-\frac{|e|}{b}\right).$$

Assuming independent samples,

$$\log L(\theta) = \text{constant} - \frac{1}{b}\sum_k |y(k) - \theta|.$$

Therefore ML minimizes

$$\sum_k |y(k) - \theta|.$$

Its minimizer is the sample median:

$$\boxed{\hat{\theta}_{\text{ML}} = 2}.$$

Thus

$$\boxed{\hat{\theta}_{\text{LS}} \approx 4.33, \qquad \hat{\theta}_{\text{ML}} = 2}.$$

The Laplacian ML estimate is less affected by the outlier 10.

Course reference: Module 3.4 and 3.6; Lecture 3, slides 21–25 and Appendix A.7–A.8.

Final 100%-Score Checklist

Before the MAP, be able to write from memory:

$$\boxed{\hat{\theta} = (\Phi^\top \Phi)^{-1} \Phi^\top \bar{y}}$$

$$\boxed{\Phi^\top \Phi \text{ invertible} \iff \Phi \text{ has full column rank}}$$

$$\boxed{\hat{\sigma}^2 = \frac{\text{SSE}}{N-p}, \qquad P = \hat{\sigma}^2 (\Phi^\top \Phi)^{-1}}$$

$$\boxed{K(t) = \frac{P(t-1)\phi(t-1)}{\lambda + \phi^\top(t-1)P(t-1)\phi(t-1)}}$$

$$\boxed{\hat{\theta}(t) = \hat{\theta}(t-1) + K(t)\left[y(t) - \phi^\top(t-1)\hat{\theta}(t-1)\right]}$$

$$\boxed{P(t) = [I - K(t)\phi^\top(t-1)]\frac{P(t-1)}{\lambda}}$$

$$\boxed{E[\phi v] = 0 \quad \text{is required to avoid LS bias}}$$

$$\boxed{\text{Gaussian ML} = \text{LS}, \qquad \text{Laplacian ML} = \ell_1 \text{ minimization}}$$

Most likely MAP-style tasks:

$$\boxed{\text{LS setup} \rightarrow \text{identifiability} \rightarrow \text{one RLS update} \rightarrow \text{stochastic/bias/ML argument}}$$

‘‘‘